Explanation: Being a strong acid , 0.5 M HCl ionizes nearly completely as follows.

$HCl\ (aq) \longrightarrow H^+\ (aq) + Cl^-\ (aq)$
(1.0 mole) (1.0 mole)

Equal number of moles of HCl and H^+ gives rise to equal concentration in both. $[H^+] = [HCl]$ .Since the concentration of HCl is equal to 0.5M ,concentration of H^+ is also 0.5.

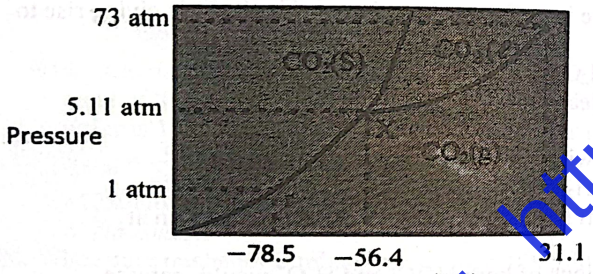
∴ Answer A

5.What is the name of a base – promoted ester hydrolysis reaction?
A .Acylation
B .Esterification
C. Condensation
D. Saponification.

Explanation: The alkaline hydrolysis of fats and vegetable oil is an important industrial process that results in the formation of soap. The process of making is known as **Saponification**.

∴ Answer D

6 .Consider the following phase diagram for $CO_2(S)$



What happens when in a CO_2 sample initially at 1 atm and -70°C, the temperature increases from -70°C to -10°C at a constant pressure of 60 atm?

A .$CO_2(g) \longrightarrow CO_2(s)$
B .$CO_2(g) \longrightarrow CO_2(\ell)$
C. $CO_2(g) \longrightarrow CO_2(g)$
D. $CO_2(g) \longrightarrow CO_2(\ell)$

Explanation: The question is based on the phase diagram of CO_2
i) At 1 atm and -70°C the solid and gaseous form of CO_2 exist in equilidrium.
ii) Between -70°C and -56°C , the CO_2 is in the solid form at 60 atm.
iii) As the temperature rises to -10°C at a constant pressure of 60 atm, the solid CO_2 liquifies.

$CO_2\ (s) \longrightarrow CO_2(\ell)$

∴ Answer B

7.What is the total number of valence –shell electrons in BrO_3^-?
A .20
B .26
C .32
D .36

Explanation: The total number of valence shell electrons in BrO_3^- is equal to 7(for Br) + 3×6(for oxygen) +1 (for the negative) that is 7+18+1 = 26 electrons.

∴ Answer B

8.The compound CuCl emits blue light having a wave length of 450 nm when heated at about 1200°C. What is the increment in energy(quantum) that is emitted at 450nm?
A .2.25×10^{-19}J
B .4.41×10^{-19}J
C .8.20×10^{-19}J
D .16.20×10^{-19}J

Explanation: The compound CuCl absorbs energy and the atoms are excited when heated .i.e
Energy absorbed to Excite = Energy of the emitted radiation.

$E_{emitted} = hV$, but $V = \frac{C}{\lambda}$

$$\Delta E = \frac{hC}{\lambda} = \frac{6.626 \times 10^{-34}\text{J.s} \times 3.0 \times 10^{8}\text{ms}^{-1}}{450\text{nm} \times 10^{-9\text{m/nm}}}$$

$$\therefore \Delta E = 4.41 \times 10^{-19}\text{J}$$

∴ Answer B

9.What is the difference between chlorine –35 and chlorine – 37?
A .Chlorine – 37 has two more protons than chlorine –35.
B .Chlorine – 37 has two more neutrons than chlorine –35.
C .Chlorine –37 has two more electrons than chlorine – 35.
D .Chlorine –37 has one more proton and one more neutron than chlorine -35.

Explanation: Construct a table to indicate the number of sub- atomic particles present in both forms of chlorine.

	P	n	E
$^{35}_{17}Cl$	17	18	17
$^{37}_{17}Cl$	17	20	17

Chlorine – 37 has two more neutrons than chlorine – 35.

∴ Answer B

10.Which of the following is **NOT** a solution?
A .milk
B .Brass
C. Whisky
D. Coca cola drink

Explanation: Milk appears to be a homogenous white liquid to the naked eye. When a drop of milk is examined under a microscope

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globules of fats in water are observed. There is no uniform composition as there is in a solution. Therefore ,**milk** is not a solution.

∴ ***Answer A***

11. Which one of the following substances is a non-conductor of electricity?

A .Graphite
C. Silver(s)
B .$MgCl_2(s)$
D. $H_2SO_4(aq)$

Explanation: **Ionic compounds** in the solid state like $MgCl_2(s)$ are not **conductors**. There is no movement of charged particles. In metals and graphite the conductivity is due to the movement of freely moving delocalized electrons. Aqueous solutions of acids are conductors.

∴ ***Answer B***

12. Which one of the following organic molecules has the highest water solubility?

A. $HOCH_2CH_2CH_2OH$
C. $HOCH_2CH_2CH_2CH_2OH$
B. $CH_3CH_2CH_2CH_2OH$
D. $CH_3CH_2CH_2CH_2CH_2OH$

Explanation: The water solubility of alcohols depends on:

i) the length of the hydrocarbon chain. The shorter the chain, the greater is the solubility.

ii) the number of functional groups. The alcohol with more number of functional groups has a greater solubility.

Each of the alcohols of choices "A" and "C" have two functional groups. But propane -1,3- diol in option A has shorter hydrocarbon chain and higher water solubility.

∴ ***Answer A***

13. The dye bromothymol blue (HBb) is a weak acid whose ionization can be represented as follows:

$HBb(aq) \rightleftharpoons H^+(aq) + Bb^-(aq)$

Which way will the equilibrium shift when NaOH is added?

A .To the left
B .To the right
C .Initially to the left and after a while to the right.
D .Initially to the right and after a while to the left.

Explanation: The color of an indicator in an acid or a base can be predicted from the ratio: $\frac{[HBb]}{[In^-]}$

The equilibrium constant expression for the reaction is :

$K_a = \frac{[H^+][Bb^-]}{[HBb]}$ ..... Multiply both sides by $\frac{[HBb]}{[In^-]} \times \frac{1}{K_a}$

$\frac{[HBb]}{[In^-]} = \frac{[H^+]}{K_a}$

i) [HBb] increases in acidic medium ,equilibrium shifts to the left.

ii) $[Bb^-]$ increases when NaOH is added indicating the shift to the right.

∴ ***Answer B***

14. Consider the following oxidation – reduction equation:

$2H_2O(\ell) + Al(s) + MnO_4^-(aq) \rightarrow AlO_4^-(aq) + MnO_2(s)$

What are the reducing and the oxidizing agents in this reaction?

A .Al(s) is the reducing agent and H_2O is the oxidizing agent.
B .Al (s) is the oxidizing agent and H_2O is the reducing agent.
C .Al(s) is the oxidizing agent and MnO_4^- (aq) is the reducing agent.
D .Al(s) is the reducing agent and MnO_4^- (aq) is the oxidizing agent.

Explanation:

i) The species whose oxidation number has increased is a reducing agent. In Al (s) $\rightarrow$ $Al(OH)_4^-$ the oxidation number of Al has increased from zero to + 3.

Therefore, **Al** is the reducing agent.

ii) The species whose oxidation number has decreased in the redox reaction is an oxidizing agent. The oxidation number of Mn in MnO_4^- has decreased from +7 to +4 in MnO_2.MnO_4^- is the oxidizing agent.

∴ ***Answer D***

15. Consider the following reaction:

$I_2O_5(S) + 5CO(g) \rightarrow I_2(s) + 5CO_2(g)$

What is the magnitude of the change in the oxidation number of the elements?

A .I, +5 to 0,C, +2 to +4
C.I, +0 to 0 , C, +2 to +4
B .I,+5 to 0 , C, +4 to +4
D. I,+4 to 0, C, +2 to +4

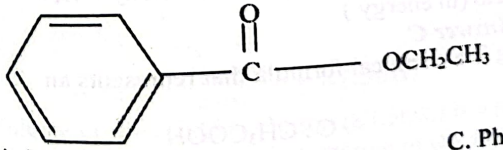
Explanation: In this redox reaction , the oxidation number of

i) I decreased from +5 in I_2O_5 to zero in I_2.

ii) C increased from +2 in CO to +4 in CO_2.

∴ ***Answer A***

16 .What is the name of the following compound?



A .Benzoate ester.
C. Phenyl butyrate.
B .Ethyl benzoate
D. Ethyl benzylketone.

Explanation: In the IUPAC or common systems of naming esters the name of the alkyl group of the alcohol portion is written first and the

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name of the acid is followed by replacing – **ic acid** with – **ate.** Therefore the name of the ester given is **ethyl benzoate.**

∴ ***Answer B***

17. Which term describes the units that make up compounds with covalent bonds?

A .Ions
B .Acids
C. Salts
D. Molecules.

Explanation:

✓ Combination of **ions** gives rise to **ionic bond.**
✓ **Acids** possess polar covalent bond consist of ions.
✓ **Salts** being ionic compounds possess ions.
✓ **Molecules** make up compounds with covalent bonds. Unlike the others do not contain ions.

∴ ***Answer D***

18. Which one of the following statements is **NOT** true about acids?

A . An acid is a proton donor.
B .An acid is an electron pair donor.
C .An acid is an electron pair acceptor.
D .An acid is a substance that is ionized in water to produce H^+ (aq).

Explanation: Give the definition of acids based on modern theories of acids and bases.

i) **Arrhenius definition :** An acid is a substance that ionizes in water to produce H^+ ions.
ii) **Bronsted – Lowry acid** is a proton donor.
iii) **Lewis concept** :An acid is a substance that forms a co-ordinate covalent bond by accepting a pair of electron into its vacant orbital.

∴ ***Answer B***

19. Which one of the following electromagnetic radiations has the shortest wavelength?

A .X –rays
B .UV rays
C. Gamma rays
D. Microwaves.

Explanation: Electromagnetic radiation is based on increasing order of energy is : Gamma ray > X –ray > UV ray >Visible ray > IR > Microwave > TV > Radio (in energy)

∴ ***Answer C***

20 .Which of the following is a chemical formula that represents an amino acid?

A .CH_4
B .CH_3NH_2
C . CH_3COOH
D .NH_2CH_2COOH

Explanation: **Amino- acids** are Organic compounds that possess both a carboxyl group(- COOH) and an amino group (-NH_2) attached to the ∝ - carbon atom.

∴ ***Answer D***

21. Which of the following compounds is unlikely to contain ionic bonds?

A .CO
B .NaF
C .LiCl
D. $MgBr_2$

Explanation: **Ionic bond** formation is favored between elements of low ionization energy and elements of high electron affinity. In choices B, C and D ionic bond formation is favored because the metals have low ionization energy and the halogens have high electron affinity. In choice A carbon and oxygen are elements with large ionization energy and high electron affinity. Therefore, ionic bond formation is not favored in CO.

∴ ***Answer A***

22. Which of the following element has the highest melting point?

A .Iodine
B .Tungsten
C. Mercury
D. Bromine

Explanation:

i) **Mercury and bromine** being liquids under normal conditions can be left out since melting is the conversion of solids to liquids.
ii) In **iodine** crystals, there is weak London or dispersion forces. But in **tungeston** there is a strong metallic bond. **Tungesten** has the highest melting point.

∴ ***Answer B***

23. How many mL of water is required to dilute 50 mL of 3.5M H_2SO_4 to 2.00M H_2SO_4?

A .37.5
B .45
C .75
D .87.5

Explanation: To determine the amount of water required to dilute, we apply the dilution formula: $\mathbf{M_1V_1 = M_2V_2}$ ***Required:*** V_2 = ?

Given : $M_1 = 3.5M$
$M_2 = 2.0\ M$
$V_1 = 50mL$
$M_2V_2 = M_1V_1$ ..... Dilution formula

$$\Rightarrow V_2 = \frac{M_1V_1}{M_2}$$

$$\therefore V_2 = \frac{3.5M \times 50mL}{2.0M} = 87.5mL$$

Volume of the diluted solution (87.5mL) is equal to the volume of the initial solution (50ML) plus volume of the water added.
87.5ML = 50ML +Volume of water needed
Volume of water required = **37.5 mL**

∴ ***Answer A***

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24. How much water has to be evaporated from 250 mL of 1M $Ca(OH)_2$ to make it 3M?

A .100mL B .150mL C. 167mL D. 200mL

Explanation:

i) the volume of the concentrated solution

Given : $M_1 = 1.0M$ ***Required:*** $V_2 = ?$

$M_2 = 3.0M$

$V_1 = 250mL$

$M_1V_1 = M_2V_2$ ..... Dilution formula

$\Rightarrow$ **$V_2 = 83.33mL$**

$V_{Solution}$ decreased from 250mL to 83.33mL

∴ Volume of water evaporated = 250mL – 83.33mL.

∴ **Vwater = 166.67 mL ≈ 167mL**

∴ Answer C

25. Consider the following acids:

I .CH_3COOH, $K_a = 1.8\times 10^{-5}$ III. HOBr, $K_a = 2.4\times 10^{-9}$

II. HCO_2H, $K_a = 1.8\times 10^{-4}$ Iv. C_6H_5OH , $K_a = 1.0\times 10^{-10}$

Which one of the following aqueous solutions will have the highest PH ?

A.0.10M NaOBr C.0.10MC_6H_5ONa

B.0.10M HCO_2Na D.0.10MCH_3COONa

Explanation: From the magnitude of their k_a values the acids can be arranged in the following order:

$C_6H_5OH < HOBr < CH_3COOH < HCOOH$. If an acid is weak its conjugate base is strong.

✓ The conjugate base of HCOOH will be the weakest and the order of the conjugate base is given as :

$HCOONa < CH_3COONa < NaOBr < C_6H_5ONa$

Weakest base Strongest base

∴ Answer C

26. Which of the following statements is **true** about esters?

A .Esters can form intermolecular hydrogen bonds.

B .Esters molecules can form intermolecular hydrogen bonds.

C .Ester molecules cannot form intermolecular hydrogen bonds.

D .Esters have higher boiling points than alcohols of comparable molecular weight.

Explanation: There is hydrogen bonding between water and the carbonyl oxygen of esters.

∴ Answer A

27. Consider the following equilibrium:

$CaCO_3(S) \rightleftharpoons CaO(S) + CO_2(g)$

Which of the following mixtures, each placed in a closed container and allowed to stand is not capable of reaching the equilibrium given above?

A .Pure $CaCO_3$

B .$CaCO_3$ and CaO

C .Some CaO and a pressure of CO_2 greater than the value of K_p.

D .Some $CaCO_3$ and a pressure of CO_2 greater than the value of K_p.

Explanation: If the mixture contains **$CaCO_3$ and CO_2** as in choice D at a higher pressure than k_p, formation of CaO is net favored.

$CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$

The equilibrium shifts to the left.

∴ Answer D

28. Which group of elements is characterized with ns^2np^2 outer –electron configuration?

A .Group 2A B .Group 4A C .Group 4B D .Group 3B

Explanation: The configuration **ns^2np^2** in the valence shell corresponds only to **Group 4A.**

∴ Answer B

29. Which of the following is the right order of the steps of a scientific method?

A .Performing experiments – Formulating hypothesis - Making observations.

B .Formulating hypothesis - Making observations – Performing experiments.

C .Making observations - Formulating hypothesis – Performing experiments.

D .Making observations - Performing experiments – Formulating hypothesis.

Explanation: The scientific method involves:

- ✓ making observation.
- ✓ collecting data/information.
- ✓ seeking patterns in the observations.
- ✓ formulating hypothesis to explain the observations.
- ✓ testing the hypothesis by further experiments.

∴ Answer C

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30. What is the number of moles of atoms and the number of atoms in a 10.0 g sample of copper?

A .0.08 mol Cu atoms and 2.16×10^{23} atoms.

B .0.16 mol Cu atoms and 9.63×10^{22} atoms.

C .0.16 mol Cu atoms and 9.63×10^{23} atoms.

D .0.31 mol Cu atoms and 4.16×10^{23} atoms

Explanation: Number of moles (n) $= \frac{m}{M} = \frac{10g}{63.5g/mole}$

n =0.157mol ≈ 0.16 mol Cu.

Number of atoms = n × N_A

Number of atoms = $0.16\times 6.02\times 10^{23}$ atoms

∴ Number of atoms $= 9.63\times 10^{22}$ atoms of Cu

∴ Answer B

31. Which of the following statement correctly describes a chemical reaction at equilibrium?

A .The concentrations of the products and reactants are equal.

B .The change in the concentrations of the products and reactants is constant.

C .The rate of the forward reaction is less than the rate of the reverse reaction.

D .The rate of the forward reaction is greater than the rate of the reverse reaction.

Explanation: For a chemical reaction at equilibrium:

i) The concentrations of reactants and products are not necessarily equal.

ii) The rate of the forward reaction is equal to the rate of the reverse reaction.

iii) It is only choice B that correctly describes a chemical reaction at equilibrium.

∴ Answer B

32. For elements in the left –most column of the periodic table, properties that have increasing values as the atomic number increases include which of the following?

I .Ionization energy II. Atomic radius III Atomic mass

A .III only C . I and II only

B .I ,II, and III D. II and III only.

Explanation: **Elements** in the left most column of the periodic table are the **Alkali metals (Group IA metals)** .

As their atomic number increases in moving down the group:

I) ionization energy decreases.

II) atomic radius increases.

III) atomic mass increases.

∴ II and III increases

∴ Answer D

33. There is a strong covalent bond between the N atoms in nitrogen gas, N_2. Why, then, does nitrogen have such a low boiling point of 196°C?

A .The bond between the N- atoms is triple.

B .N is very electronegative, only next to F and O .

C .The strong bond, an intra molecular one, determines the boiling point of the substance.

D .Boiling point is determined by intermolecular force, which in this case is weak as the molecule is non-polar.

Explanation: **Nitrogen** is a non-polar substance. The intermolecular force of attraction between N_2 molecules is the weak dispersion or London forces. The low boiling point is due to the weak intermolecular force.

∴ Answer D

34. What is the half- life, $t_{\frac{1}{2}}$ for a zero order reaction A → B, (k is rate constant)?

A .ln2k B. $\frac{[A]_o}{2k}$ C. ln$k[A]_o$ D. ln2$[A]_o$k

Explanation: The **half life of a reaction** is the time taken for the concentration of a reactant to fall to half of its initial concentration, that is , from $[A]_o$ to $\frac{1}{2}[A]_o$.

It can be derived from its integrated rate law:

$$[A] = -kt + [A]_o \text{ and } [A] = \frac{1}{2}[A]_o$$

$$\frac{1}{2}[A]_o = -kt + [A]_o$$

$$[A]_o = 2kt \Longrightarrow t = \frac{[A]_o}{2k}$$

∴ Answer B

35. Which of the following quantum number/s determine the energy of an electron in a hydrogen atom?

A .n B. n and ℓ C. n , ℓ and m D. n , ℓ,m and s

Explanation: The energy of an electron is determined by the principal quantum number (n).The quantum numbers ℓ, and m_ℓ determine the shape and the orientation of the orbitals respectively.

∴ Answer A

36. Which of the following statements is true?

A .A battery is rechargeable.

B .A battery produces electricity spontaneously.

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C .A battery has generally no liquid components.
D .A battery produces the same amount of electricity, regardless of composition.

Explanation: A **spontaneous redox** reaction produces electricity in a battery. A car battery has liquid component, that is dilute solution of sulfuric acid. **Secondary battery** is rechargeable but a primary battery cannot be rechargeable.

∴ ***Answer B***

37. A solution at 25℃ contains the metal ions Ni^{2+}, Pt^{2+} and Pd^{2+}, all at 1.0M concentrations. Consider the following standard reduction potentials:

$Ni^{2+} + 2e^- \longrightarrow Ni \quad E^0 = -0.23\ V$
$Pt^{2+} + 2e^- \longrightarrow pt \quad E^0 = 1.20V$
$Pd^{2+} + 2e^- \longrightarrow pd \quad E^0 = 0.99\ V$

Which metal(s) will plate out first when the solution is electrolyzed?
A .Ni B. Pd C. Pt D. Ni and Pd

Explanation: The metal ion with more positive standard reduction potential will plate out first. Because it has a greater tendency to be reduced. Pt^{2+} is prefentially discharged

∴ ***Answer C***

38. Under what conditions will the enthalpy change of a process equals the amount of heat transferred into or out of the system?
A .Under constant volume.
B .Under constant pressure.
C .Under constant pressure and volume.
D .Under constant pressure and temperature.

Explanation: According to the first law of thermodynamics the heat (q) gained or lost by a system at constant pressure is equal to the change in enthalpy (ΔH) or $\Delta H = q_p$.

∴ ***Answer B***

39. For an electron that has quantum numbers n= 4 and m $\ell = 0$, which of the following is true?
A .It must have the quantum number n = 0.
B .It must have the quantum number $\ell = 0$.
C .It must have the quantum number ms = +1/2.
D .It may have the quantum number $\ell = 0,1,2,3$.

Explanation: If n =4 , ℓ can have the values 0,1,2 and 3. The value of m_ℓ is determined by the value of ℓ and it goes from - ℓ to +ℓ including 0 (zero) .
For each value of ℓ. m_ℓ - can have the value 0 for any value of ℓ.

∴ ***Answer D***

40. Which of the following electron transitions requires the smallest energy to be absorbed by the hydrogen atom?
A .From n= 1 to n =2 C. From n = 3 to n =4
B .From n = 2 to n = 3 D. From n = 4 to n = 5

Explanation: For each transition of an electron, the energy absorbed can be calculated using: $\Delta E = 2.18 \times 10^{-18} \left(\frac{1}{n_i^2} - \frac{1}{n_f^2}\right)$

n_i : initial energy level and n_f: final energy level.
For the transition of an electron from
i) n =1 to n=2 $\Delta E = 1.635 \times 10^{-18} J$
ii) n = 2 to n=3 $\Delta E = 0.302 \times 10^{-18} J$
iii) n = 3 to n =4 $\Delta E = 0.105 \times 10^{-18} J$
iv) n = 4 to n =5 $\Delta E = 0.0225 \times 10^{-18} J$

The smallest energy absorbed is for the transition from n =4 to n =5.

∴ ***Answer D***

41. What is the hydroxide ion concentration for a solution with a pH of 10 at 25°C?
A .10^{-14} M B .10^{-10}M C .10^{-7}M D .10^{-4}M

Explanation: From PH = 10 , we can determine $[H_3O^+]$.

$PH = -\log [H_3O^+]$
$\Rightarrow [H_3O^+] = \text{Antilog}(-P^H)$
$\Rightarrow [H_3O^+] = 10^{-PH} = 10^{-10}$
$\Rightarrow$ From $k_w = [H_3O^+][OH^-]$,
$\therefore \quad [OH^-] = \frac{K_w}{[H_3O^+]} = 10^{-4}$

∴ ***Answer D***

42. All of the following when added to water will produce an electrolytic solution **EXCEPT**.
A .$N_2(g)$ C. HCl(g)
B .NaI(s) D. KOH (s)

Explanation: Nitrogen gas being a non-polar substance cannot produce negative and positive ions. That is it cannot produce an electrolytic solution when added to water. The other three ionize when added to water and form electrolytic solution.

∴ ***Answer A***

43. Which of the statement below best explains why atoms react chemically with each other?
A .When atoms react, they gain protons and are more stable.
B .When atoms react, they lose all their electrons and become more stable.

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C .When atoms react, they lose, gain, or share electrons and are then less stable.

D .When atoms react, they lose , gain or share electrons to attain a full outer energy level and are then more stable.

Explanation: There is a general tendency in nature for the potential energy to be at a minimum. Energy is given off in bond formation when atoms lose, gain or share electrons in their valence shell. As a result energy is minimized and the system becomes more stable.

∴ Answer D

44.What did Rutherford's alpha particle experiment show?

A .Electrons have a negative charge.

B .A proton is a hydrogen atom without electron.

C .Electrons circle the nucleus of an atom in orbits.

D .Most of the mass and all of the positive charge of an atom is found in a tiny nucleus.

Explanation: From **Rutherford's alpha** particle experiment the following points are obtained.

i) The majority of the alpha particles passed through the metal foil.

ii) Only very few alpha particles bounced back indicating the presence of positive charge occupying a very small volume of the atoms of the metal foil.

∴ Answer D

45.Which law relates the concentration of a dissolve gas, Cg ,to its partial pressure?

A .Henry's law C. Boyle's law

B .Raoult's law D. Ideal gas law

Explanation: At a given temperature, the solubility of a gas dissolved in a liquid is directly proportional to the partial pressure of the gas over the solution.

$C_g \propto p_g \Rightarrow c_g = K_H p_g$

This is the mathematical expression of **Henry's law.**

∴ Answer A

46.The solubility of sodium selenate, Na_2SeO_4, is 84g/100g of water at 35°C. If a solution is obtained by dissolving 92g of Na_2SeO_4 in 200 g of water at 35°C, what do you call this solution?

A .Diluted C. Unsaturated

B .Saturated D. Supersaturated.

Explanation: Compared to the saturated solution of sodium selenate which is 84g/100g of water, 92g/200g of water at the same temperature of 35°C is **unsaturated solution.**

∴ Answer C

47. How many atoms are present in 22g CO_2?

A .3.01×10^{23} C. $2\times6.02\times10^{23}$

B .6.02×10^{23} D. $5\times6.02\times10^{23}$

Explanation:

✓ 44g of CO_2 (1.0 mole) contains 6.02×10^{23} molecules.

✓ 22g of CO_2(0.5 mole) contains 3.01×10^{23} molecules.

✓ Each carbon dioxide molecule (CO_2) contains 3 atoms.

∴ The total number of atoms in 3.01×10^{23} molecules is 3 times as much , as that is 9.03×10^{23} atoms.

This is equal to $1.5\times6.02\times10^{23}$ atoms.

∴ Answer D

48.Which of the following titrations will have an equivalent point at a PH< 7?

A .Strong acid with weak base. C. Strong acid with strong base.

B .Weak acid with weak base D. Weak acid with strong base.

Explanation: When a strong acid is titrated with a weak base a salt whose cation hydrolyzes is formed. A cationic hydrolysis of a salt gives an acidic solution and the P^H at the equivalent point will be less than 7. ***Example*** .NH_4Cl

Cation hydrolysis: $NH_4^+ + H_2O \rightleftharpoons NH_3 + H_3O^+$

∴ Answer A

49. Which molecule listed below has two sigma(σ) and two pi (π) bonds?

A .N_2 C .N_2F_2

B .C_2H_4 D .HCN

Explanation:

- N_2 has one sigma and two π bonds.
- C_2H_4 has five sigma bonds, One π bonds.
- N_2F has three sigma bonds and one π bond.
- HCN has two sigma bonds and two pi (π) bonds.

∴ Answer D

50.Which of the following explains why, at room temperature, I_2 is a solid, Br_2 is a liquid and Cl_2 is a gas?

A .Ionic bonding C. Hydrogen bonding

B .Hybridization D. London dispersion forces.

Explanation: **Cl_2, Br_2 and I_2** are all non-polar molecular substances. The type of intermolecular forces in such substances is dispersion force or London force. The strength of the dispersion force increases with increasing molecular size in moving down the group from $Cl_2(g)$ to I_2

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(s).The larger I_2 molecules polarize and form temporary induced dipoles.I_2 is solid, Br_2(liquid) and Cl_2 (gas) under normal condition.

∴ ***Answer D***

51.Which molecule has a Lewis structure that does **NOT** obey the octet rule?

A .NO B .CS_2 C .PF_3 D .HCN

Explanation: The central atoms C and P, obey the octet rule in CS_2, PF_3 and HCN. But N in NO has odd number of electrons and does not obey the octet rule.

Bond type	Number of sigm and pi bond
Single bond (—)	Has only one σ bond
Double bond (=)	Has one σ and one π bond
Triple bond (≡)	Has one σ and two π bonds

∴ ***Answer A***

52.The organic compound $CH_3C(O)CH_3$ is:

A .aldehyde B .ester. C. Carbonyl D. ketone

Explanation: The organic compound $CH_3C(O)CH_3$ is ketone.

∴ ***Answer D***

53.What is the hybridization of phosphorus atom in PCl_5?

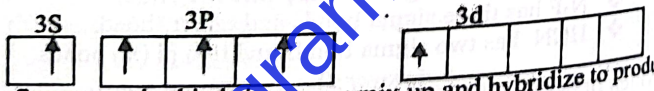
A .sp^3d C. sp^3

B .sp^3d^2 D. sp^2

Explanation:

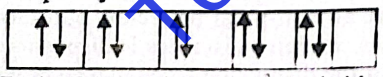
i) Ground state electronic structure of p

ii) Phosphorus atom- excited state.



These five unequal orbitals in energy mix up and hybridize to produce five equivalent sp^3d hybrid orbits. S+ P + P + P+ d ≈ sp^3d.

iii) Phosphorus having gained five electrons from chlorine atoms into the five sp^3d hybrid orbitals.



Each of the orbitals is sp^3d hybrid orbital.

∴ ***Answer A***

54. Which of the following species has the smallest H –X –H bond angle where X is the central atom?

A .OH_2 C. CH_4

B .NH_3 D .BH_3

Explanation:

Species	Shape	Bond angle
❖ OH_2	V – shape or Bent	104 .5^0
❖ NH_3	Trigonal pyramid	107^0
❖ CH_4	Tetrahedral	109.5^0
❖ BH_3	Trigonal planar	120^0

Therefore,H_2O has the smallest H –X –H bond angle which is104^0.

∴ ***Answer A***

55.The reaction between NO and I_2 is second – order in NO and first order in I_2.What change occurs in the rate of the reaction if the concentration of NO is doubled and I_2 left unchanged?

A .Double C. Eight times

B .Quadruple D. Three times.

Explanation: If the concentration of NO is doubled , it will be [2NO].Substituting this into the rate law expression for the reaction yields the following result.

Rate = K $[2NO]^2$ .....second order in NO .

Rate = 4K $[NO]^2$

The rate increases by a factor of 4 that is quadruples.

∴ ***Answer B***

56.If the following reaction is at equilibrium, which one of the following changes will shift the equilibrium to the left?

$N_2 + 3H_2 \longleftrightarrow 2NH_3 + heat$

A .Increasing pressure

B .Decreasing temperature

C .Adding more N_2 and H_2

D .Increasing the volume of the reaction container.

Explanations: The changes that shift the equilibrium position to the left in the reaction given are:

i) decrease in pressure (increase in volume).

ii) increase in temperature.

iii)removing the reactants N_2 or $3H_2$ and adding the product $2NH_3$

The changes in A,B, and C shift the equilibrium position to the right.

∴ ***Answer D***

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57. The reaction 2X+ Y → Z was studied and the following data were obtained.

Experiment	[X]	[Y]	Rate (mole. $L^{-1}.S^{-1}$)
1	3.0	3.0	1.8
2	3.0	1.5	0.45
3	1.5	1.5	0.45

What is the proper rate expression?

A .Rate = k[X]
B .Rate = k [Y]
C. Rate =K $[Y]^2$
D .Rate = k$[X]^2[Y]$

Explanation: The rate of this particular reaction remained constant at 0.45 mol $L^{-1}S^{-1}$ when the concentration of X changed from 3.0 to 1.5 in experiments 2 and 3.The rate of the reactant is independent of the change in concentration of X. We treat the rate in terms of change in concentration of Y. Take experiments 1 and 2 for which X is constant.

Experiments 1: Rate = $K[Y]^P$ .....P: order in Y.

$1.8 = K(3)^P$

Experiment 2 : Rate = $K[y]^P$

$0.45 = k(1.5)^P$

Divide the results of experiment 1 by the results of experiment 2.

$$\frac{1.8}{0.45} = \frac{k(3)^P}{k(1.5)^P}$$

$4 = (2)^P \Longrightarrow P = 2$ .....Second –order in y.

∴**Rate = $k[y]^2$**

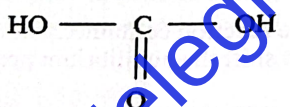
∴ Answer C

58.All of the following can act as Bronsted –Lowry bases **EXCEPT**

A .I^-
B .NH_3
C. HCO_3^-
D. H_2CO_3

Explanation: **A Bronsted – Lowry base is a proton acceptor .**

H_2CO_3 has the following Lewis structure.



Carbon in this structure cannot accept a proton. Being a second period element, carbon cannot have a 5th covalent bond. But I^- ,NH_3 and HCO_3^- can accept proton.

∴ Answer D

59. Consider the following reaction:

$2S_2O_3^{2-}(aq) + I_2(aq) \longrightarrow S_4O_6^{2-}(aq) + 2I^-(aq)$

If, in an experiment, 0.05 mol $S_2O_3^{2-}$ is consumed in 1.0L of solution each second, at what rates are $S_4O_6^{2-}$ and I^- produced in this solution?

A .$S_4O_6^{2-} = 0.025$; $I^- = 0.025$
B .$S_4O_6^{2-} = 0.05$; $I^- = 0.05$
C. $S_4O_6^{2-}=0.025$; $I^- =0.05$
D. $S_4O_6^{2-} = 0.05$;$I^- = 0.025$

Explanation: From the coefficients of the balanced equation we get the following facts.

i) The rate at which $S_2O_3^{2-}$ is consumed is twice as much as the rate at which $S_4O_6^{2-}$ is produced.

$$\frac{\Delta[S_2O_3]^{2-}}{\Delta t} = 2\,\frac{\Delta[S_4O_6]^{2-}}{\Delta t}$$

$$0.05 = 2\,\frac{\Delta[S_4O_6]^{2-}}{\Delta t}$$

$$\Rightarrow \frac{[S_4O_6]^{2-}}{\Delta t} = 0.025$$

ii) The rate at which $S_2O_3^{2-}$ is consumed is the same as the rate at which I^- is produced .That is 0.05 mol $L^{-1}S^{-1}$

∴ Answer C

60.What mass of aluminum is produced in one hour by the electrolysis of molten $AlCl_3$ with a current of 10A?

A .1.5g
B. 2.5g
C. 3.4g
D. 4.3g

Explanation: The mass of Al produced is directly proportional to Q Which is equal to IT.

Q = It

Q = 10A ×1.0hr× 60min /hr ×60sec/min = 36,000 A.sec

∴Q = 36,000 C.

Convert this in to the number of moles of electrons.

96,500 C..... 1 mole of electrons.

36,000 C..... X ?

$$\frac{X}{1\text{mole}} = \frac{36{,}000\text{C}}{96{,}500\text{C}} = 0.373 \text{ moles of electrons}$$

Relate the number of moles electrons to the cathode reaction.

$Al^{3+} + 3e^- \rightarrow Al.$

3 moles of electrons deposit 1 mole of Al.

0.373 moles of electrons deposite X mole of Al.

$$\frac{x}{1.0\text{mole}} = \frac{0.373}{3}$$

⇒ **0.124** mole of Al is produced at the cathode.

The mass of Al in 0.124 mol of Al can be determined using the

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relation :$n = \frac{m}{M}$

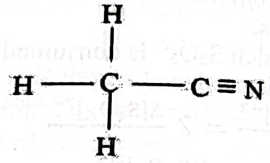
$m = n \times M$

$m = 0.124 \text{ mole} \times 27 \text{g/mol}$

$\therefore$ m= 3.357 g Aluminum $\approx$**3.4 g Aluminum**

∴ Answer C

61.What is the hybridization of the carbon atom attached to nitrogen in acetonitrile shown below?



A .sp

B. sp^2

C. sp^3

D. sp^4

Explanation: For the carbon atom attached to the N-atom there are only two sets of electrons corresponding to two hybrid orbitals. An S-orbital and a p – orbital mix and hybridize to give two equivalent **sp- hybrid orbitals.**

∴ Answer A

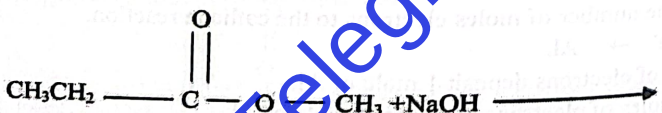
62. For which of the following elements is **Hund's rule** used in writing the electron configuration?

A .C B .B C .Be D. Li

Explanation: **Hund's rule:** orbital's of equal energy are singly occupied before pairing .The rule applies to species which have two or more electrons in their last p, d or f orbital's. From the 4 elements only carbon has two electrons in its p orbital's .The two electrons occupy separate p orbital's. The rule is not used for the other. None of them has two or more electrons in p subshell.

∴ Answer A

63.Consider the following reaction:



What are the products of this reaction?

A .Sodium acetate and ethanol.

B .Sodium acetate and methanol.

C. Sodium propionate and methanol.

D. Methyl propionic acid and methanol.

Explanation: This is the reaction of an ester with a base .The alkaline hydrolysis of esters yields the salt of the carboxylic acid and the corresponding alcohol .The acid portion of the ester is propanoic acid and its salt is sodium propanoate or sodium propionate and the alcohol is methanol.

∴ Answer C

64.Suppose reactions A $\longrightarrow$ B and B $\longrightarrow$ A are both elementary processes with rate constants of $8 \times 10^2 \text{ s}^{-1}$ and $4 \times 10^4 \text{ s}^{-1}$, respectively. What is the value of the equilibrium constant for the equilibrium: A $\rightleftharpoons$ B

A 2×10^{-2} B. 0.5×10^2 C .4×10^2 D .4×10^{-2}

Explanation: According to the law of chemical equilibrium the value of the equilibrium constant is equal to the rate constant for the forward reaction divided by the rate constant for the reverse reaction.

$$k_{eq} = \frac{k_f \text{(rate constant for forward reaction}}{k_r \text{ (rate constant for reverse reaction)}}$$

$$\therefore K_{eq} = \frac{8 \times 10^2 S^{-1}}{4 \times 10^4 S^{-1}} = 0.02 = 2.0 \times 10^{-2}$$

∴ Answer A

65.A reaction is 50% complete in 2 hours and 75% complete in 4 hours. What is the order of this reaction?

A .0 B. 1 C .2 D .3

Explanation: The following table can be constructed from the given information.

Time (hr)	0	2	4	6	8	10	12
concn [A]	A_o	$\frac{1}{2}A_o$	$\frac{1}{4}A_o$	$\frac{1}{8}A_o$	$\frac{1}{16}A_o$	$\frac{1}{32}A_o$	$\frac{1}{64}A_o$

Plot of $\frac{1}{A}V_s$ time gives a straight line with a positive slope k, confirming that it is a second – order reaction.

∴ Answer C

66.Consider the following balanced equation :

$3Ba(NO_3)_2\,(aq) + Fe_2(SO_4)_3(aq) \longrightarrow 3BaSO_4(s) + 2Fe(NO_3)_3\,(aq)$

The net ionic equation to describe this balanced equation is

A .$3Ba^+(aq) + SO_4^{2-}(aq) \longrightarrow 3BaSO_4(s)$

B .$3Ba^{2+}(aq) + 3SO_4^{2-}(aq) \longrightarrow 3BaSO_4(s)$

C .$6NO_3^-(aq) + 2Fe^{2+}(aq) \longrightarrow 2Fe(NO_3)_3(aq)$

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D .$3Ba^{2+}(aq) + 2NO_3^-(aq) + 2Fe^{3+}(aq) + 3SO_4^{2-}(aq) \rightarrow 3BaSO_4(s) + 2Fe^{3+}(aq) + 6NO_3^-(aq)$.

Explanation: $3Ba^{2+}$ ions and $3SO_4^{2-}$ ions precipitate as $3BaSO_4$ (s). Fe^{3+} and $3NO_3^-$ are spectator ions that appear on both sides. They cancel out and the net ionic equation is :

$3Ba^{2+}(aq) + 3SO_4^{2-}(aq) \longrightarrow 3BaSO_4(s)$

∴ Answer B

67. Which of the following is most likely to deviate from ideal gas behavior?

A .He B. Ar C. Cl_2 D. CCl_2F_2

Explanation: The strength of the intermolecular forces increases with increasing molecular mass. Therefore , the **chlorofluoro carbon gas** has the greatest tendency to deviate from ideal behavior.

∴ Answer D

68. Which of the following is a conjugate acid/base pair?

A .HCl/OCl^- C. NH_4^+/NH_3

B .H_3O^+/OH^- D .H_2SO_4/SO_4^{2-}

Explanation: An acid and its conjugate base or a base and its conjugate acid differ only by one hydrogen, that is , one hydrogen less or one hydrogen more. Accordingly only $NH4^+/NH_3$ is a conjugate acid – base pair.

∴ Answer C

69. For the reaction A+ B ⟶ C +D, ΔH^O= +40kJ and ΔS^O= +50J/K. Therefore, the reaction under standard conditions is:

A .spontaneous at all temperatures.

B .non- spontaneous at all temperatures.

C .spontaneous at temperatures greater than 800K.

D .spontaneous only at temperatures between 10K and 800K.

Explanation: The **spontaneity** of a reaction can be predicted from the value of change in free energy (ΔG).

i) If **ΔG > 0** , the reaction is non- spontaneous but the reverse reaction is spontaneous.

ii) If **ΔG < 0** , the reaction is spontaneous.

iii) If **ΔG = 0** , the system is at equilibrium. Use **Δ G = ΔH -TΔS**

Insert the given values into this equation:

Δ H= +40kJ , ΔS = +50J/K or (50× 10^{-3}KJ/k)

T = 800K or 10 K

Δ G =40KJ – 800 K ×50 ×10^{-3}KJ/K

Δ G = 40KJ – 40 KJ

∴ Δ G = 0 , when T = 800 K .....Equilibrium

Spontaneous above 800K.

Non – Spontaneous below 800K.

The reaction will proceed spontaneously at temperatures greater than 800K.

∴ Answer C

70. Which of the following compounds would give the lowest freezing point depression when 100g of each are dissolved in 1 kg of water (K_f for water = 1.86°C/m)? Assume complete dissociation.

A .NaCl C. $(NH_4)_2SO_4$

B .NH_4NO_3 D. glucose,$C_6H_{12}O_6$

Explanation: When the given 100g mass for each substance is converted to the number of moles ,it will be 1.7, 1.25,0.76 and 0.56 moles respectively for NaCl,NH_4NO_3,$(NH_4)_2SO_4$ and $C_6H_{12}O_6$.The freezing point depression: $\Delta T_f = k_f m$ will be the smallest for $C_6H_{12}O_6$.The larger the number of moles ,the larger is the molality, the higher is ΔT_f.

∴ Answer D

71. Which of the following ions is the most abundant in sea water?

A .Na^+ B. Ca^{2+} C. Cl^- D. HCO_3^-

Explanation: **Chloride ion (Cl^-)** is the most abundant in sea water in the form of NaCl, KCl or $MgCl_2$.The water of the sea and oceans contain an average of 3.6% of salts, the highest concentration being in the dead sea.

∴ Answer C

72. Which element is added to natural rubber to make it harder and reduce its susceptibility to oxidation and chemical attacks?

A .Sulfur B .Silicon C. Carbon D. Nitrogen

Explanation: Adding **sulfur to rubber** and heating the mixture makes the rubber harder and reduces its susceptibility to oxidation or other chemical attack.

∴ Answer A

73. Which of the following is **NOT** a carbohydrate?

A .Starch B. Glucose C. Glycine D. Cellulose.

Explanation: **Starch, glucose and cellulose** are different forms of carbohydrates. But **glycine** is a component of **protein** and it is an amino acid.

∴ Answer C

74. Which catalyst is used in the hydrogenation of vegetable oils?

A .Iron B. Nickel C. Platinum D. Molybdenum

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Explanation: Hydrogen molecules may be dissociated into atoms. It can react more rapidly on nickel surface than on Pt, Mo or iron surface. Nickel has more active sites to which the hydrogen gas is absorbed.

∴ ***Answer B***

75. How can silver can be plated onto nickel?

A .Electricity to a nickel anode in a solution of silver ions.
B .Electricity to a silver anode in a solution of nickel ions.
C .A solution of nickel ions to react with a piece of silver.
D .Electricity to a nickel cathode in a solution of silver ions.

Explanation: To cover the surface of Nickel with silver electrolytic ally, Nickel is made the cathode. The electrolyte must contain silver ions.

∴ ***Answer D***

76. Why does a pressure cooker reduce cooking time?

A .The heat is more evenly distributed.
B .A stronger flame is used for cooking.
C .The high pressure tenderizes the food.
D .The boiling point of water inside the cooker is increased.

Explanation: In a pressure cooker water boils at a temperature greater than its normal boiling point. As pressure increases, the boiling point increases. More heat energy in the pressure cooker reduces cooking time.

∴ ***Answer D***

77. For the conversion of $C_{(diamond)} \longrightarrow C_{(Graphite)}$, $\Delta H = -3$ KJ. What does this mean?

A .Both are equally stable.
B .Graphite is more stable than diamond.
C .Diamond is more stable than graphite.
D .Graphite has more stable energy than diamond.

Explanation: Diamond $\longrightarrow$ Graphite; $\Delta H = -3$KJ. There is a general tendency in nature for the potential energy to be at a minimum. In the conversion of diamond to graphite, there is flow of energy from the system to its surroundings and the system becomes more stable in the form of graphite.

∴ ***Answer B***

78. What is the change in internal energy, ΔE of a system if it absorbs 300KJ of heat from the surroundings and does 500kJ of work on the surroundings?

A .100KJ B. −200KJ C. 400KJ D .500KJ

Explanation: The question is based on the first law of thermodynamics:

$$\Delta E = Q + W$$

Q is positive if heat is absorbed.
Q is negative if heat flows out of the system.
Work is positive if work is done on the system.
Work is negative if work is done by the system.
According to this convention, Q is +300KJ and work is -500KJ.

$$\Delta E = Q + W$$
$$\Delta E = +300 - 500$$
$$\therefore \Delta E = -200KJ$$

∴ ***Answer B***

79. The PH at room temperature of a 0.1M solution of formic acid ($HCHO_2$) was measured to be 4.50. What is the hydrogen ion concentration?

A 3.16×10^{-5}M C. 6.32×10^{-3}M
B 3.16×10^{-12}M D. 6.32×10^{-4}M

Explanation: $PH = -\log [H^+]$

$$[H^+] = \text{antilog}(-PH)$$
$$[H^+] = 10^{-PH}$$
$$\therefore [H^+] = 10^{-4.5} = 3.16 \times 10^{-5}M$$

∴ ***Answer A***

80. What is the molarity of a solution containing 10g of sulfuric acid in 500mL of solution?

A .0.02 B .0.03 C .0.12 D .0.2

Explanation: To calculate the, molarity convert the 10g mass of H_2SO_4 to the number of moles of the acid.

$$n = \frac{m}{M} = \frac{10g}{98\ g/mole} = 0.102\ mol$$

$$\text{Molarity} = \frac{\text{Number of moles of solute}}{\text{volume of solution in liters}}$$

$$\therefore \text{Molarity} = \frac{0.102\ mol}{0.5L} = 0.2M$$

∴ ***Answer D***

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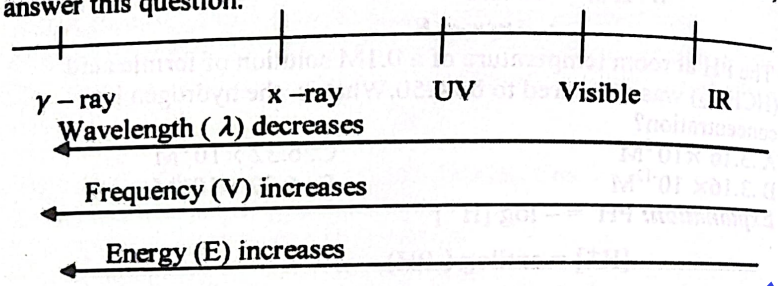
1. Which of the following statements is **TRUE**?
   A .Ultraviolet light has longer wavelength than visible light.
   B .The energy of radiation decreases as the wavelength decreases.
   C .The frequency of radiation increases as the wavelength decreases.
   D .Wave number of an electromagnetic radiation increases as wave length increases.

Explanation : Use a portion of the electromagnetic radiation (EMR) to answer this question.



- A is false. Because UV has a shorter wavelength .
- B is false .Because the energy of radiation increases as the wavelength decreases in moving from right to left.
- C is true according to the explanation given in B.
- D Wave number of an EMR decreases.

∴ ***Answer C***

2. Which of the following ionic compounds is formed from the reaction between magnesium and nitrogen?
   A .MgN_2
   B .Mg_2N_2
   C. Mg_3N_2
   D. Mg_2N_3

Explanation : Being in Group II A Mg has a valency of 2 (2,8,2) N – in group VA has a valency of 3 (2, 5) criss- cross their valences and place under the corresponding symbol as subscripts.

$Mg^{2} \times N^{3} = Mg_3N_2$

(magnesium nitride) is the ionic compound formed from the reaction between magnesium and nitrogen

∴ ***Answer C***

3. Which of the following molecules represents a non-polar covalent bond?
   A .B-Cl B. C-Cl C. Cl –Cl D.Mg –Cl

Explanation: There is an equal sharing of bonding electron pair between two atoms with equal electro negativities.

Example :H –H (in H_2), Cl – Cl (in Cl_2) and O = O (in O_2).The resulting bond is non-polar covalent bond.

∴ ***Answer C***

4. Which one of the following groups in the periodic table has paramagnetic atoms?
   A .Group zero
   B .Group IIA
   C .Group IIB
   D. Group IVA

Explanation : **Paramagnetic atoms** are atoms that have unpaired electrons in their valence shell. The ground state valence shell electronic configuration for the Groups zero , IIA ,IIB and IVA are given in the following table.

Group	Valence shell electronic configuration	Orbital diagram
VIII A(O)	ns^2np6	ns np (n-1)d All paired
IIA	ns^2	All paired
IIB	$ns^2(n-1)d^{10}$	All paired
IVA	ns^2np^2	Two unpaired

∴ Group IVA has paramagnetic atoms.

∴ ***Answer D***

5. Which of the following is **NOT** a basic SI unit?
   A .Candela B. Gram C. Mole D. Second

Explanation: **Basic SI units** are the units of the fundamental physical quantities length, mass, time, electric current, Amount of substance temperature and luminous intensity.

Fundamental quantities	**Basic SI unit**
Length	Meter (m)
Mass	kilogram (kg)
Time	Second (s)
Temperature	Kelvin (k)
Electric current	Ampere (I)
Amount of substance	Mole (mol)
Luminous intensity	Candela(cd)

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∴ **Answer B**

6. What is the process that converts liquid vegetable oils to solid fats?
A .Hydration
B . Hydrogenation
C. Hydrolysis
D. Saponification.

Explanation: The process that converts liquid vegetables oils.

- **Hydration** is interaction of ions of a solute with water. In the dissolution of NaCl, the negative end of water molecules attract Na^+ and the positive hydrogen end pulls the Cl^- ions causing solvation of NaCl.
- **Hydrogenation** a chemical reaction with hydrogen. Particularly it is an addition reaction in which hydrogen adds to unsaturated compounds like vegetables oils, alkenes or alkynes in the presence of **nickel catalyst.**

∴ **Answer B**

7. What would be the solubility of $HOCH_2(CH_2)_6CH_2OH$ compared to $CH_3(CH_2)_6CH_2OH$?
A .Less soluble in water.
B .More soluble in water
C .The same solubility in water.
D .More soluble in a non-polar solvent such as dichloroethane.

Explanation: The degree of **solubility** of alcohols in water depends on the number of functional groups and the length of the hydrocarbon chain. The shorter the hydrocarbon, chain the greater is the solubility. Alcohols with more number of functional group are more soluble in water. Therefore, the dihydric alcohol $HOCH_2(CH_2)_6CH_2OH$ is more soluble than the monohydric alcohol $CH_3(CH_2)_6CH_2OH$

∴ **Answer B**

8. Which of the following gases is manufactured using the Haber process?
A .Ammonia
B .Nitric oxide
C. Nitrogen
D. Nitrogen dioxide.

Explanation: **Haber process** is a process employed for the industrial production of ammonia (NH_3).

∴ **Answer A**

9. Which one of the mixture of the following pairs will **NOT** give a buffer solution?
A .HCN and NaCN
B .NH_3 and NH_4Cl
C. H_3PO_4 and KH_2PO_4
D. HNO_3 and $NaNO_3$

Explanation: **Buffers** contain either a weak acid and its conjugate base or a weak base and its conjugate acid. The combination is according to the definition for options A,B and C. But in choice D, there is a combination of a strong acid (HNO_3) and its conjugate base NO_3^- in $NaNO_3$.

∴ **Answer D**

10. How many types of cubic unit cells are known?
A .2
B.3
C.4
D.5

Explanation: Three types of cubic unit cells are known. These are:
i) simple cubic.
ii) face centered cubic and
iii) body – centered cubic.

∴ **Answer B**

11. Which of the following metals is NOT obtained by commercial electrolytic process?
A .Ag
B. Al
C. Cu
D. Na

Explanation:
- Al and Na are obtained only by electrolytic process .
- Cu and Ag are isolated from their ores by chemical reduction method.
- Copper is further purified by electrolysis. Production of silver (Ag) does not involve electrolysis.

∴ **Answer A**

12. Which of the following metals has the highest electrical and thermal conductivities?
A .Ag
B. Co
C. Cu
D. Ni

Explanation: All metals are good conductors of heat and electricity.
- Silver (Ag) is a metal with the highest known thermal and electrical conductivities.

∴ **Answer A**

13. Enthalpy is defined as the heat content of the system at constant:
A .heat
B .moles
C. pressure
D. volume

Explanation: **Enthalpy** is defined as the heat content of a system at constant pressure.

∴ **Answer C**

14. Which of the following elements is the second most abundant element in the earth's crust?
A .Aluminum
B .Iron
C. Oxygen
D. Silicon

Explanation : The first eight most abundant elements are given in the table below in the order of their relative percent abundance.

Element	Oxygen	Silicon	Aluminum	Iron	Calcium	Sodium	Potassium	Magnesium
Relative abundance	49 %	26 %	7.5%	4.6 %	3.0 %	2.6 %	2.4%	2.0%

∴ **Answer D**

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15. Which one of the following is **TRUE** for salts formed from strong acids and strong bases?

A .No hydrolysis takes place.

B .Produces ions that are proton donors.

C .Produces ions which are proton acceptors.

D .Depends on the pk_a and pk_b of the parent acids and bases respectively.

Explanation : We have to define hydrolysis in order to answer this question. **Hydrolysis** is the reaction of ions of salts to give an acidic or basic solution. Salts formed from strong acids and strong bases are neutral , that is, they do not hydrolyze. The reason for this is that the anions derived from strong acids are weak conjugate bases and do not undergo hydrolysis. The cations of strong bases being weak conjugate acids also do not hydrolyze. Salts like **NaCl, $NaNO_3$, KNO_3** do not hydrolyze.

∴ Answer A

16. What is the number of significant figures in 0.0030050?

A .4 B. 5 C. 7 D .8

Explanation: To determine whether zeros are significant figures or not we use the following guide lines.

i) Zeros preceding the first non-zero digit are not significant. The first two zeros right after the decimal point preceding 3 are not significant.

ii) Zeros between non-zero digits are significant. The two zeros between 3 and 5 are significant.

iii) Zeros after a non-zero digit preceded by a decimal point are significant figures. Therefore ,0.0030050 has five significant figures.

∴ Answer B

17. Which of the following properties of a substance does **NOT** represent an intensive physical property?

A .Boiling point C. Density

B .Color D. Volume

Explanation: There are two kinds of physical properties, namely extensive and intensive.

i) **Extensive physical properties** are those properties which depend on the amount of substance present in a sample.
Example: Length , mass ,volume.

ii) **Intensive physical properties** are properties which do not depend on the amount of substance.
Example: density, colour ,melting point ,hardness

∴ Answer D

18. What is the mass of one molecule of water?

A .3.0×10^{-23}g C. 0.0003 g

B .1.8×10^{-22}g D. 18.0g

Explanation : Apply the mole concept to answer this question. The mass of one mole of water is 18 grams and contains an Avogadro number of molecules. Use this relation to calculate the mass of one molecule of water.

6.02×10^{23} molecules (one mole) = 18g of water(1 mole)

One molecule = ? (x)

$$\frac{x}{18g} = \frac{1}{6.02\times10^{23}}$$

$$x = \frac{1}{6.02\times10^{23}} \times 18g = 2.99\times10^{-23}$$

$$\therefore x = 3.0\times10^{-23}g$$

∴ Answer A

19. Which of the following quantum number (s) is (are) related to the size and energy of an electron in a hydrogen atom?

A .n B. n, ℓ C. n, ℓ,m D. n, ℓ,m,s

Explanation: Give the description for n , ℓ and m_ℓ to answer this question.

- **n** is known as the principal quantum number and it describe the size and energy of the orbital.
- ℓ is subsidiary quantum number that gives number of subshells present.It describes the shape of the orbital and is sometimes called the **orbital shape quantum number.**
- m_ℓ is known as the magnetic quantum number. Determines the number of orientation of the orbitals in 3 – dimensional space.
- For $\ell = 0 \longrightarrow$ only one orientation is possible.
- For $\ell = 1 \longrightarrow$ 3 possible orientations,corresponding to the three orbitals of p –sub shell.
- For $\ell = 2$, five possible orientations related to the 5 orbitals of d – subshell.

The size of the orbital and the energy of the electron is related to the **principal quantum number (n).**

∴ Answer A

20. Which of the following plant nutrient will be produced as a result of nitrogen – fixation?

A .Carbohydrate C. Mineral

B .Cellulose D. Protein

Explanation : **Nitrogen – fixation** is a process of converting elemental N_2 to nitrogen compounds like ammonia and nitrates. Most plants

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absorb nitrogen from the soil in the form of nitrates and synthesize the complex organic polymers called **proteins**.

∴ ***Answer D***

21. Which of the following molecules has a trigonal bipyramidal structure?

A .SF_4 B. IF_5 C. ICl_4^- D.BrF_5

Explanation : Construct a table of steric number and lone pair to determine the structure.

Molecule or Ion	StericNo	Lone pair	Molecular geometry
SF_4	5	1	Seesaw
IF_5	6	1	Square pyramidal, tetragonal pyramid
ICl_4^-	6	2	Square planar
BrF_5	6	1	Square pyramidal ,tetragonal pyramid

None of the species has a **trigonal bipyramidal** structure. It is only SF_4 that has a trigonal bipyramidal electronic geometry.

∴ ***NO Answer***

22. Which of the following represents the general configuration of the transition elements?

A .ns^2np^6 C. ns(n−2) f

B .ns(n -1) d D.$ns^2np^6(n-1)d^{10}$

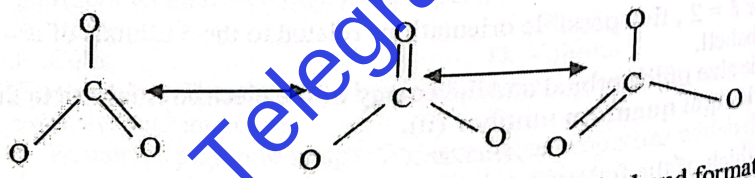
Explanation: For transition elements which are also known as **d- block elements** the last electrons enter the d- orbital's, that are in the (n-1)Shell .

∴ ***Answer B***

23. The total number of electrons participating in the bond formation of carbonate anion,CO_3^{2-} ,in the molecule of carbonic acid are:

A .16 B. 10 C .8 D. 5

Explanation : There are three possible Lewis structures for the carbonate ion as shown below.



The total number of electron pairs participating in the bond formation is equal to the number bonds.

∴ 4 covalent represent 4 pairs of electrons which is equal to 8 electrons.

∴ ***Answer C***

24. Which of the following crystals possess high electrical and thermal conductivities?

A .Ionic crystals C. Molecular crystals.

B .Metallic crystals D. Covalent network crystals.

Explanation: Metallic crystals possess high electrical and thermal conductivities because of the delocalized mobile electrons. The Three other crystals here are not conductors.

∴ ***Answer B***

25. An electron has a spin quantum number, $s = +\frac{1}{2}$ and a magnetic quantum number, $m_\ell = +1$.In which of the following orbital will it **NOT** be present?

A .S –orbital C. d - orbital

B .P –orbital D. f –orbital

Explanation : Use the following table to answer

n	ℓ	m_ℓ	Sub shell
1	0	0	S
2	0,1	-1,0,+1	P
3	0,1,2	-2,-1,0,+1,+2	d
4	0,1,2,3	-3,-2,-1,0,+1,+2,+3	f

For S –sub shell, m_ℓ cannot be +1 . Since the Value of ℓ is 0.

∴ ***Answer A***

26. Which of the following statements is **TRUE** about equilibrium reaction?

A .No more reactants are transformed into products.

B .There are equal amounts of reactants and products.

C .The rate constant for the forward reactions equals that of the reverse reaction.

D .The rate for the forward reactions equals that of the reverse reactions.

Explanation: Characteristic of Chemical Equilibrium:

✓ The rate of the forward reaction is equal to the rate of reverse reaction.
✓ The equilibrium is dynamic in nature.
✓ All reactants and products are present at equilibrium.
✓ The concentration of reactants and products are generally not equal.

∴ ***Answer D***

27. Three gases are in equilibrium in a closed chamber sealed with a piston. The following equilibrium is established:

$$2NH_3(g) \rightleftharpoons N_2(g) + 3H_2(g)$$

What will happen if the piston is pushed into the chamber?

A . The mole fraction of N_2 increases.

B .The mole of N_2 decreases.

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C .The mole fraction of N_2 remains the same.
D .The mole fraction of N_2 increases and then decreases.
Explanation : If the piston is pushed into the chamber, the pressure on the equilibrium system increases. An increase in pressure causes a shift of the equilibrium position to the side that has fewer number of moles of gases. In this case the equilibrium position shifts to the left consuming more N_2 and H_2.Therefore, the moles of N_2 and H_2 decreases .

∴ ***Answer B***

28. Which of the following reactions will produce an alkyl carboxylic acid?
A .Heating a methyl ketone with acid and iodine.
B .Reacting an alkyl halide with hydrogen gas and platinum
C .Reacting an alcohol with ozone.
D .Oxidation of a primary alcohol with hot permanganate or chromate.
Explanation: The most important method for the preparation of alkyl carboxylic acid is by the oxidation of the primary alcohol using hot potassium permanganate ($KMnO_4$) or potassium dichromate.

∴ ***Answer D***

29. Which of the following statements is **NOT TRUE**?
A Naturally derived soaps consist of a soluble salt of a long chain fatty acid.
B .Triacylglycerols are esters of glycerol and long chain carboxylic acids.
C .Long chain carboxylic acids are also known as fatty acids.
D .The major acidic component of vinegar is formic acid.
Explanation: A ,B and C are true but D is false because the major acidic component of vinegar is acetic or ethanoic acid not formic acid .

∴ ***Answer D***

30. What is a valid rate expression for the following reaction?

$$2NO + 2H_2 \longrightarrow N_2 + 2H_2O$$

A . $\frac{1\Delta[H_2]}{2\Delta t}$
B . $-\frac{1\Delta[H_2O]}{2\Delta t}$
C. $-\frac{1\Delta[NO]}{2\Delta t}$
D. $-\frac{\Delta[N_2]}{\Delta t}$

Explanation : The rate expression for the general reaction

$$aA + bB \longrightarrow cC + dD \text{ is given by:}$$

$$\text{Rate} = -\frac{1}{a}\frac{\Delta[A]}{\Delta t} = -\frac{1}{b}\frac{\Delta[B]}{\Delta t} = \frac{1}{C}\frac{\Delta[C]}{\Delta t} = \frac{1}{d}\frac{\Delta[D]}{\Delta t}$$

Applying this general expression of **rate to the reaction** given, we get the following:

$$\text{Rate} = -\frac{1}{2}\frac{\Delta[NO]}{\Delta t} = -\frac{1}{2}\frac{\Delta[H_2]}{\Delta t} = \frac{\Delta[N_2]}{\Delta t} = \frac{1}{2}\frac{\Delta[H_2O]}{\Delta t}$$

∴ ***Answer C***

31. Which of the following is the most important type of solute –solvent interaction in a solution of n –butanol in water?
A .Dispersion
B .Ion- dipole
C. Dipole - dipole
D. Hydrogen bonding
Explanation: The most important type of solute –solvent interaction in a solution of n- butanol with water.
✓ Is not **dispersion** force. Because this is a type of interaction in non-polar solute – solvent.
✓ Is not **ion- dipole** interaction .Because none of the components is ionic compound to release ion.
✓ Is a **dipole – dipole** but is not the most important type for this system.
✓ Is a special type of **dipole-dipole** interaction containing hydrogen bonding .

∴ ***Answer D***

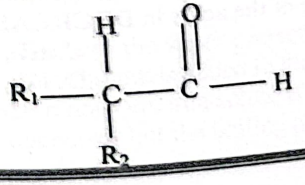
32. What is the product of the hydrolysis of esters in the presence of a mineral acid catalyst?
A .Alcohol
B .Carbon dioxide
C. Ether
D. Ketones
Explanation: **Acid- catalyzed hydrolysis** of esters yields the corresponding carboxylic acid and alcohol.

Ester + water $\overset{H_3O^+}{\rightleftharpoons}$ carboxylic acid + Alcohol

$$R-\overset{O}{\overset{\|}{C}}-OR' + H-OH \rightleftharpoons R-\overset{O}{\overset{\|}{C}}-OH + R'-OH$$

∴ ***Answer A***

33. To which organic functional group does the following molecular representation, i.e, R_1R_2CHCOH belong?(R_1 and R_2 represent different alkyl chains)
A .Amides
B .Aldehyde
C. Ethers
D. Organic acids.
Explanation: Structure of the molecule given in the question is



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Since its functional group is CHO it is an aldehyde.

∴ ***Answer B***

34. What is the quantity of water, in mL , required to prepare 0.5M of HCl from a concentrated solution of 3.5M in 50mL is ?

A .50mL B. 100mL C. 300mL D. 350mL

Explanation : The quantity of water, in mL , required to dilute 50mL of 3.5M to 0.5M can be determined using the dilution formula $M_1V_1 = M_2V_2$, where: M_1 = (initial concentration) = 3.5 M

V_1 = (initial volume) = 50mL

M_2 = (concentration after dilution) = 0.5M

V_2= (total volume of solution (after dilution) = V_2

From $M_1V_1 = M_2V_2$

$$\Rightarrow V_2 = \frac{M_1V_1}{M_2}$$

$$V_2 = \frac{3.5M \times 50mL}{0.5M} = 350mL$$

$$V_{H_2O\ added} = V_2 - V_1 = 350\ mL - 50\ mL$$

$$V_{H_2O\ added} = \mathbf{300mL}$$

∴ ***Answer C***

35. What is the name of triacylglycerol (triglycerides) that is solid at room temperature?

A .Lecithin B. Fat C. Wax D. Oil

Explanation:

- **Fats and oils** are triglycerides or triacylglycerols .
- **Fats** are esters of glycerol and saturated fatty acids. They are solids at room temperature.
- **Oils** are esters of unsaturated fatty acids and glycerol and they are liquids at room temperature.

∴ ***Answer B***

36. Given the following equilibria and equilibrium constants;

I) $HC_2H_3O_2 + H_2O \rightleftharpoons H_3O^+ + C_2H_3O^-_2$ $k_a = 1.80\times 10^{-5}$

II) $H_2CO_3 + H_2O \rightleftharpoons H_3O^+ + HCO_3^-$; $k_a = 4.20\times 10^{-7}$

III) $NH_4^+ + H_2O \rightleftharpoons H_3O^+ + NH_3$; $k_a = 5.6\times 10^{-10}$

IV) $HCOOH + H_2O \rightleftharpoons H_3O^+ + COOH^-$; $k_a = 1.80\times 10^{-4}$

What is the strength of the acids in **DECREASING** order?

A .I,IV,II and III C. III , II , I and IV

B .II,III,IV and I D. IV , I II and III

Explanation: Strength of the acid in decreasing order can be obtained by arranging the K_a values in decreasing order. The Larger the value of K_a ,the Stronger is the acid.

Number	Acid	K_a values	Rank
I	$H_2C_2H_3O_2$	1.80×10^{-5}	2....Intermediate
II	H_2CO_3	4.20×10^{-7}	3...Intermediate
III	NH_4^+	5.6×10^{-10}	4...Weakest
IV	HCOOH	1.80×10^{-4}	1... Strongest

∴ the strength of the acids in decreasing order: IV > I > II > III.

∴ ***Answer D***

37. Given the reaction:

$H_2PO_4^- + H_2O \longrightarrow H_3O^+ + HPO_4^{2-}$

Which of the following represents a conjugate acid – base pair?

A .$H_2PO_4^-$ and H_2O C. $H_2PO_4^-$ and H_3O^+

B .$H_2PO_4^-$ and HPO_4^{2-} D. H_2O and HPO_4^{2-}

Explanation : .

i) when an **acid donates a proton** it forms its conjugate base. In the reaction given $H_2PO_4^-$ donates a proton to H_2O forming the conjugate base HPO_4^{2-}.As a result the conjugate acid – base pair **$H_2PO_4^-/HPO_4^{2-}$** is formed

ii) When **a base accepts a proton** it forms its conjugate acid. In the reaction given H_2O a base accepts a proton and produces its conjugate acid H_3O^+ resulting a conjugate acid – base pair ,H_3O^+/H_2O

∴ ***Answer B***

38. Which of the following statements is **TRUE** about colligative properties?

A .Both vapor pressure and freezing point increase when a non-volatile solute is added to a solvent.

B .Both freezing point and boiling point increase when a non-volatile solute is added to a solvent.

C .Both vapor pressure and boiling point decrease when a non- volatile solute is added to a solvent.

D .Colligative properties depend only upon the number of solute particles in a solution and not upon their identity.

Explanation: **Colligative properties** are properties of solutions which depend upon the concentration of solute particles and not upon their identity.

- A is false. Because both the vapor pressure and freezing point decrease when a non- volatile solute is added to the solvent.
- B is false .When a non- volatile solute is added to the solvent the freezing point decreases but the boiling point increases.

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❖ C is false .The boiling point increases but the vapor pressure is lowered when a non- volatile solute is added to the solvent.

❖ D is true. According to the definition.

∴ ***Answer D***

39. What is the PH of 0.005 M solution of $Ca(OH)_2$?

A .2.3 B .10 C.12 D .14

Explanation: Assuming complete dissociation:

$Ca(OH)_2(aq) \longrightarrow Ca^{2+}(aq) + 2OH^-(aq)$

The number of moles of OH^- is twice as much as the number of moles of $Ca(OH)_2$ solution. That is

$[OH^-] = 2[Ca(OH)_2]$

$[OH^-] = 2\,(0.005M)$

$[OH^-] = 0.01M$

$P^{OH} = -\log [OH^-]$

$^{OH} = -\log (0.01)$

$P^{OH} = -\log 1.0\times 10^{-2}$

$P^{OH} = 2 \Longrightarrow P^H = 12$ Since $P^H + P^{OH} = 14$

∴ ***Answer C***

40. What is the equivalent weight of HNO_3,as an oxidizing agent, in the following balanced reaction?

$3Fe^{2+} + 4H^+ + NO_3^- \longrightarrow 3Fe^{3+} + NO + 2H_2O$

A .10.50 B .15.75 C .21.00 D. 31.50

Explanation: **Equivalent mass** of an oxidizing agent or a reducing agent is equal to the molar mass of the oxidizing agent or reducing agent divided by the number of moles of electrons gained or lost in the redox reaction.

For HNO_3 as an oxidizing agent.

In the reaction the N in HNO_3 is reduced from the +5 oxidation state to the +2 oxidation state in NO by accepting 3 moles of electrons.

∴ Equivalent mass of $HNO_3 = \frac{\text{molar mass of } HNO_3}{3 \text{ moles of electrons}} = \frac{63}{3} = 21.$

∴ ***Answer C***

41. How many kilojoules of heat are absorbed when 20 g of NaCl (s) is decomposed into Na(s) and $Cl_2(g)$ at constant pressure according to the following reaction?

$2Na(s) + Cl_2(g) \longrightarrow 2NaCl(s)\ \Delta H = -821.8\ KJ$

A .-281.0 B .-140.5 C .+140.5 D. +281.0

Explanation: The change in enthalpy (ΔH) for a reaction is equal to the heat absorbed or released at constant pressure.

$2Na(s) + Cl_2(g) \longrightarrow 2NaCl(s);\ \Delta H = -821.8\ KJ.$

The enthalpy change (ΔH) for the reverse reaction is equal to +821.8KJ.

$2NaCl(s) \longrightarrow 2Na(s) + Cl_2(g)\ ;\ \Delta H = +821.8\ KJ.$

Now ,convert 20 g of NaCl(s) to the number of moles using the relation

Number of moles $= \frac{\text{Given mass}}{\text{Molar mass}} \Rightarrow n = \frac{m}{M}$

$n = \frac{20g}{58.5g/mole} = 0.34$ moles of NaCl in 20 g

To decompose 1 mole of NaCl (s) into Na(s) and $Cl_2(g)$ 821.8KJ of heat energy is required. How many kilojoules is required to decomposes 0.34 moles of NaCl(s).

1 mole 821.8

0.34 mole..... ? x

$\frac{x}{821.8} = \frac{0.34\ \text{mole}}{1.0\ \text{mole}}$

$x = \frac{0.34 \times 821.8}{1.0} = 280.967$

∴**x = 281 .0 KJ of heat**

∴ 281.0KJ of heat is required (absorbed) to decompose 20 g of NaCl to Na and Cl_2.

∴ ***Answer D***

42. In the electroplating of nickel from a solution containing Ni^{2+} ion, what will be the weight of the metal deposited on the cathode by a current of 8A flowing for 500 minutes?

A .73g B. 103g C. 117.4g D. 145g

Explanation: The question is on the application of Faraday's first Law of electrolysis in industrial process of coating (covering) metals for decoration or protection against corrosion.

Electroplating is the electrolytic deposition of a less active metal on the surface of a more active metal to prevent the corrosion of the active metal. The mass of the metal deposited is directly proportional to Q (amount of charge) that passes the electrolyte. Q = It.

$Q = 8A \times 500 \text{minutes} \times 60 \text{sec/min} = 240{,}000\ A.s$

∴**Q = 240,000 C**

Calculate the moles of electrons using $n = \frac{Q}{F}$

$n = \frac{Q}{F} = \frac{240{,}000\ C}{96{,}500\ C/\text{mole}} = 2.487$ moles of electrons

$n \approx 2.50$ moles of electrons

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2 moles of electrons deposit 1.00 mole of Ni

$Ni^{2+} + 2e^- \longrightarrow Ni$

How many moles of nickel is deposited by 2.5 moles of electrons?

2 moles of electrons deposit ⟶ 1 mole of Ni

2.5 mole of electrons deposit ⟶ x =?

$\frac{x}{1} = \frac{2.5}{2} \Longrightarrow$ 1.25 mole of nickel

Use the relation $n = \frac{m}{M} \Longrightarrow n \times M = m$

The molar mass of nickel (58.7) should have been given

$m = 1.25 \text{ mole} \times 58.7 \text{g mol}^{-1} = 73.3 \text{ g}$

∴ ***Answer A***

43. A 1M solution of $Cu(NO_3)_2$ is placed in a beaker with a strip of Cu metal. A 1M of $SnSO_4$ is placed in a second beaker with a strip of Sn metal. The two beakers are then connected by a salt bridge and the two metal electrodes are connected by wires to a voltammeter. Which of the following electrode serves as the anode and which electrode gains mass?

Given that $E^oCu^{2+}/Cu = 0.34V$ and $E^o Sn^{2+}/Sn = -0.14V$

A .Anode, Sn ,Sn electrode gains mass.

B .Anode Sn, Cu electrode gains mass.

C .Anode , Cu , Sn electrode gains mass.

D .Anode Cu, Cu electrode gains mass.

***Explanation*: A half- cell** with the more active metal is made the anode because it oxidizes easily. Here Sn is more active than copper and it is made the anode.

❖ **Oxidation at the anode:** $Sn \longrightarrow Sn^{2+} + 2e^-$

The anode electrode dissolves in the process of its oxidation.

Transition of Sn^{2+} ions to the solution.

❖ **Reduction at the cathode :** $Cu^{2+} + 2e^- \longrightarrow Cu$

Copper ions (Cu^{2+}) in the solution are reduced and deposited on the copper electrode. This causes an increase in the mass of the copper electrode.

∴ ***Answer B***

44. Which of the following metal alloys does **NOT** contain tin?

A .Brass B. Bronze C. Pewter D. Plumber's solder

***Explanation*:** To answer this question construct a table of the alloys and their components.

Alloy	Components
❖ Brass	Cu, Zn
❖ Bronze	Cu, Sn
❖ Pewter	Pb, Sn ,Cu,Sb
❖ Plumber's solder	Pb, Sn

∴ ***Answer A***

45. The indicator bromthymol Blue (HBb) is a weak acid with $K_a = 1.0 \times 10^{-7}$ ionizes as follows:

HBb (aq, yellow) $\rightleftharpoons H^+$ (aq, colourless) + In^- (aq, blue)

Which way will the equilibrium shift when NaOH is added and what will the colour of NaOH solution be containing this indicator?

A .Equilibrium will shift to the right and the color of NaOH solution will turn green.

B .Equilibrium will shift to the right and the color of NaOH solution will turn blue.

C .Equilibrium will shift to the left and the color of NaOH solution will turn yellow.

D .Equilibrium will shift to the left and the color of NaOH solution will turn blue.

***Explanation*:** The dissociation of the indicator HB_b must be correctly written as follows:

$HBb(aq) \rightleftharpoons H^+(aq) + Bb^-(aq)$

(yellow) blue

The indicator is yellow in the molecular form HBb and blue in the Bb^- form: $K_a = \frac{[H^+][Bb^-]}{[HBb]}$ multiply both sides by $\frac{[HBb]}{[Bb^-]} \times \frac{1}{k_a}$

$\frac{[HBb]}{[Bb^-]} ka \times \frac{1}{ka} = \frac{[H^+]}{ka} \Longrightarrow \frac{[HBb]}{[Bb^-]} = \frac{[H^+]}{ka}$

$\frac{[HBb]}{[Bb^-]} = \frac{[H^+]}{ka}$

Use this equation to predict the color change , when the concentration of H^+ changes and Shift to the right .

i) If the concentration of $[H^+] = 10^{-5}$

$\frac{[H^+]}{ka} = \frac{[HBb]}{[Bb^-]}$

$\frac{10^{-5}}{10^{-7}} = 100 = \frac{[HBb]}{[Bb^-]}$

$\Rightarrow$ concentration [HBb] is 100 times $[Bb^-]$.

The indicator is mostly in molecular form and the color is yellow. In acidic medium the equilibrium shifts to the left.

ii). Take concentration of H^+ 10^{-9} .....Basic

$\frac{[H^+]}{ka} = \frac{[HBb]}{[Bb^-]}$

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$$\frac{10^{-9}}{10^{-7}} = \frac{[HBb]}{[Bb^-]} = \frac{1}{100}$$

$\Rightarrow 100[HBb] = [Bb^-]$

If a base is added , the indicator is mostly in the ionized form $[Bb^-]$ and the color is blue.

iii) Take $[H^+]$ as 10^{-7}

$$\frac{[H^+]}{ka} = \frac{10^{-7}}{10^{-7}} = \frac{[HBb]}{[Bb^-]}$$

$\Rightarrow [HBb] = [Bb^-]$

Blue and yellow in equal concentrations results in green color.

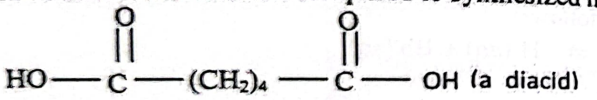
∴ ***Answer B***

46. What are the raw materials required to synthesize nylon 66, a specific kind of nylon?

A .Diacids
B .Diamines
C. Diacids and Diamines
D. Polyethylene.

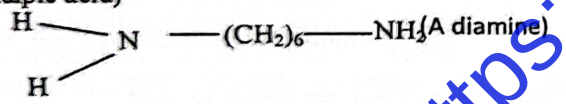
Explanation: The raw materials required to synthesized nylon 66 are:

i)



(Adipic acid)

ii)



Hexamethylene diamine

Each monomer has two functional groups of the same type corresponding the terms **diacids and diamines**

∴ ***Answer C***

47. Which of the following substances are added to natural rubber to toughen it?

A .Calcium
B .Carbon
C. Nitrogen
D. Sulfur

Explanation: Heating natural rubber with sulfur establishes cross-links between linear polymer chains. This makes the rubber hard and tough.

∴ ***Answer D***

48. What is the quantity of heat required to raise the temperature of 80g of ethanol from 25 °C to 75°C? (specific heat of ethanol = $2.46 Jg^{-1}K^{-1}$).

A . 2.46KJ
B .4.0KJ
C.9.84KJ
D. 18.68KJ

Explanation: ***Given:*** Use the equation $Q = c\ m\ \Delta T$

Where q = the quantity of heat

m = mass of substance heated

ΔT = change in temperature

Required: Heat = Q = ?

C = specific heat = $2.46 Jg^{-1}K^{-1}$

$Q = C.m.\Delta T$

$$Q = \frac{2.46\,J}{gk} \times 80g \times 50k = 9840J$$

∴ Q = 9.84 KJ

∴ ***Answer C***

49. Consider the following gaseous reaction at 25°C:

$CH_4(g) + 2O_2(g) \longrightarrow CO_2(g) + 2H_2O(g)$ $\Delta H = -802$ KJ

which energy change would occur if 3.2 moles of CH_4 is completely combusted?

A .2.57×10^2 KJ will be released.
B .2.57×10^2 KJ will be absorbed .
C. 6.43×10^3 KJ will be released.
D. 6.43×10^3KJ will be absorbed.

Explanation: The combustion of one mole of methane , CH_4 releases 802KJ. Therefore 3.2 moles of methane releases 3.2 times as much as 802 KJ , that is , 3.2 moles× 802 KJ mol^{-1} which gives of 2566 .4 KJ or 2.57×10^3 kilojoules. All the four options are different from this calculated amount.

∴ ***No Answer***

50.What structural feature is usually needed to be present in order for an addition polymer to be produced?

A .A carbon – carbon sigma bond .
B .A carbon –oxygen pi bond .
C. A carbon-oxygen sigma bond .
D.A carbon-carbon pi bond.

Explanation: For an addition polymer to be produced carbon- carbon double bond must be present in the structure of the monomers(s).In a double bond one bond is always a pi bond that cleaves in the polymerization process.

∴ ***Answer D***

51. Which of the following is the most important ingredients used for production of DAP fertilizer?

A .Ammonia and phosphoric acid.
B .Nitric acid, urea and phosphoric acid
C .Phosphoric acid , urea and ammonia.

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D .Sulphuric acid , ammonia and urea.

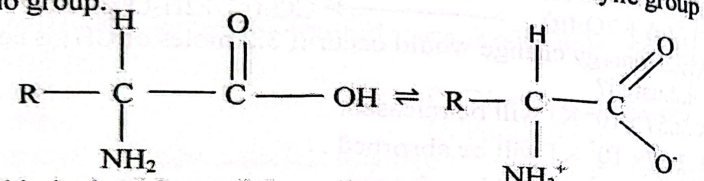
***Explanation*: In DAP fertilizer**, D.A.P stands for Di ammonium phosphate fertilizer. The ingredients are ammonia and phosphoric acid.

∴ ***Answer A***

52. Which of the following bio molecules forms a zwitterion at higher or lower PH?

A .Cellulose
B .Glucose
C. Protein
D. Starch

***Explanation*:** Because of the presence of both acidic and basic functional groups , amino acids can undergo an internal acid- base reaction in which a proton is transferred from the carboxylic group to the amino group.



This doubly ionized form of the amino acid is called **Zwitterion**. They are formed at normal physiological P^H (neutral P^H) and not at higher or lower PH. The bio molecule that forms a **zwitterions** is protein at normal physiological PH

∴ ***Answer C***

53. Which of the following form of synthetic rubbers can be vulcanized to greatly enhance its mechanical strength?

A .Noeprene
B .Isoprene
C. Butyl rubber
D. Styrene – butadiene rubber.

***Explanation*: Butyl rubber** can hold air better than natural rubber when it is vulcanized. It is used for inner tubes.

∴ ***Answer C***

54. What is the range in the number of carbon atoms of the monosaccharide's that are found in nature?

A .3 to 7 B .4 to 10 C. 4 to 12 D. 5 to 12

***Explanation*: Monosaccharide** are white crystalline solids that contain a single aldehyde or ketone functional group. They are classified as triose, tetrose ,pentose , or hexose corresponding to 3,4,5 and 6 carbon atoms respectively.

∴ The range is from 3 to 6 carbon atoms

∴ ***NO Answer***

55. What is the number of chloride ions (Cl^-) present in 1.0×10^{-5} mol of $AlCl_3$?

A .1.80×10^{19}
B .6.02×10^{18}
C. 6.02×10^{23}
D. 6.02×10^{28}

***Explanation* :** Write down the dissociation equation:

$AlCl_3 \longrightarrow Al^{3+} + 3Cl^-$

The number of moles of chloride ions is 3 times as much as the number of moles of $AlCl_3$. The number of moles of $AlCl_3$ is 1.0×10^{-5}.The number of moles of chloride ions is 3.0×10^{-5} moles.

One mole of Cl^- ion $=6.02\times 10^{23}$ ions.

3.0×10^{-5} moles $=$?

Number of Cl^- ions $= 3\times 10^{-5}\times 6.02\times 10^{23}$

$= 18.06\times 10^{18}$

$= 1.80\times 10^{19}$ Cl^- ions

∴ ***Answer A***

56. What is the Kelvin scale(K) corresponding to the temperature readings when the degree Celsius (°C) is identical to the degree Fahrenheit(°F)?

A .0K B .37K C 233K D .273K

***Explanation*:** First of all , we have to be able to calculate the value of the temperature at which the readings on the Fahrenheit and Celsius scale are identical using any one of the two conversion equations:

$°F = \frac{9}{5}\,°C + 32$

Let $x = °C = °F$ .....Identical readings.

$x = \frac{9}{5}x + 32$

$x - 32 = \frac{9}{5}x$

$5(x-32) = 9x$

$5x - 160 = 9x$

$-160 = 9x - 5x$

$-40 = x$

∴ Both are identical at -40 , that is -40°F = -40°C

Kelvin = °C + 273

Kelvin = -40°C +273 = 233k = -40°F

∴ ***Answer C***

57. What is the wavelength associated with an electron of mass $m = 9.11\times 10^{-28}$ g , travelling at 40% of the velocity of light?

A .6.06×10^{-15}m
B .2.42×10^{-15} m
C. 6.06×10^{-12}m
D. 2.42×10^{-11}m

***Explanation*:** The wavelength associated with the electron can be calculated using **deBroglie's equation** : $\lambda = \frac{h}{mV}$

Given: $h = 6.626\times 10^{-34}$ $kg.m^2\,s^{-1}$

$V = 40\,\%(3\times 10^8)$ $ms^{-1} = 1.2\times 10^8$ ms^{-1}

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$m = 9.11 \times 10^{28}\,g\ (9.11 \times 10^{-31}\,kg)$

Required : the wavelength $= \lambda = ?$

$$\lambda = \frac{h}{mv}$$

$$\lambda = \frac{6.626 \times 10^{-34}\,kg.m^2s^{-1}}{9.11 \times 10^{-31}\,kg \times 1.2 \times 10^{8}\,ms^{-1}}$$

$$\therefore \lambda = 6.06 \times 10^{-12}\,m$$

∴ Answer C

58. The quantum numbers listed below are meant for four different electrons in an atom:

I : $n = 4, \ell = 0, m_\ell = 0, m_s = +\frac{1}{2}$

II: $n = 3, \ell = 1, m_\ell = 1, m_s = +\frac{1}{2}$

III : $n = 4, \ell = 2, m_\ell = 0, m_s = +\frac{1}{2}$

IV . $n = 4, \ell = 1, m_\ell = 0, m_s = -\frac{1}{2}$

When these sets of quantum numbers are arranged in order of increasing energy , one may get:

A .I < II < III < IV
C .II < I < III < IV
B .I < III < II < IV
D .IV < III < II < I

Explanation: The question is treated using the following simple table for the four electrons.

Electron	n	ℓ	Type of subshell
I	4	0	4s
II	3	1	3p lowest energy
III	4	2	4d highest
IV	4	1	4p

∴3p < 4s < 4p < 4d or II < I < IV < III

∴ No Answer

59. Which of the following hybrid orbitals is favoring the formation of trigonal bipyramidal?

A .sp^3d B. sp^3 C. sp^3d^2 D. sp^3d^3

Explanation: We can use the following table to answer the question

Type of hybrid orbital	Number of hybrid orbitals	Electronic geometry
❖ Sp^3	4	Tetrahedral
❖ Sp^3d	5	Trigonal bipyramidal
❖ Sp^3d^2	6	Octahedral
❖ Sp^3d^3	7	Pentagonal bipyramidal

∴ Answer A

60. What are the two principal polysaccharide forms of starch?

A .Aldohexose and ketopentose. C. Maltose and cellobiose
B .Amylase and amylopectin D. Sucrose and lactose.

Explanation: Plants store food as polysaccharides called starch. There are two basic kinds of starch. These are:

i) **Amylose:** water soluble. Found in algae and other lower forms of plants. It is a linear polymer of glucose residues.

ii) **Amylopectin:** is water insoluble. Dominant form of starch in higher plants. It is a branched polymer of glucose residues.

Therefore , the two principal polysaccharide forms of starch are **amylose and amylopectin.**

Amylase is an enzyme. None of the four options is the answer .

∴ No Answer

61. For the reaction: $2A + B \longrightarrow C$

The following experimental results were obtained:

Experiment	[A]	[B]	Rate (mol $L^{-1}s^{-1}$)
1	0.50	0.50	0.300
2	0.50	0.25	0.075
3	0.25	0.25	0.075

What is the value of the rate constant?

A .0.6mol $L^{-1}s^{-1}$ C. 1.2 $Lmol^{-1}s^{-1}$
B .0.6 L $mol^{-1}s^{-1}$ D. 2.4 mol $L^{-1}s^{-1}$

Explanation: As we can be observed from the kinetic data given , the rate remained to be 0.075 even if[A] is halved from 0.5M to 0.25M in trials 2 and 3 .The rate of the given reaction is independent of the change in concentration of A. The reaction is zero order in A.

Calculate the order of the reaction in B using trials 1 and 2 for which the concentration of A is constant.

Trial 1: Rate $= k[B]^x \Longrightarrow 0.3 = k(0.5)^x$

Trial 2: Rate $= k[B]^x \Longrightarrow 0.075 = k(0.25)^x$

Divide the result of trial 1 by the result of trial 2 as follows.

$$\frac{0.3}{0.075} = \frac{k(0.5)^x}{k(0.25)^x} \Longrightarrow 4 = (2)^x \Longrightarrow 2^2 = (2)^x \Longrightarrow x = 2$$

The reaction is thus second –order in B.

The rate law for the reaction is :**Rate** $= k[B]^2$

Calculate K using any one of the three rates and the corresponding concentration. Using trial 1 ,rate $= k[B]^2$

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$0.3 = k(0.5)^2 \Rightarrow 0.3\ mol\ L^{-1}S^{-1} = k(0.25)\ mol^2\ L^{-2}$

$\Rightarrow k = \frac{0.3\ mol\ L^{-1}S^{-1}}{0.25\ mol^2L^{-2}} = \mathbf{1.2\ L\ mol^{-1}s^{-1}}$

∴ *Answer C*

62. Increase in temperature of rate of a given reaction is due to the increase in the:

A .extent of molecular dissociation.
B .activation energy of the reaction.
C .frequency of collision of the reacting species.
D .numerical value of the rate constant of the reaction.

Explanation :The question must be modified as follow to be meaningful. Increase in the rate of a given reaction as temperature increases is due to the increase in the – frequency of collision of the reacting species.

∴ *Answer C*

63. If a piece of aluminum (Al) foil measuring 24 cm by 31 cm has a mass of 10.35 g, (density of Al = 2.70 g cm^3) .What is thickness of the foil in millimeters?

A .5.15×10^{-3}
B .5.15×10^{-2}
C. 3.833
D. 744

Explanation : Use the known (given) quantities to determine the thickness of the foil in millimeters.

Given quantities : $\ell = 31$ cm , h = 24 cm , m = 10.35 g

Required: thickness

$\text{Density} = \frac{\text{mass}}{\text{volume}}$ ..... Volume is unknown.

$\text{Volume} = \frac{\text{mass}}{\text{density}} = \frac{10.35\ g}{2.70\ g\ cm^{-3}} = 3.833\ cm^3$

$\text{Volume} = 3.833 cm^3$

$\text{Volume} = \ell \times h \times \text{thickness}$

$\Rightarrow \text{Thickness} = \frac{\text{volume}}{\ell \times h}$

$x(\text{thickness}) = \frac{3.833 cm^3}{744\ cm^2} = 5.15 \times 10^{-3}\ cm \times 10mm/cm$

∴ x= 5.15×10^{-2} millimeter since 1 cm = 10mm.

Answer B

64. What sizes of particles and velocities can one consider quantum effect?

A .Particles with very large mass and large velocities.
B .Particles with large mass and small velocities.
C .Particles with very small mass and large velocities.

D .Particles with small mass and small velocities.

Explanation: Quantum effect include:
i) quantization of energy (E = hv) , $E = mc^2$
ii) wave nature of particles.
From the particle nature of energy and the wave nature of particle, **de Broglie** derived the relation: $\lambda = \frac{h}{mv}$.
This relation is meaningful (valid) for particles with very small mass and large velocities.

Answer C

65.Which one of the following molecules/molecular ions is paramagnetic according to the molecular orbital theory?

A .O_2^{2-} B .O_2 C .F_2 D .O_2^{2+}

Explanation: Use the following order of energy of molecular orbital's to work out the electronic configuration of the species given in the options.

	No of electrons	σ 1s	σ^* 1s	σ 2s	σ^* 2s	σ $2p_x$	$\pi 2p_y$	$\pi 2p_z$	π^* $2p_y$	π^* $2p_z$
O_2^{2-}	18e-	2	2	2	2	2	2	2	2	2
O_2	16e-	2	2	2	2	2	2	2	1	1
F_2	18e-	2	2	2	2	2	2	2	2	2
O_2^{2+}	14e-	2	2	2	2	2	2	2	-	-

Note : $O_2^{2-} \Rightarrow$ all paired
- $O_2 \Rightarrow$ two unpaired
- $F_2 \Rightarrow$ all paired
- $O_2^{2+} \Rightarrow$ all paired

i) A species in which all the electrons are paired exhibits dia magnetism. Such species are slightly repelled by an external magnetic field.F_2,O_2^{2+} and O_2^{2-} **are diamagnetic.**
ii) A species with unpaired electrons exhibits paramagnetic property, that is it will be slightly attracted by an external magnetic field

∴ O_2 is paramagnetic

Answer B

66. The reaction for the formation of nitrosyl chloride

$2NO(g) + Cl_2(g) \rightleftharpoons 2NOCl(g)$

was studied at 25°C .The value of k_p for this reaction at 25°C is $1.9\times 10^3\ atm^{-1}$.What is the value of K at 25°C?

A .1.9×10^{-3}L/mol
B .3.8×10^{-3}L/mol
C. 4.6×10^4L/mol
D. 4.6×10^5L/mol

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$0.3 = k(0.5)^2 \Rightarrow 0.3\ mol\ L^{-1}S^{-1} = k(0.25)\ mol^2\ L^{-2}$

$\Rightarrow k = \frac{0.3\ mol\ L^{-1}S^{-1}}{0.25\ mol^2L^{-2}} = 1.2\ L\ mol^{-1}s\text{-}1$

∴ ***Answer C***

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A .extent of molecular dissociation.
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C .frequency of collision of the reacting species.
D .numerical value of the rate constant of the reaction.
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∴ ***Answer C***

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A .5.15×10^{-3} C. 3.833
B .5.15×10^{-2} D. 744
Explanation : Use the known (given) quantities to determine the thickness of the foil in millimeters.
Given quantities : $\ell = 31$ cm , h = 24 cm , m = 10.35 g
Required: thickness

Density = $\frac{mass}{volume}$ ..... Volume is unknown.

Volume = $\frac{mass}{density} = \frac{10.35\ g}{2.70\ g\ cm^{-3}} = 3.833\ cm^3$

Volume = $3.833 cm^3$

Volume = $\ell \times h \times$ thickness

$\Rightarrow$ Thickness = $\frac{volume}{\ell \times h}$

$x(thickness) = \frac{3.833 cm^3}{744\ cm^2} = 5.15\times 10^{-3}$ cm × 10mm /cm

∴ $x = 5.15\times 10^{-2}$ millimeter since 1 cm = 10mm.

∴ ***Answer B***

64. What sizes of particles and velocities can one consider quantum effect?
A .Particles with very large mass and large velocities.
B .Particles with large mass and small velocities.
C .Particles with very small mass and large velocities.

D .Particles with small mass and small velocities.
Explanation: Quantum effect include:
i) quantization of energy (E = hv) , $E = mc^2$
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From the particle nature of energy and the wave nature of particle, ***de Broglie*** derived the relation: $\lambda = \frac{h}{mv}$.
This relation is meaningful (valid) for particles with very small mass and large velocities.

∴ ***Answer C***

65.Which one of the following molecules/molecular ions is paramagnetic according to the molecular orbital theory?
A .O_2^{2-} B .O_2 C .F_2 D .O_2^{2+}
Explanation: Use the following order of energy of molecular orbital's to work out the electronic configuration of the species given in the options.

	No of electrons	σ 1s	σ^* 1s	σ 2s	σ^* 2s	σ $2p_x$	$\pi 2p_y$	$\pi 2p_z$	π^* $2p_y$	π^* $2p_z$
O_2^{2-}	18e-	2	2	2	2	2	2	2	2	2
O_2	16e-	2	2	2	2	2	2	2	1	1
F_2	18e-	2	2	2	2	2	2	2	2	2
O_2^{2+}	14e-	2	2	2	2	2	2	2	-	-

Note : O_2^{2-} $\Rightarrow$ all paired
- $O_2 \Rightarrow$ two unpaired
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∴ **O_2 is paramagnetic**

∴ ***Answer B***

66. The reaction for the formation of nitrosyl chloride
$2NO(g) + Cl_2(g) \rightleftharpoons 2NOCl(g)$
was studied at 25°C .The value of k_p for this reaction at 25°C is $1.9\times 10^3\ atm^{-1}$.What is the value of K at 25°C?
A .1.9×10^{-3}L/mol C. 4.6×10^{4}L/mol
B .3.8×10^{-3}L/mol D. 4.6×10^{5}L/mol

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Explanation: Use the relation : $K_P = K_C(RT)^{\Delta n}$

Where Δn is sum of coefficients of products minus sum of coefficient of reactants.For the reaction given in the equation, $\Delta n = 2 - (2+1) = -1$

$\therefore K_P = K_C (R_T)^{-1} \implies K_P = \frac{K_C}{R_T} \implies K_P R_T = K_C$

$K_C = K_P . R . T$

$K_C = 1.9 \times 10^3\ atm^{-1} \times 0.0821 L.atm .mol^{-1}k^{-1} \times 298K$

$\therefore K_C = 4.6 \times 10^4\ L\ mol^{-1}$.

∴ Answer C

67. Which of the following reactions will convert carboxylic acids to primary amines?

A .Decarboxylationwith HBr and peroxide, then reaction of the alkyl bromide with ammonia.

B .Reduction of the acid to the alcohol with NaOH / formaldehyde then reaction with ammonium chloride and heat.

C .A two- step conversion, first to the amide with ammonia and heat, and then by reduction with lithium aluminum hydride or hydrogen plus catalyst.

D .Lactase fermentation in the presence of ammonia atmosphere.

Explanation: Amines are ammonia derivatives in which the hydrogen atoms are replaced by hydrocarbon radicals. Depending on the number of hydrocarbon radicals amines are divided into .These are:

i) **primary :** $R\text{-}NH_2$

ii) **secondary :** $R\text{–}NH\text{–}R'$ and

iii) **tertiary :** $R\text{—}\underset{\displaystyle R^2}{\overset{\displaystyle |}{N}}\text{—}R'$

In the presence of HBr and a peroxide, the carboxylic undergoes **decarboxylation** producing alkyl bromide ($R\text{-}CH_2\text{–}Br$).

✓The alkylation of ammonia yields the amine

$$R\text{-}CH_2\text{–}Br + NH_3 \longrightarrow \left[R\text{—}\underset{\displaystyle NH_3}{\overset{\displaystyle |}{CH_2}}\right]^+Br^- \xrightarrow{HBr^-} R\text{–}CH_2\text{–}NH_2$$

The product is a primary amine.

∴ Answer A

68. A solution was prepared by adding 48 g of methanol (CH_3OH) into 81g of water (H_2O).What is the mole fraction of methanol in this solution?

A .0.25 B. 0.75 C .1.5 D .4.5

Explanation: The mole fraction (X) of a component in a solution is the number of moles of the component divided by the total number of moles of all the components present in the solution. To calculate the mole fraction, we first convert the given masses to the number of moles.

i) Moles of methanol $= \frac{\text{Given mass}}{\text{molar mass}} = \frac{48g}{32g/mole} = 1.5$ moles

ii) Moles of water $= \frac{\text{Given mass of water}}{\text{molar mass of water}} = \frac{81g}{18g/mole} = 4.5$ moles

Now determine the mole fraction of methanol (CH_3OH)

$X_{methanol} = \frac{n_{methanol}}{n_{water} + n_{methanol}} = \frac{1.5}{4.5+1.5} = 0.25$

Mole fraction of methanol in the solution is 0.25.

∴ Answer A

69. A solution was prepared by dissolving 3.75g of pure hydrocarbon in 95.0 g of cyclohexane. The boiling point of pure cyclohexane was observed to be 80.70°C and that of the solution was 81.45°C. What is the approximate molecular weight of the hydrocarbon? (k_b for cyclohexane = 2.79°C/m)

A .71.0g/mol C. 147g/mol

B .105g/mol D. 312g/mol

Explanation: The boiling point elevation (ΔT_b) is defined as the boling point of the solution (T_L) minus the boiling point of the pure solvent ($T°_b$).

$\Delta T_b = T_b - T_b° \implies \Delta T_b = 81.45 - 80.7 = 0.75°C$

ΔT_b is directly proportional to the molality of the solution.

$\Delta T_b \propto m \implies \Delta T_b = K_b m$

Where K_b is the molal boiling point elevation constant and it is equal to 2.79°C/m for cyclohexane.

From $\Delta T_b = K_b m$, m (molality) $= \frac{\Delta T_b}{K_b} = \frac{0.75°C}{2.79°C/m}$

$\therefore m = 0.2688$

$\text{Molality} = \frac{\text{moles of solute}}{\text{kg of solvent}}$

$0.2688\ m = \frac{n}{0.095kg}$

$\implies n = 0.2688 \times 0.095$

$\therefore$ Moles of solute = 0.0255 mol = n

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From $n = \frac{m}{M}$, M (molar mass) $= \frac{m}{n} = \frac{3.75\ g}{0.0255\ mole}$

∴**M (molar mass) =147 g mol^{-1}**

∴ Answer C

70. Which of the following molecules has a dipole moment?
A .XeF_4 B. H_2S C. SO_3 D.CH_4

Explanation: For **CH_4 ,XeF_4 , SO_3** dipole moment is equal to zero because all are non-polar while **H_2S** is polar.

∴ Answer B

71. To 0.2 M solution of a weak monoprotic acid, HA, enough quantity of its sodium salt, NaA , was dissolved to give a concentration of 0.2M of the salt. What will be the acid concentration, $[H_3O^+]$,in the final solutions? (k_a of HA $= 1.80\times 10^{-5}$).

A .3.60×10^{-6} M C .1.80×10^{-5} M
B .1.00×10^{-5} M D .1.90×10^{-3} M

Explanation: Dissociation equation of the weak acid , HA

$HA(aq) + H_2O(\ell) \rightleftharpoons H_3O^+(aq) + A^-(aq)$

When a solution of NaA is added to this equilibrium system, the dissociation of HA decreases due to the common ion A^- which shifts the equilibrium to the left. Use the following concentration table to solve for the concentration of the ions.

$HA(aq) + H_2O(\ell)$	$\rightleftharpoons$	$H_3O^+(aq)$	+	$A^-(aq)$
Initial 0.2M		0		0.2M
Change -x		+x		+x
Equil 0.2 –x		x		0.2 + x

$K_a = \frac{[H_3O^+][A^-]}{[HA]} = \frac{X(0.2+X)}{(0.2-X)}$ ..... Assume $0.2 + x = 0.2 - x = 0.2$

$K_a = \frac{(0.2)x}{(0.2)} = k_a$ which is given as 1.8×10^{-5}

∴ **$x = [H_3O^+] = 1.8\times10^{-5}$M**

∴ Answer C

72. Gaseous petrol in a combustion system has done 375KJ of work during its expansion in the piston. Simultaneously, it absorbed 586 KJ of heat from the engine. What is the internal energy change during the process?

A .+211KJ C. -211 KJ
B .+961 KJ D. -961KJ

Explanation: When a system does work on its surroundings , its internal energy decreases by an amount equal to the work done

(-375 KJ) .If the system absorbs heat energy from its surroundings, its internal energy increases by an amount equal to the heat absorbed (+586 KJ).

$E = Q + w$ .....First Law of thermodynamics.

For this particular system $W < 0$ and $Q > 0$

$E = Q + W \Longrightarrow E = 586kJ - 375kJ = +211$ **KJ**

∴ Answer A

73. Which of the following reactions is expected to have negative value of entropy change(ΔS)?

A .$C_6H_6(S) + 6O_2(g) \longrightarrow 6CO_2(g) + 6H_2O(\ell)$
B .$CaCO_3(S) \longrightarrow CaO(S) + CO_2(g)$
C .$N_2O_4(g) + Cl_2(g) \longrightarrow 2NOCl(g) + O_2(g)$
D .$2SO_2(g) + O_2(g) \longrightarrow 2SO_3(g)$

Explanation: **Entropy (s)** is a thermodynamic property of a system that is related to its degree of randomness or disorder. If the particles of a system are disorder, as in the following change, the change in entropy (ΔS) is positive .

Solid ⟶ Liquid ⟶ Gas
Increase in entropy

System becomes more disordered

If the arrow is reversed, the system will be in a more ordered situation and ΔS is negative. Use this guide line to analyses each choice.

- ✓ Choice A change from the more ordered solid form to the less ordered liquid and highly disordered gas and **$\Delta S > 0$.**
- ✓ Choice B highly ordered crystal lattice of $CaCO_3$ to a less ordered CaO (s) and highly disordered CO_2(g) making and **ΔS positive.**
- ✓ **Choice C two moles** of particles are transformed in to three moles of particles, that is , more disordered and **$\Delta S > 0$.**
- ✓ **Choice D 3moles** of particles combined to give 2 moles of particle , that is , more ordered and **$\Delta S < 0$.**

∴ Answer D

74. Consider the following unbalanced redox reaction in acidic solution:

$MnO_4^- + Fe^{2+} \longrightarrow Mn^{2+} + Fe^{3+}$

What is the change in oxidation state for both the substances oxidized and reduced, and the coefficients of Fe^{2+} and Mn^{2+}, respectively, after balancing?

A .2 and 7 ,and 2 and 5 C. 1 and 5 , and 5 and 1
B .3 and 2 , and 5 and 1 D. 2 and 5 , and 5 and 2

Explanation: We can determine the coefficient by balancing the equation using either the half – reaction or change in oxidation number method. Let us use change in oxidation number method.

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i) $\overset{+7}{MnO_4^-} + \overset{+2}{Fe} \longrightarrow Mn^{2+} + Fe^{3+}$

5 electrons gained

Loss of electron

ii) Equalize the number of electrons gained and lost. Multiply the oxidized species by 5

$MnO_4^- + 5Fe^{2+} \longrightarrow Mn^{2+} + 5Fe^{3+}$

iii) Balance the charge by adding H^+ to the less positive side since it is in acidic medium: **-1 +10 ≠ 2+15**

Add $8H^+$ to the left hand side: **9+8 = 15 + 2**

$8H^+ + MnO_4^- + 5Fe^{2+} \longrightarrow Mn^{2+} + 5Fe^{3+}$

iv) Balance oxygen atoms by adding H_2O to the side deficient in oxygen.

$8H^+ + MnO_4^- + 5Fe^{2+} \longrightarrow Mn^{2+} + 5Fe^{3+} + 4H_2O$

✓ Change in oxidation for the substance oxidized and reduced respectively is 1 and 5.

✓ Coefficient for the substance oxidized and reduced are respectively 5 and 1 .The answer is **1, 5, 5, 1** .

∴ ***Answer C***

75. A 50mL solution of H_2SO_4 of 0.205 M is titrated with NaOH solution of unknown concentration. The endpoint against phenolphthalein indicator was signaled when 41.0 mL of NaOH was added. What is the concentration of NaOH solution?

A .0.10M
B .0.25M
C .0.41M
D .0.50M

Explanation: One equivalent of an acid neutralizes , one equivalent of a base. The number of equivalent of an acid or a base is equal to normality times volume. Convert the molarity to normality using the relation : $N = M \times n$ where **n** : No of ionizable hydrogen

∴ $NH_2SO_4 = 0.205 \times 2 = 0.41N$

Equivalent of acid = Equivalent of base $\Rightarrow N_AV_A = N_BV_B$

$$0.41\,N \times 50\,mL = N_B \times 41\,mL$$

$$\Rightarrow N_B = \frac{0.41N \times 50mL}{41mL} = \frac{50}{100} = \mathbf{0.5\ N}$$

For NaOH , Normality = molarity since n =1 .....In M× n

Therefore ,0.5N NaOH is the same as 0.5 M

∴ ***Answer D***

76. What is the PH of an aqueous solution prepared to contain 1.3×10^{-3} M sodium nitrite ($NaNO_2$) if the acid dissociation equilibrium constant ,K_a for nitrous acid (HNO_2) is 5.1×10^{-4} ? $K_w = 1.0 \times10^{-14}$

A .3.1 B.) 5.1 C .6.0 D .7.3

Explanation: Being a salt of the weak acid, HNO_2 ,$NaNO_2$ undergoes hydrolysis to give a slightly basic solution.

Hydrolysis of $NaNO_2$

$NO_2^-(aq) + H_2O(\ell) \rightleftharpoons HNO_2(aq) + OH^-(aq)$

K_b (hydrolysis constant) $= \frac{[HNO_2][OH^-]}{[NO^-_2]}$

K_b = (hydrolysis constant) $= \frac{K_W}{K_a} = 1.96\times 10^{-11}$

	$NO_2^-(aq) + H_2O(\ell)$	$\rightleftharpoons HNO_2(aq)$	$+ OH^-(aq)$
Initial:	0.0013	0	0
Change:	-x	+x	+x
Equil:	0.0013 –x	x	x

$$K_b = 1.96\times10^{-11} = \frac{[HNO_2][OH^-]}{[NO^-_2]} = \frac{(x)(x)}{0.0013-x}$$

Since k_b is very , very small x in (0.0013 –x) is negligible and can be ignored.

$$1.96\times 10^{-11} = \frac{x^2}{0.0013}$$

$$\Rightarrow x^2 = 1.3\times10^{-3}\times1.96\times10^{-11}$$

$$x^2 = 2.5\times10^{-14}$$

$$\therefore x = 1.58\times 10^{-7} = [OH^-]$$

$$P^{OH} = -\log[OH^-]$$

$$P^{OH} = -\log(1.58\times10^{-7})$$

$$P^{OH} = 7 - \log 1.58$$

$$P^{OH} = 6.7$$

$$P^H + P^{OH} = 14$$

$$P^H + 6.7 = 14$$

$$\therefore P^H = 7.3$$

∴ ***Answer D***

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77. What reactions occur at the anode and cathode when an aqueous solution of Na_2SO_4 is electrolyzed?

		E^o red
I) $S_2O_8^{2-} + 2e^-$	$\longrightarrow 2SO_4^{2-}$	2.01V
II) $O_2 + 4H^+ + 4e^-$	$\longrightarrow 2H_2O$	1.23 V
III) $2H_2O + 2e^-$	$\longrightarrow H_2 + 2OH^-$	−0.83V
IV) $Na^+ + e$	$\longrightarrow Na$	−2.71V

A. H_2 at cathode and O_2 at anode.
B. Na at cathode and $S_2O_8^{2-}$ at anode.
C. H_2 at cathode and $S_2O_8^{2-}$ at anode
D. Na at cathode and O_2 at anode

Explanation:

i) The species present at the anode, are H_2O and SO_4^{2-}. Oxidation of H_2O requires less energy and H_2O is preferentially oxidized.

ii) The species present at the cathode are Na^+ ions and H_2O and water reduced easily.

Anode reaction : $2H_2O \longrightarrow O_2 + 4H^+ + 4e$ (oxidation)

Cathode reaction: $2H_2O + 2e^- \longrightarrow H_2 + 2OH^-$ (reduction)

Reduction of H_2O yields H_2 at the cathode.

Oxidation of H_2O produces O_2 at the anode.

∴ Answer A

78. Which synthetic polymer is produced from Caprolactam?

A. Nylon – 6
B. Nylon 6,10
C. Teflon
D. Terylene

***Explanation*: Caprolactum** is a cyclic monomer that polymerizes to produce **nylon 6.**

∴ Answer A

79. For the following hypothetical equation, in aqueous solution, what is the correct representation of the cell notation?

$A(s) + B^{2+}(aq) \longrightarrow A^{2+}(aq) + B(s)$

A. $A(s)\,|A^{2+}(aq)\,\|\,B^{2+}(aq)|\,B(s)$
B. $B^{2+}(aq)\,|\,B(s)\,\|A^{2+}(aq)|\,A(s)$
C. $A^{2+}(aq)\,|\,A(s)\,\|B(s)\,|\,B^{2+}(aq)$
D. $B(s)\,|B^{2+}(aq)\,\|\,A^{2+}(aq)\,|\,A(s)$

Explanation : $A(s) + B^{2+}(aq) \longrightarrow A^{2+}(aq) + B(s)$

$A(s) \longrightarrow A^{2+}(aq) + 2e^-$ .....oxidized (Anode)

$B^{2+}(aq) + 2e^- \longrightarrow B$ ..... Reduced (cathode)

Cell notation: also known as cell diagram is the representation of a voltaic cell in which the anode half – cell is written to the left. The cell notation for the above cell is represented as follows:

$A(s)\,|\,A^{2+}(aq)\,\|\,B^{2+}(aq)\,|\,B(s)$

∴ Answer A

80. How many mL conc. HNO_3 and how many mL of water are required to prepare 500ML of 0.1M HNO_3 from a conc. 13M HNO_3?

A. 1mL HNO_3 and 496.15 mL H_2O
B. 3mL HNO_3 and 500mL H_2O
C. 3.85 mL HNO_3 and 500 mL H_2O
D. 3.85 mL HNO_3 and 496.15 mL H_2O.

***Explanation*:** A 13 M concentrated nitric acid is diluted to 500mL of 0.1M HNO_3. For dilution we use the relation : $M_1V_1 = M_2V_2$

$M_1 = 13M$ $\quad M_2 = 0.1M$

$V_1 = ?$ $\quad V_2 = 500ml$

$M_1V_1 = M_2V_2$

$$V_1 = \frac{M_2V_2}{M_1}$$

$$V_1 = \frac{0.1M \times 500mL}{13M}$$

$\therefore V_1 = 3.85$ mL

Volume of concentrated nitric acid is 3.85 mL.

The total volume of the diluted solution is 500 mL. This is equal to the volume of the concentrated nitric acid (3.85 mL) plus the volume of the water added (496.15 mL.)

∴ Answer D

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1 .Commercially , liquid vegetable oils are converted to solid fats such as margarine by:

A . hydrogenation. C. Saponification
B .hydration D. Oxidation.

***Explanation*: Liquid vegetables oils** can be converted to solid fats such as margarine by the addition of hydrogen (hydrogenation) at high pressure in the presence of **nickel or palladium** catalyst.

∴ Answer A

2.What is the chemical name for Aspirin ?

A .Acetyl salicylic acid C. Methyl salicylate
B .Salicylic acid D. Sodium salicylate

***Explanation*: Aspirin** , the common pain killer , is prepared by the reaction of salicylic acid with acetic acid. Its chemical name is acetyl **salicylic acid (ASA).**

∴ Answer A

3.Which of the following statement(s) is (are) true of an ideal liquid – liquid solution?

I . It obeys PV = nRT
II. It obeys Raoult's law.
III .Solute – solute , solvent – solvent , and solute – solvent interactions are very similar.
IV . Solute – solute , solvent – solvent , and solute – solvent interactions are quite different.

A .I ,II and III B. I ,II and IV C .II and III D .II and IV

***Explanation*:** For an ideal **liquid – liquid** solution:

i) heat of solution (ΔH) is zero.
ii) volumes of components are additive.
iii) the vapor pressures are additive.
iv) obey Raoult's Law.
v) solute – solute , solute – solvent and solvent – solvent interactions are very similar.

∴ Answer C

4 .Which of the following is correct?

A .1L= $1dm^3$ C. 10L=$1dm^3$
B. 1L=$10dm^3$ D. 1L=$1m^3$

***Explanation*:** 1L= 1000mL = $1000cm^3$ (1mL = $1cm^3$)
$1dm^3 = 1000cm^3$ implies that 1L = $1dm^3$

∴ Answer A

5.Which of the following is **NOT** a conjugate acid – base pair?

A .HNO_3 /NO_3^- C .NH_3/NH_2^-
B .H_2SO_4 /HSO_4^- D. H_3O^+/OH^-

***Explanation*:** An acid has one more H than its conjugate base. A base has one fewer hydrogen than its conjugate acid.

∴ Answer D

6. Which electron transition in a hydrogen atom releases the largest energy?

A .n =2 ⟶ n =1 C. n = 6 ⟶ n =3
B .n = 4 ⟶ n =2 D. n =7 ⟶ n =6

***Explanation*:** For the transition of electrons between energy levels, the energy released is calculated by using the equation:

$\Delta E = 2.18\times 10^{-18}\left(\frac{1}{n_i^2}-\frac{1}{n_f^2}\right)$ where n_i – initial energy level and n_f – final energy level

i) For the transition of an electron from n =2 to n=1.

$\Delta E = 2.18 \times 10^{-18}\ \left(\frac{1}{4}-\frac{1}{1}\right)J$

$\Delta E = -0.75\times 2.18 \times 10^{-18} J$

$\therefore \Delta E = -1.635\times 10^{-18}$ J

✓ Using the same equation, we get the following values for the other transitions

ii) For the transition from n = 4 to n = 2

$\Delta E = -0.409 \times 10^{-18} J$

iii) For the transition from n= 6 to n= 3

$\Delta E = -0.1816 \times 10^{-18} J$

iv) For the transition from n= 7 to n = 6

$\Delta E = -0.0161\times 10^{-18} J$

✓ From the different four transitions , the electron transition from n= 2 to n=1 releases the largest energy.

∴ Answer A

7. The simplest formula for a compound containing Mn^{4+} and O^{2-} is:

A .MnO C .Mn_2O_4
B .MnO_2 D. Mn_4O_2

***Explanation*:** To write the formula of a compound obtained from two elements (components) we cris –cross the charges (valences) on each symbol or radical and place under each symbol as subscripts and simplicity to the smallest counting numbers.

$Mn^{4+}\ O^{2-} \Longrightarrow Mn_2O_4 \Rightarrow MnO_2$

∴ Answer B

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8. The unit cell in a certain lattice consists of a cube formed by an anion at each corner, an anion in the center , and a cation at the center of each face. How many cations and how many anions does the unit cell have?

A .5 anions and 6 cations. C. 2 anions and 3 cations

B .5 anions and 3 cations D. 3 anions and 4 cations.

Explanation **: A unit cell** is the smallest fraction of a space lattice. Its unit translations make up the entire crystal lattice.

✓ The unit cell of a simple cubic ionic crystal consists of anions and cat ions.

✓ In a cubic crystal lattice , **eight unit cells** share one vertex. An anion at one vertex occupies one eighth$\left(\frac{1}{8}\right)$ of each unit cell. Since each cubic unit cell has eight vertices each unit must have the equivalent of one whole$\left(\frac{1}{8}\times 8\right)$,that is one anion.

✓ The unit cell of this question gets a share of **one anion** from the vertices and **one anion** in the center , that is ,two anions per unit cell.
The faces of two unit cells share one cation and half of a cation belongs to a unit cell. Therefore , of the six cations in the centers of the faces $\left(\frac{1}{2}\times 6\right)$ =3 belong to a unit cell. The unit cell of this question has 2 anions and 3 cations.

∴ Answer C

9. In a reaction , A + B ⟶ product, the rate is doubled when the concentration of B is doubled , and the rate increases by a factor of 8 when the concentrations of both the reactants (A and B) are doubled, the rate law for the reaction can be written as:

A .Rate = k [A][B] C. Rate = $k[A]^2[B]^1$

B. Rate = k [A][B]² D. Rate = $k[A]^1[B]^2$

Explanation **:** A + B⟶ product .Its rate law (rate equation) is

rate = $k[A]^x[B]^y$

i) When the concentration of B is doubled the rate is doubled. This implies that the reaction is first –order in B and y =1.

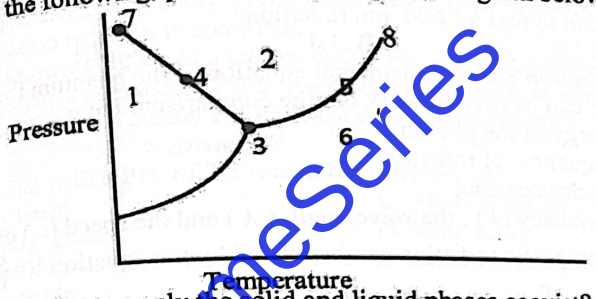
ii) The rate increases by a factor of 8 when the concentrations of both A and B are doubled .The rate must increase by a factor of 4 when [A] is doubled. The reaction is second –order in A and x =2 .

The rate law for the reaction is :

Rate = $k[A]^2[B]^1$

∴ Answer C

10. Answer the following question using the phase diagram below.



At which point can only the solid and liquid phases coexist?

A .1 B.2 C.3 D. 4

Explanation**: Point 1** : represents solid – phase.

- **Point 2:** indicates liquid phase.
- **Point3:** represents the triple point.
- **Point 4** : the solid and liquid phases coexist in equilibrium at point 4.

∴ Answer D

11. Butane burns in oxygen according to the equation below.

$C_4H_{10}(g)+13O_2(g) \longrightarrow 8CO_2(g)+10H_2O(\ell)$.If 11.6 g of butane is burned in 11.6 g of oxygen ,which is the limiting reagent?

A .Butane C. Neither

B .Oxygen D. Both oxygen and butane

Explanation **:** From the balanced chemical equation 2 moles of butane (116g of butane) requires 13 moles of oxygen (416 g of oxygen)for its complete combustion. By the same reasoning 11.6 g of butane requires 41.6 g of oxygen. But the amount of oxygen available is only 11.6 g. Therefore, oxygen is the limiting reactant.

∴ Answer B

12. A beaker filled to the 100 mL mark with salt (the salt has a mass of 100g) and another beaker to the 100mL mark with water (the water has a mass of 100g)are mixed together in a bigger beaker until the salt is completely dissolved. What will be the mass of the solution?

A .It will be much more than 200g.

B .It will be much smaller than 200g.

C .It will be exactly 200g.

D .It will be slightly more than 200g.

Explanation: Mass being the amount of matter in a substance (or system of solution) is constant and the same everywhere. Mass is conserved and the mass of the solution is exactly 200g.

∴ Answer C

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13. What is the ratio of the energy of a photon of 300 nm wavelength radiation to that of 600 nm radiation?

A .1:2 B .1:1 C .2:1 D .3:1

Explanation : The fundamental equation of the quantum theory, that is $E = h\nu$ can be used to work out the comparison. Here

E : Energy of the photon.

ν : Frequency of radiation.

h : planck's constant

The frequency (ν), the wavelength (λ) and the speed (c) of electromagnetic radiation can be related by the equation $\nu = \frac{c}{\lambda}$

Substitute $\frac{c}{\lambda}$ for V in the equation $E = h\nu$.

Thus energy of a photon whose $\lambda = 300nm$ is given as

$$E = \frac{hc}{300nm}$$

In the same manner the energy of a photon whose $\lambda =$ 600 nm is expressed as $E = \frac{hc}{600nm}$

Take the ratio as $\frac{hc}{300nm} \div \frac{hc}{600nm}$

This ratio is $\frac{hc}{300nm} \times \frac{600nm}{hc} = \frac{600nm}{300nm} = \frac{2}{1}$

Therefore, **the ratio is 2:1**

∴ Answer C

14. Which one of the following is **NOT** an intensive property?

A .Mass B. Temperature C. Colour D. Density

Explanation : There are two kinds of physical properties. These are:

i) **Extensive physical properties** are properties which depend on the amount or quantity of a sample. ***Example :*** Length , mass ,volume.

ii) **Intensive physical properties** are properties which do not depend on the amount of substance present. ***Example:*** Density ,colour, melting point ,temperature and hardness.

∴ Answer A

15. During the titration of a known volume of a strong acid with a strong base ,there is

A .a steady increase in PH.

B .a sharp increase in PH around the end point.

C .a steady decrease in PH.

D .a sharp decrease in PH around the end point.

Explanation: **Titration** is a technique in which a solution of known concentration is used to determine the concentration of an unknown solution. The titrant (**the known solution**) is added from a burette to a

known quantity of the analyte (**the unknown solution**) until the neutralization reaction is complete.

✓ When a known volume of a strong acid is titrated with a strong base ,there is steady increase in PH at the start ,a sharp increase in PH near the end point and a steady PH increase after the end point

∴ Answer B

16. A solution with a PH of 7.5 would be described as:

A .Very basic C. Slightly acidic

B .Slightly basic D. Very acidic

Explanation: A solution with a PH of 7.5 would be described as **slightly basic .**

∴ Answer B

17. The electrolysis of molten NaCl is an industrial process. What does the electrolysis produce?

A .Na and Cl_2 C .Na^+ and Cl^-

B .H_2 and O_2 D. NaOH and Cl_2

Explanation: In the electrolysis of molten NaCl the following redox reactions take place at the electrodes.

✓ **Anode reaction (oxidation):** $2Cl^- \longrightarrow Cl_2(g) + 2e^-$

✓ **Cathode reaction (reduction) :** $2Na^+ + 2e^- \longrightarrow 2Na$

✓ **Overall reaction :** $2Na^+ + 2Cl^- \longrightarrow 2Na + Cl_2$

✓ Sodium metal and chlorine gas are the products.

∴ Answer A

18. Which of the following metals forms a volatile compound during the extraction process?

A .Fe B. Co C. Ni D. Cu

Explanation : In a process known as the **Mond process** ,carbon monoxide at 80°C is passed over impure nickel to form a volatile compound called **Nickel carbonyl**. This volatile liquid is isolated from impurities by distillation. The purified nickel carbonyl is heated to about 200°C. $Ni(CO)_4 \longrightarrow Ni + 4CO$.

∴ Answer C

19. Which one of the following chemicals is used to disinfect water?

A .Fluorine C. Oxygen

B .Nitrogen D. Chlorine

Explanation: **Chlorine** is used to disinfect water. Because chlorine is strong oxidant enough to destroy bacteria and there by acts as a germicide (a disinfectant). It is widely used for producing safe drinking water.

∴ Answer D

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20. Which of the following is natural polymer?
A .Nylon B. PVC C .Cotton D. Dacron

***Explanation*: Polymers** are broadly classified into two groups. These are (1)natural (ii) synthetic

✓**Natural polymers** include proteins, nucleic acids , starch , rubber (polyisoprene). **Cotton** is a naturally occurring polymer

***Examples* of synthetic polymers** :Nylon ,PVC , Dacron where Dacron is one type of polyester .

∴ ***Answer C***

21. Why are metals soft and malleable?
A .Because they are very shiny.
B .Because they experience electrostatic repulsion.
C .Because of the presence of mobile electrons.
D .Because the metal cations can slip over each other fairly easily.

***Explanation*:** The ability of metals to deform (their malleability and ductility) can be explained by the following facts.

i) The metal cations are capable of slipping with respect to one another when force is applied.
ii) The **bonding mobile electrons** are relatively free to move throughout the three dimensional structure and new metallic bonds can be formed.

∴ ***Answer C***

22. Which molecule or ion does **NOT** have a tetrahedral shape?
A .XeF_4 B .$SiCl_4$ C. BF_4^- D. NH_4^+

***Explanation*:** The shape of a molecular or an ion is **tetrahedral** if there are 4 bonding pairs of electrons around the central atom. This is the case for NH_4^+ ;BF_4^- , $SiCl_4$ but not for XeF_4 molecule. There are 4 bonding pairs and 2 lone pairs of electrons around Xe in XeF_4 molecule.

∴ ***Answer A***

23. How many π bonds are present in CO_2?
A .One B. Two C. Three D. Four

***Explanation*:** The Lewis structure for CO_2 is $\ddot{O} = C = \ddot{O}$ .In a double bond there is one **pi-bond and one sigma bond.** Therefore , in a molecule of CO_2 there are **two sigma and two pi- bonds.**

∴ ***Answer B***

24. In which region of the periodic table would the element with the electronic structure below be located? $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6 4d^4 5s^2$
A .Group 6 C. S -block
B .Noble gases D. D -block

***Explanation*:** The position of a representative element (S –block and p-block element) can be determined from its ground state electronic configuration. The number of valence electrons for the particular element is 2 and the element is in **group IIA and period 5,** Since electrons are filled up to 5^{th} shell. Since the last two electrons entered the S –sub shell, the element is an s –block element.

∴ ***Answer C***

25. Which factors will influence the rate of the reaction shown below?
$NO_2(g) + CO(g) \rightleftharpoons NO(g) + CO_2(g)$
I .The number of collisions per second.
II . The energy of the collisions.
III. The geometry with which the molecules collide
A .I only C. I and II only
B .II only D. I, II and III

***Explanation*:** The alternatives are based on the postulates of the **collision theory**. According to the collision theory a chemical reaction occurs if:

i) the reacting particles collide frequently which is influenced by **concentration , temperature and pressure** for gases.
iii) the reacting particles collide with an energy that is greater than or equal to the **activation energy**. The smaller the **activation energy** , the faster is the reaction.
iv) The collisions have proper geometrical orientation. All the factors can influence the rate.

∴ ***Answer D***

26. Which set of quantum numbers (n, ℓ, m_ℓ ,m_s) is not possible?
A .1,0,0, $\frac{1}{2}$ C. 1,0,0, $-\frac{1}{2}$
B .1,1,0, $\frac{1}{2}$ D. 2,1,-1,$\frac{1}{2}$

***Explanation*:** The question is based on the four types of quantum numbers which are briefly defined below:

i) **principal quantum number (n)** is a positive integer having values ; n = 1,2,3 ,...
ii) **The azimuthal quantum number (ℓ)** is also known as angular momentum quantum number or subsidiary quantum number.It has integral values ranging from 0 to n- 1 .
Example : when n =1 $\ell = 0$
when n = 2 $\ell = 0, 1$
iii) **The magnetic quantum number (m_ℓ)** is also known as the orbital orientation quantum number. It is an integer having values from $-\ell$ to $+\ell$.
Example: if $\ell = 1$, $m_\ell = -1 , 0 , +1$ 3 .....orientations of **p –orbital.**

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$\ell = 2$, $m_\ell = -2, -1, 0, +1, +2$ ..... 5 orientations of d- orbitals.

Iv) **Spin quantum number (m_s)** has only two possible values $+\frac{1}{2}$ (represented by the arrow ↑) and $-\frac{1}{2}$ (represented by the arrow ↓).

Some permitted values of the quantum numbers.

n	ℓ	m_ℓ	m_s
1	0	0	½
1	0	0	-1/2
2	1	0	½

If n=1 , the value of ℓ is only zero.

∴ Answer B

27.Which of the following particles contains more electrons than neutrons?

I. $^{1}_{1}H$ II. $^{35}_{17}Cl^-$ III $^{39}_{19}K^+$

A .I only
B .II only
C .I and II only
D. II and III only

Explanation: The number of subatomic particles contained in each species is given in the following table.

		Number of proton	Number of electrons	Number of neutrons
I	$^{1}_{1}H$	1	1	0
II	$^{35}_{17}Cl^-$	17	18	18
III	$^{39}_{19}K^+$	19	18	20

II and III have equal number of electrons that is 18 electrons each . But only I contains more electrons than neutrons.

∴ Answer A

28.Which statement is true about chemical reactions at equilibrium?

A .The forward and backward reactions proceed at equal rates.
B .The forward and backward reactions have stopped.
C .The concentrations of the reactants and products are equal.
D .The forward reaction is exothermic.

Explanation: For a chemical reaction at equilibrium :

i) The rate of the forward reaction is equal to the rate of the reverse reaction.
ii) All the reactants and products are present.

iii) The equilibrium is dynamic in nature that is , the reaction does not stop.
v) The free energy change of the reaction is zero at equilibrium.
vi) The concentration of reactants and products remain constant if all the reaction parameters are kept constant.

∴ Answer A

29.Which one of the following atoms in its ground state has the greatest number of unpaired electrons?

A .$_{13}Al$ B. $_{14}Si$ C. $_{15}P$ D. $_{16}S$

Explanation: **The number of unpaired electrons** for an element in its ground state can be shown using orbital diagram. **An orbital diagram** is diagram that consists of a box (or a circle or a line) and arrow which indicate the direction of electron spin (↑ or ↓) .For some elements of period 3 , the ground state electronic configuration can be worked out using Neon core and orbital diagram for the valence electrons as follows

$_{13}Al$: [Ne] ↑↓ | ↑ | | ..... One unpaired electron
3S 3P

$_{14}Si$: [Ne] ↑↓ | ↑ | ↑ | ....Two unpaired electrons
3S 3P

$_{15}P$: [Ne] ↑↓ | ↑ | ↑ | ↑ .. Three unpaired electrons.
3S 3P

$_{16}S$: [Ne] ↑↓ | ↑↓ | ↑ | ↑ ... Two unpaired electrons
3S 3P

Phosphorus has the greatest number of unpaired electrons.

∴ Answer C

30.Which compound contains both covalent and ionic bonds?

A .Sodium carbonate , Na_2CO_3
B .Magnesium bromide , $MgBr_2$
C .Dichloromethane , CH_2Cl_2
D .Ethanoic acid , CH_3COOH

Explanation :The bond in sodium carbonate ,Na_2CO_3 .

a) The bond between $2Na^+$ and CO_3^{2-} is **ionic.**
b) The bond in CO_3^{2-} is **covalent bond** between the carbon and the oxygen atoms.
∴ Na_2CO_3 contains both covalent and **ionic bonds.**
c) All the bonds in CH_2Cl_2 and CH_3COOH are **covalent.**

∴ Answer A

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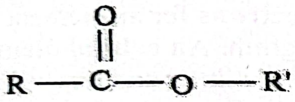
31. Which compound is an ester?

A .CH_3COOH C .C_2H_5CHO

B .$CH_3OC_2H_5$ D. $HCOOCH_3$

Explanation : Esters are among the most widely occurring compounds in nature. Many esters are pleasant smelling substances and responsible for the flavor and fragrance of many fruits.

Example: Banana, pineapple, strawberry, Oils, fats and waxes of plants or animals are esters. Esters have the following basic structure.



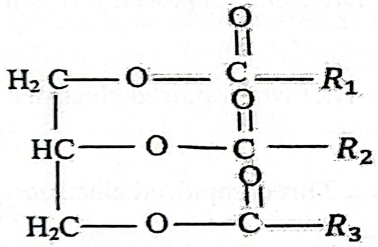
Where R = hydrogen ,alkyl or aryl group and R' = Alkyl or aryl group. Choice D has a structure, that is similar to the above structure.

∴ ***Answer D***

32. A triacylglycerol that is solid at room temperature is called:

A .Lecithin B. Fat C. Wax D. Oil

Explanation :



A triacylglycerol also known as **triglyceride** is an ester of glycerol.It is the major constitutent of **fats and oils. Fats** are solid at room temperature but **oils** may be liquids.

∴ ***Answer B***

33. How many moles of sodium hydroxide are present in 2.5L of 0.5 M aqueous solution?

A .0.2 B .0.5 C .1.25 D .12.5

Explanation : **Concentration** of a solution in moles per liter (molarity) is equal to the number of moles of solute divided by the volume of the solution is given by : $C = \frac{n}{V}$ Where

C : concentration in mol/L or Molarity , M

n : Number of moles of solute.

V : volume of solution in liters.

Data given in the question:

C = 0.5M or 0.5 mol/L V =2.5 L

Required: n (number of moles of solute)

Solution: $C = \frac{n}{V}$

$n = C \times V$

$n = C\,V$

∴ **n = 0.5 mol/L × 2.5 L = 1.25 moles**

∴ ***Answer C***

34. If a solute dissolves in an endothermic process,

A .H bonds must exist between solvent and solute.

B .Strong ion- dipole forces must exist in the solution.

C .The entropy of the solution must be greater than of its pure components.

D .The solute must be a gas.

Explanation : **Solution process** is affected by two factors. These are:

i) enthalpy (ΔH) and

ii) entropy (ΔS)

iii) temperature

iv) Intermolecular force

a) processes with $\Delta H < 0$ are exothermic proceed spontaneously. The situation favors dissolution process.

b) **Endothermic dissolution** ($\Delta H > 0$) can occur spontaneously when the entropy of the solution is greater than the entropy of the components. **Entropy** is a measure of the disorder of the system. Process in which the disorder of the system increases, tend to occur spontaneously.

∴ ***Answer C***

35.A solution is made by dissolving 250.0g of potassium chromate crystals(K_2CrO_4, molar mass ,194.2 g) in 1.00 kg of water. What will be the freezing point of the solution? (K_f for water is 1.86°Cm^{-1}) .

A .–8.87°C B .–7.18°C C .–5.73°C D .–1.86°C

Explanation: To work out this question

i) determine the number of moles of potassium chromate.

$$n = \frac{m}{M} = \frac{250\ g}{194.2 g/mole} = 1.287 \text{ moles of } K_2CrO_4$$

ii) Molality of a solution $= \frac{\text{number of moles of solute}}{\text{kg of solvent}}$

$= 1.287$ mole per kg

iii) Calculate the freezing point depression (ΔT_f) for the electrolyte solution ,K_2CrO_4 (aq) .The Van't Hoff factor (i) is equal to 3 assuming complete dissociation of K_2CrO_4 .Thus $\Delta T_f = iK_f m$

$= 3 \times 1.86°C/m \times 1.287\ m = 7.1814$

iv) Calculate the freezing point (T_f) of the Solution using the relation

$\Delta T_f = T - T_f$.

$7.1814 = 0 - T_f$

$\Rightarrow T_f = -7.18°C$

The freezing point of the solution is – 7.18°C

∴ ***Answer B***

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36. The mechanism of a reaction is shown below.

$HOOH + I^- \longrightarrow HOI + OH^-$ (Slow)

$HOI + I^- \longrightarrow I_2 + OH^-$ (fast)

$2OH^- + 2H_3O^+ \longrightarrow 4H_2O$ (fast)

What is the rate law based on this mechanism?

A .Rate = k [HOOH][I^-]
C .Rate = k [HOOH][I^-]^2
B .Rate = k [HOOH]^2[I^-]
D .Rate = K [HOOH]

Explanation: For a given mechanism, the rate law based on the mechanism is equal to the rate law of the slowest step. The rate law for this elementary step is equal to its molecularity. The rate law for the slow step is :**Rate = K [HOOH][I^-].**

∴ ***Answer A***

37. What is the equilibrium constant expression for the following reaction?

$2Hg(g) + O_2(g) \rightleftharpoons 2HgO(s)$

A .k =1/([Hg]^2[O_2])
C. k = [HgO]^2/([Hg]^2[O_2]
B. k = [Hg]^2[O_2]
D .k = [2HgO]/([2Hg][O_2])

Explanation : The equilibrium constant expression for a reaction, places concentration terms of the products in the numerator and concentration terms of reactants in the denominator. A straight forward application of the law of mass action leads to the following expression.

$$K_c = \frac{[HgO]^2}{[Hg]^2\,[O_2]}$$

However , experimental results show that the position of a heterogeneous equilibrium does not depend on the amount of solid or pure liquid present. Therefore , the equilibrium constant expression for this reaction does not contain the solid term HgO in the numerator.

$$K_C = \frac{1}{[Hg]^2[O_2]}$$

∴ ***Answer A***

38. Which changes will increase the amount of SO_3 (g) at equilibrium?

$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$ $\Delta H^o = -197KJ$

I . Increasing the temperature
II. Decreasing the volume
III. Adding a catalyst

A .I only
C. I and III only
B .II only
D. I ,II and III

Explanation: For the given reaction , changes which cause the equilibrium position to shift to the right are:

i) decrease in temperature at equilibrium shifts the equilibrium position to the product side for exothermic reaction.

ii) increase in pressure (decrease in volume).
iii) adding any of the reactants.
iv) frequent removal of the product.

∴ ***Answer B***

39. Which pair of monomers forms polyesters?

A .Difunctional alcohol, difunctional organic acid.
B .Difunctional alcohol, difunctional amino acid.
C .Ethylene, ethylene.
D .Amino acid, amino acid

Explanation:

i) Difunctional alcohol and difunctional organic acid polymerize to yield a poly ester.
ii) A difunctional alcohol and an amino acid give a polyamide.
iii) Ethylene with ethylene undergoes addition polymerization to produce the polymer polyethylene.
iv) Amino acid with amino acid undergo condensation polymerization to give protein polymers. It only the first reaction that produces a polyester .

∴ ***Answer A***

40.Which compound is a carboxylic acid?

A .CH_3COOH
C .$(CH_3)_2$ $CHOOCH_3$
B .$(CH_3\ CO)_2O$
D .$(CH_3)_2O$

Explanation:

- A= Ethanoic acid (acetic acid).
- B = Ethanoic anhydride (Acetic anhydride)
- C = Not a relevant option.
- D =Dimethyl ether.

∴ ***Answer A***

41. What volume of 0.500 M NaOH is required to neutralize 25.0mL of 1.2 M H_2SO_4 ?(Assume complete ionization of the acid).

A .60 mL
B .90mL
C .100 mL
D .120 mL

Explanation:

i) **One equivalent of an acid** is the mass of the acid in grams that can release one mole of H^+ ions.

ii) **One equivalent of a base** is the mass of the base in grams that can release one mole of OH^- ions.

$H^+ + OH^- \longrightarrow H_2O$

Therefore, one equivalent of an acid neutralizes one equivalent of a base.

iii)Since the number of equivalents is equal to the **normality** times volume it is convenient to convert concentrations given in molarity to normality using the following relation :N =M× n

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NaOH : N = M× n

N = 0.5M× 1 = 0.5N

H_2SO_4 : N = M × n

N = 1.2 × 2 = 2.4 N

$$\text{Normality} = \frac{\text{Number of equivalent of solute}}{\text{volume of solution}}$$

Normality × Volume = No of equivalents

Equivalent of acid = equivalent of a base

$N_aV_a = N_bV_b$..... a ,b are acid and base respectively.

$2.4\ N \times 25mL = 0.5\ N \times V_b$

$$\frac{2.4\ N \times 25\ mL}{0.5N} = V_b = 120\ mL$$

∴ ***Answer D***

42. For which conversion is an oxidizing agent required ?

A .$2H^+(aq) \longrightarrow H_2(g)$ C. $SO_3(g) \longrightarrow SO_4^{2-}(aq)$

B .$2Br^-(aq) \longrightarrow Br_2(aq)$ D. $MnO_2(s) \longrightarrow Mn^{2+}(aq)$

Explanation:

i) $2H^+(aq) \longrightarrow H_2(g)$. Hydrogen is reduced from the +1 oxidation state to zero .Therefore , **reducing agent** is involved.

ii) $2Br^-(aq) \longrightarrow Br_2(aq)$ .Bromide ions are oxidized from -1 to zero. An **oxidizing agent** is involved for this conversion.

iii) $SO_3(g) \longrightarrow SO_4^{2-}(aq)$.In the conversion of SO_3 gas to SO_4^{2-} (aq) , there is no change in the oxidation number of sulfur since it is +6 in both cases. Neither oxidizing agent nor a reducing agent is required.

iv) $MnO_2(s) \longrightarrow Mn^{2+}(aq)$.The substance which causes the reduction of Mn from +4 in MnO_2 to Mn^{2+} is **a reducing agent.**

∴ ***Answer B***

43. Which statement about the following reaction is correct?

$2FeS(s) + 3CO_2(g) \longrightarrow Fe_2O_3(s) + 3CO(g) \quad \Delta H^o = +26.6\ KJ$

A .26.6 KJ of energy are released for every mole of Fe reacted.

B .26.6 KJ of energy are absorbed for every mole of Fe reacted.

C .53.2 KJ of energy are released for every mole of Fe reacted.

D .13.3 KJ of energy are absorbed for every mole of Fe reacted.

Explanation: The equation does not obey the law of conservation of matter because sulfur is present only on the reactant side. But regarding the enthalpy change (ΔH) the following answer can be suggested .If ΔH > 0 (positive) it is an endothermic process, that is ,energy is absorbed. We can automatically reject Choices "A" and "C" The enthalpy change (ΔH)is given in kilojoules per mole (KJ/mol.)

∴ ***Answer B***

44. A gas is confined to a cylinder under constant atmospheric pressure. When the gas undergoes a particular chemical reaction it releases 135 KJ of heat to its surroundings and does 63 KJ of P -V work on its surroundings. What are the values of ΔH and ΔE for the process?

A .ΔH = 135 KJ , ΔE =63 KJ

B .ΔH = -135 KJ , ΔE = -63 KJ

C .ΔH = 135 KJ , ΔE = 198 KJ

D . ΔH = -135 KJ , ΔE = -198 KJ

Explanation: The question is based on the first law of thermodynamics which deals with energy transfer between a system and its surroundings. The energy transfer is either in the form of heat (Q) or work (W) which can be represented as: $\Delta E = Q - W$

A summary of the first law of thermodynamics is given below .

Quantity	algebraic Sign	Meaning
Heat q	+	..... Heat is absorbed by the system
	−	..... Heat is released by the system
W	+	··· Work is done on the system
	−	··· Work is done by the system
ΔE	+	··· Energy of the system increases
	−	··· Energy of the system decreases

(ΔH) 135 KJ of heat is released (-135 KJ).

(W) 63 KJ of P-V work is done (-63 KJ).

The internal energy of the system decreases by -198KJ (ΔE) .

ΔH = -135 KJ heat is released .

∴ **Δ E = -198 KJ**

∴ ***Answer D***

45 .The oxidation numbers of nitrogen in NH_3 ,HNO_3 and NO_2 are ——— respectively

A .-3 ,-5 , + 4 C . -3 ,+5 , -4

B .+3 , +5 , +4 D. -3 , +5 , +4

Explanation: Apply the following rules to determine the oxidation number of hydrogen in each compound.

i) For a compound , the sum of the oxidation number of the constituent elements is zero.

ii) The oxidation number of oxygen in most compounds is -2 ,-1 in peroxide and - $\frac{1}{2}$ in super oxides.

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iii) The oxidation number of hydrogen in most compounds is +1 and -1 in metal hydrides .Now calculate the oxidation number of nitrogen in each compound and.

let x represent the oxidation number of nitrogen

a) In NH_3 : $x + 3(1) = x + 3 \Longrightarrow x = -3$

b) In HNO_3 : $1 + x + 3(-2) = 0 \Longrightarrow x = +5$

c) In NO_2 : $x + 2(-2) = 0 \Longrightarrow x = +4$

∴ ***Answer D***

46. If the solute –solvent interactions are greater than the solute- solute and solvent –solvent interactions, What will be the total vapor pressure of the solution ?

A .Greater than that calculated from Raoult's law.

B .Less than that calculated from Raoult's law.

C .The same as calculated from Raoult's law.

D .Raoult's law cannot be applied for such interactions.

Explanation: Solutions that obey **Raoult's law** are said to be ideal solutions. Many solutions do not behave ideally. They deviate from Raoult's law and are said to be non- ideal. Such solutions can be explained by the following two cases.

Case i : If the **solute –solvent** attractions are weaker than those between solute-solute and those between solvent –solvent , there is a greater tendency for molecules to leave the solution-compared to the case of an ideal solution. Consequently the vapor pressure of the solution is greater than the vapor pressure predicated by **Raoult's law** (positive deviation from Rauolt's law).

Case ii : If solute molecules attract solvent molecules more strongly than they do their own kind then the escaping tendency decreases and the vapor pressure of the solution is less than the sum of the vapor pressures as predicted by Rauolt's law .(Negative deviation from Rauolt's law).Case 2 provides an answer to 46.

∴ ***Answer B***

47. Which of the following procedures will produce a buffered solution?

I . Equal volumes of 0.5M NaOH and 1M HCl solutions are mixed.

II. Equal volumes of 0.5M NaOH and 1M CH_3 COOH solutions are mixed.

III .Equal volumes of 1M $NaCH_3CO_2$ and 1M CH_3COOHsolutions are mixed.

IV .Equal volumes of 1.0M NaOH and 1M HCl solutions are mixed.

A .I B .III C.I and II D. II and III

Explanation: **A buffer solution** can be prepared by mixing;

i) a weak acid and its conjugate base (CH_3COOH,CH_3COONa).

ii) Or a weak base and its conjugate acid (NH_3 ,NH_4Cl).

- A strong acid (HCl) and the strong base (NaOH) do not produce a buffer.
- A strong base like NaOH cannot be used .
- The weak acid CH_3COOH and its conjugate base CH_3COO^- from $NaCH_3COO$ make a buffer .
- Strong acid HCl and the strong base NaOH do not produce.

∴ ***Answer B***

48. Which of the following statements is true?

A .A universal indicator is a mixture of indicators that will give a different colour for a different PH.

B .Phenolphthalein is a universal indicator.

C .A universal indicator can only be used in either strongly acidic or basic solution.

D .The colour of a universal indicator is red in a weak acid.

Explanation:

- **A universal indicator** is a mixture of acid –base indicators that change colour over a range of PH its color change is in the order. **(red ,yellow ,orange , green , blue)** red in solution with the lowest PH and blue in solutions with the highest PH.
- **Phenolphthalein** is a dye used as an acid – base indicator. It is colorless in solutions with PH less than 8 and pink (red) in solutions with greater PH .Phenolphthalein is not a universal indicator.
- **A universal indicator** can be used over a range of PH for weak, moderately strong or strong acids and bases.
- The color of universal indicator is red in strong and yellow in a weak acid.

∴ ***Answer A***

49. Which species CANNOT act as a Lewis acid?

A NH_3 C .Fe^{2+}

B .BF_3 D. $AlCl_3$

Explanation: **A Lewis acid** is a species that has a vacant orbitals to accommodate a pair of electrons or more.

A Lewis acid is the electron pair acceptor. : NH_3 has a lone pair of electrons. It does not possess a vacant orbital. Therefore it cannot act as a Lewis acid. It is an electron pair donor and a Lewis base .

But B in BF_3 ,Al in $AlCl_3$ and Fe^{2+} all have vacant orbital .Therefore , all the three are Lewis acids

∴ ***Answer A***

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50 .Three acids , HA , HB ,HC have the following K_a values. K_a (HA) = 1×10^{-5} K_a (HB) =2×10^{-5} K_a (HC) = 1×10^{-6}.What is the correct order of increasing acid strength (weakest first)?

A .HA ,HB ,HC
B ,HC , HB ,HA
C .HC ,HA ,HB
D .HB ,HA ,HC.

Explanation:

❖ HA	1×10^{-5}	Intermediate
❖ HB	2×10^{-5}	Strongest
❖ HC	1×10^{-6}	Weakest

The magnitude of the dissociation constant of an acid ,k_a is a measure of the strength of the acid. The larger the value of k_a, the Stronger is the acid .Therefore , the correct order of acid strength is

HC < HA < HB

∴ ***Answer C***

51.Which of the following is a natural polymer?

A .polythene
B .polysaccharides
C. Nylon
D. Terylene

Explanation:

i) **A natural polymer** as the name indicates is a large or macromolecule that occurs naturally. This includes **proteins, nucleic acids , starch, polysaccharides cellulose , rubber.**

ii) **A synthetic polymer** is a polymer made artificially by man. This includes ,**Nylon, terylene, Polyethene, rayon, rubber , pvc ,polystyrene**

∴ ***Answer B***

52.An aqueous solution is 70.0 % nitric acid (HNO_3) by mass. What is the concentration of HNO_3 expressed in molality?

A .0.559
B .8.62 m
C. 11.1 m
D. 37.0 m

Explanation: Assume that we have 100 g of solution of which 70 g is the mass of the solute and 30 g mass of the solvent.

i) convert 70 g HNO_3 to moles.

$$n=\frac{m}{M}=\frac{70g}{63g/mole}=1.11 \text{ moles of } HNO_3$$

ii) Convert 30 g solvent (H_2O) to kg, $\frac{30g}{100g/kg}=0.03kg$

30g = 0.03 kg

iii) Calculate molality

$$\text{Molality}=\frac{\text{Number of moles of solute}}{\text{kg of solvent}}=\frac{1.11 \text{ moles}}{0.03 \text{ kg}}=37.02 \text{ m}$$

∴ **Its concentration in molality is 37m.**

∴ ***Answer D***

53. When 0.68 is divided by 14.364 , the actual answer is 0.0473405.What will be the correct answer?

A .0.05
B. 0.047
C. 0.0473
D. 0.04734

Explanation : For multiplication and division, a result can only be as accurate as the factor with the least number of significant figures.

$\frac{0.68}{14.364}=0.04734057$

According to the rule, the answer should contain only two significant figures, that is , the same as 0.68 which has only two significant figures.

∴ ***Answer B***

54. What is the ionization energy of an iron atom if it requires a radiation of 276nm to completely remove its outer most electron in the gaseous state?

(plank's constant , h = 6.626×10^{-34} Js ,speed of light , c =$3\times10^{8}ms^{-1}$).

A .7.21×10^{-19} J
B .7.21×10^{-19}KJ
C. 7.21×10^{19}J
D .7.21×10^{19}KJ

Explanation: Use the equation **E = hV** to solve the problem.

From the relation $\lambda V = C \Rightarrow V=\frac{c}{\lambda}$

Substitute $\frac{c}{\lambda}$ for the frequency in the first equation : $E=\frac{hc}{\lambda}$

$$E=\frac{6.626\times10^{-34}\ JS\times3\times10^{8}mS^{-1}}{276\times10^{-9}\ m}$$

$$E=\frac{19.878\times10^{-26}J}{2.76\times10^{-7}}$$

$$E=\frac{19.9}{2.76}\times10^{-19}\ J$$

∴ **E = 7.21×10^{-19} J**

∴ ***Answer A***

55.Which of the electron configurations describes the ground state electron configuration of Cl^-?

A .$1s^2 2s^2 2p^6 3s^2 3p^6$
B .$1s^2 2s^2 2p^6 3s^1$
C. $1s^2 2s^2 2p^6 3p^1$
D. $1s^2 2s^2 2p^6 3s^2 3p_x^2 3p_y^1$

Explanation: $17Cl^o$ has 17 protons and 17 electrons .But $17Cl^-$ has one more electron than the number of protons because of the minus one charge, that is ,$18e^-$.

∴ ***Answer A***

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56. Which one of the following reactions is **NOT** a redox reaction?

A. $Ag^+(aq) + Cl^-(aq) \longrightarrow AgCl(s)$

B. $2Na(s) + Cl_2(g) \longrightarrow 2NaCl(s)$

C. $Mg(s) + 2HCl(aq) \longrightarrow MgCl_2(aq) + H_2(g)$

D. $Cu^{2+}(aq) + Zn(s) \longrightarrow Cu(s) + Zn^{2+}(aq)$

Explanation : **A redox reaction** is a reaction in which there is a decrease or increase in the oxidation number of the reactants as they are converted to products. But for choice A, there is no change in the oxidation number of Ag^+ and Cl^-

∴ Answer A

57. The half –reaction for formation of magnesium metal upon electrolysis of molten $MgCl_2$ is: $Mg^{2+} + 2e^- \longrightarrow Mg$

What is the mass of magnesium formed upon passage of a current of 60.0 A for a period 0f 2.00×10^3s?

A. 5.0g B. 10.0g C. 15.1 g D. 30.2 g

Explanation: To get the mass of Mg that is produced, we apply **faraday's first law of electrolysis** which is stated as follows. The mass of a substance oxidized or reduced at an electrode is directly proportional to the amount of electricity that passes through the electrolyte. Next calculate the amount of charge from the given time and current using the relation: $Q = It$

$Q = 60\ A \times 2 \times 10^3\ sec$

$Q = 120{,}000\ A.sec$

$\therefore Q = 120{,}000\ C$

Now calculate moles of electrons using: $n = \frac{Q}{F}$

$n = \frac{120{,}000\ C}{96{,}500\ C/mol} = 1.24$ moles of electrons

From the equation $Mg^{2+} + 2e^- \longrightarrow Mg$

2 moles of electrons reduce 1 mole of Mg^{2+} to 1 mole of Mg metal

1.24 moles of electrons reduce 0.62 mole of Mg^{2+} to 0.62 mole of Mg

Using the relation: $n = \frac{m}{M}$ we get $m = n \times M$

$\therefore m = 0.62\ mol \times 24.3\ g/mol = 150.066\ g \approx 15.1\ g$

∴ Answer C

58. The two standard electrode potentials involved in the nickel-cadmium rechargeable cell are given below. Calculate the ΔG^o in KJ of the cell.

$NiO_2(s) + 2H_2S(\ell) + 2e^- \longrightarrow Ni(OH)_2(S) + 2OH^-$ $E^o = +0.49V$

$Cd(OH)_2(s) + 2e^- \longrightarrow Cd(s) + 2OH^-(aq)$ $E^o = -0.76\ V$

A. -184 B. -153 C. -241 D. -206

Explanation :

i) Cadmium with a negative standard potential will serve as the anode and oxidation takes place at this electrode during the spontaneous redox reaction of the cell.

ii) The oxidation potential is 0.76 V the opposite of its reduction potential – 0.76 V.

iii) Calculate the cell potential E^o_{cell}

E^o_{cell} = Oxidation potential + Reduction potential

$E^o_{cell} = 0.76\ V + 0.49\ V = 1.25\ V$

iv) Calculate ΔG^o of the cell using the reaction:

$\Delta G^o = -nFE^o_{cell}$

$\Delta G^o = -2mol \times 96500C/mol \times 1.25\ V$

$\Delta G^o = -241500\ J = -241.25\ KJ$

∴ Answer C

59. Which of the following metals is extracted by thermal reduction process?

A. Cu B. Fe C. Al D. Mg

Explanation: Both Al and Mg being very reactive can be produced only by electrolysis. But only thermal reduction for Fe.

∴ Answer B

60. Which of the following is a synthetic rubber produced from caprolactam (CPL)?

A. Nylon 6, 10 C. Teflon

B. Terlyene D. Nylon 6

Explanation: Caprolactum polymerizes to the polymer nylon 6.

∴ Answer D

61. In zero –order reaction for every 10^0 rise of temperature, the rate is doubled. If the temperature is increased from 10°C to 100°C, the rate of the reaction will become.

A. 64 times C. 256 times

B. 128 times D. 512 times

Explanation: For every 10°C rise in temperature the rate doubles. From 10°C upto 100°C, there are 9 internals of 10°C. Therefore the rate of the reaction increases by a factor of 2^9 (512 times).

∴ Answer D

62. The half life for the first order decomposition of nitromethane, CH_3NO_2, at 500K is 650 seconds. If the initial concentration of CH_3NO_2 is 0.500M, what will its concentration be (M) after 1300 seconds have elapsed?

A. 0.125 B. 0.140 C. 0.250 D. 0.425

Explanation: The half life of a reaction is the time it takes for the concentration of the reactant to decrease to half of its initial

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concentration. The definition works not only for a first order reaction. There is no need of mentioning the order.

✓After 650 seconds , the concentration of the nitromethane decreases to half of 0.5M which is equal to 0.25M

✓After another 650 seconds, its concentration decreases from 0.25M to 0.125 M

∴ After 1300 seconds the concentration of nitromethane becomes 0.125M.

∴ Answer A

63. The value of K_C for the following equilibrium reaction is 4.0 at a temperature of 373 K.

$CH_3COOH + C_2H_5OH \rightleftharpoons CH_3COOC_2H_5 + H_2O$

What mass of ethyl ester ($CH_3COOC_2H_5$) would be present in the equilibrium mixture if 15 g of acetic acid and 11.5 g of ethanol were mixed and equilibrium was established at this temperature?

A .5.2 B .10.1 C .12.6 D .14.1

Explanation : If the volume of the reaction Vessel is not Mentioned we can assume it to be 1.0 L, since molar concentration is given in moles per liter.

	CH_3COOH +	C_2H_5OH	$\rightleftharpoons$ $CH_3COOC_2H_5$ +	H_2O
Initial	0.25	0.25	0	
Change	-x	-x	+x	+x
Equil	0.25 – x	0.25 – x	x	x

$$K_C = \frac{[CH_3COOC_2H_5]}{[CH_3COOH][C_2H_5OH]}$$

$$K_C = \frac{x}{\left(\frac{1}{4} - x\right)\left(\frac{1}{4} - x\right)}$$

$$4 = \frac{x}{x^2 - \frac{1}{2}x + \frac{1}{16}}$$

$$x = 4x^2 - 2x + \frac{1}{4}$$

$$4x^2 - 3x + \frac{1}{4} = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{3 \pm \sqrt{9 - 4 \times 4 \times \frac{1}{4}}}{8}$$

x = 0.0955 mol /L of ester

0.25 mol/L – 0.0955 mol /L = 0.1545 mole /L

Mass of ester = 0.0955 mol ×88 g/mole =3.4 g

Therefore, 8.4 g of ester which is not in the choices.

∴ No Answer

64. Which of the following mathematical relationship between K, K_1 and K_2 is correct ?

$CO_2(g) + H_2(g) \rightleftharpoons CO(g) + H_2O(g)$.....K

$Fe(s) + CO_2(g) \rightleftharpoons FeO(s) + CO(g)$.....$K_1$

$Fe(s) + H_2O(g) \rightleftharpoons FeO(s) + H_2(g)$.....$K_2$

A .$K = K_1 + K_2$ C .$K = K_1/K_2$

B .$K = K_1 \times K_2$ D .$K = K_2/K_1$

Explanation: The target equation , $CO_2(g) + H_2(g) \rightleftharpoons CO(g) + H_2O(g)$ can be obtained from the elementary steps first by reversing equation 2 and then add.

$Fe(s) + CO_2(g) \rightleftharpoons FeO(s) + CO(g)$..... K_1

$FeO(s) + H_2(g) \rightleftharpoons Fe(s) + H_2O(g)$... Reciprocal of K_2

Add these two.

$CO_2(g) + H_2(g) \rightleftharpoons CO(g) + H_2O(g)$.....K

Rule: For a reaction that proceeds in a sequence of elementary steps, the equilibrium constant K for the overall reaction is equal to the product of the equilibrium constant for the elementary steps.

$$\therefore K = K_1 \times \frac{1}{K_2} = \frac{K_1}{K_2}$$

∴ Answer C

65. Which of the following gives the correct order of decreasing acidity of carboxylic acids?

A .Cl_3CCOOH, $Cl_2CHCOOH$, FCH_2COOH, CH_3COOH

B .FCH_2COOH, CH_3COOH, $Cl_2CHCOOH$, Cl_3CCOOH.

C .CH_3COOH, FCH_2COOH, $Cl_2CHCOOH$, Cl_3CCOOH

D.$Cl_2CHCOOH$, CH_3COOH, FCH_2COOH, Cl_3CCOOH

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Explanation: The Substitution of electronegative atom like Cl and F for hydrogen withdraws electron from the molecule making it more acidic or less basic.

∴ ***Answer A***

66. How many 3d electrons are present in the ground state of chromium atom?

A .4 B. 5 C .6 D .10

Explanation : The atomic number of Cr is 24 and a neutral Cr atom has 24 electrons distributed as $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^4$. But half–filled d–orbital is more stable than unfilled d –orbital so that Cr attains the following stable electron configuration in its ground state.

$1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^5$

∴ Five 3d electrons are present in the ground state of chromium.

∴ ***Answer B***

67. When the following substances are arranged in order of increasing melting point (lowest melting point first), the correct order is:

A .$CH_3CH_2CH_3, CH_3COCH_3, CH_3CH_2CH_2OH$

B .$CH_3CH_2CH_3, CH_3CH_2CH_2OH, CH_3COCH_3$

C.$CH_3COCH_3, CH_3CH_2CH_2OH, CH_3CH_2CH_3$

D.$CH_3CH_2CH_2OH, CH_3CH_2CH_3, CH_3COCH_3$

Explanation: The **melting point of a compound** is directly related to the strength of the intermolecular forces. The stronger the IMF of attraction the higher is the melting point. Because of the presence of **H – bonding** which is stronger than the other types of intermolecular forces of attraction in alcohols, alcohols have the highest melting and the corresponding alkanes has the lowest melting point.

∴ $CH_3 - CH_2 - CH_3 < H_3C - CO - CH_3 < CH_3 - CH_2 - CH_2OH$

∴ ***Answer A***

68. Which of the following molecules or ions will exhibit delocalized bonding? NO_2^-, NH_4^+, N_3^-

A .NO_2^- and N_3^- C .NO_2^-

B .NH_4^+ and N_3^- D .NO_2^- and NH_4^+

Explanation: **A molecule or ion** that can be represented by two or more Lewis structure exhibits delocalized bonding.

i .NO_2^- has **delocalized bond.**

ii.NH_4^+ has **localized bond** and represented by a single Lewis Structure.

iii. N_3^- is represented by a single Lewis Structure and localized bond

Therefore, only NO_2^- has delocalized bond.

∴ ***Answer C***

69. What is the correct molecular electronic configuration for the molecular ion , B_2^+?

A .$\sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 \sigma 2p^2$

B . $\sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 \pi 2p_{y1}^2$

C .$\sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 \pi 2p_{x1}$

D .$\sigma 1S^2 \sigma^* 1S^2 \sigma 2S^2 \sigma^* 2S^2 \pi_{2Px} \pi_{2Py}^1$

Explanation: The atomic number of boron (B) is 5 .For B_2^+ there are10 electrons minus one because of the one positive charge and only 9 electrons are available. For a molecule or ion that has 14 electrons or less the following order of orbital energy is used.

$\sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 \begin{cases} \pi 2P_Y^1 \\ \pi 2P_Z \end{cases} \sigma 2P_X$

Choices A ,B and D have a total of 10 electrons in their orbital's.

∴ ***Answer C***

70. Based on molecular orbital theory ,the bond orders of H_2, H_2^+ and H_2^- are _____, respectively.

A .1 , 0 and 0 B .1 ,$\frac{1}{2}$ and 0 C. 1,0 and $\frac{1}{2}$ D. 1 ,$\frac{1}{2}$, and $\frac{1}{2}$

Explanation: The number of electrons for the species H_2 ,H_2+ and H_2– are respectively 2,1, and 3.

Bond order is equal to the number of electrons in bonding molecular orbitals minus the number of electrons in anti bonding molecular orbitals divided by two.

Electron configuration

- $H_2 : \sigma 1S^2\ \sigma^* 1S^0$ ⇒ **Bond order** $= \frac{2-0}{2} = \frac{2}{2} = 1$
- $H_2^+ : \sigma 1S^1\ \sigma^* 1S^0$ ⇒ **Bond order** $= \frac{1-0}{2} = \frac{1}{2} = 0.5$
- $H_2^- : \sigma 1S^2\ \sigma^* 1S^1$ ⇒ **Bond order** $= \frac{2-1}{2} = \frac{1}{2} = 0.5$

Therefore, the bond orders are respectively 1 ,$\frac{1}{2}$,$\frac{1}{2}$.

∴ ***Answer D***

71. The kinetic data below are for the reaction: A +B ⟶ C

From these data what are the orders of the reaction with respect to A and B ?

[A]	[B]	Initial Rate ($mol\ dm^{-3} sec^{-1}$)
0.1	0.1	1×10^{-5}
0.1	0.2	4×10^{-5}
0.2	0.1	1×10^{-5}

A . Order of A = 1, order of B = 0

B .Order of A = 0, order of B =2

C .Order of A = 0, order of B = 4

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D .Order of A = 1, order of B =2

Explanation: The rate law for the reaction is rate = $K[A]^x[B]^y$

x and y are orders of A and B respectively.

(i) use trials 1 and 3 to determine the value of x from the variation of rate with change in concentration of A keeping the concentration of B constant.

For experiment 3 : Rate = $K[A]^x$

$$1\times10^{-5} = K(0.2)^x$$

From experiment 1 : rate = $K[A]^x$

$$1\times10^{-5} = K(0.1)^x$$

Divide the results of experiment 3 by the result of experiment1.

$$\frac{k(0.2)^x}{k(0.1)^x} = \frac{1.0\times10^{-5}}{1\times10^{-5}}$$

$$2^x = 1$$

$$\Longrightarrow 2^x = 2^0$$

$$\therefore x = 0$$

ii) Use experiment 1 and 2 to determine y

For experiment 2: rate = $k[B]^y$

$$4\times10^{-5} = K(0.2)^y$$

For experiment 1 : rate = $k[B]^y$

$$1\times10^{-5} = K(0.1)^y$$

Divide the results of experiment 2 by the result of experiment1.

$$\frac{4\times10^{-5}}{1\times10^{-5}} = \frac{(0.2)^y}{(0.1)^y}$$

$$4 = 2^y$$

$$\Longrightarrow 2^2 = 2^y$$

$$\therefore y = 2$$

Therefore, order of A =0 and order of B = 2

∴ Answer B

72. Which of the following is **NOT** a characteristic of the electrolytic cell containing aqueous solution of NaCl used in the manufacture of sodium hydroxide?

A .The sodium hydroxide solution is produced in the electrolytic cell.

B .The electrolyte must be a dilute solution of NaCl.

C .Hydrogen is produced at the cathode.

D .The production of chlorine gas occurs at the anode.

Explanation: It is concentrated solution of sodium chloride that is used in the manufacture of sodium hydroxide. The electrolysis of dilute solution of sodium chloride gives H_2 at the cathode and O_2 at the anode but not sodium hydroxide

∴ Answer B

73. Which of the following is the most important source for the extraction of iron?

A .Hematite C. Chalcopyrite

B .Bauxite D. Sphalerite.

Explanation:

- **Hematite :** chief ore of iron.
- **Bauxite :** the principle ore from which aluminum metal is extracted.
- **Chalcopyrite :** the major ore of copper metal.
- **Sphalerite :**is a mineralogical name for zinc sulfide from which zinc is isolated .

∴ Answer A

74. Which of the following metals has the highest electrical and termal conductivities of any metal?

A .Ag B. Cu C. Ni D. Co

Explanation: **Silver (Ag)** – is a soft malleable metal with the highest known **electrical and thermal conductivities.**

∴ Answer A

75. Which of the following is a linear polymer?

A .High density polyethene (HDPE) C. Bakelite

B .Low density polyethene (LDPE) D. Vulcanized rubber.

Explanation:

- **HDPE :**Hard , flexible ,tough and linear molecule.
- **LDPE :** soft and light branched polymer.
- For **Bakelite and vulcanized rubber** polymer chains are cross – linked.

∴ Answer A

76. Which of these compounds is the ester formed from the reaction of acetic acid and 1 –propanol?

A. $CH_3\underset{\displaystyle OH}{\underset{|}{C}}OH$

B. $\underset{\displaystyle OH}{\underset{|}{CH_3CH_2C}}OH$ $\quad OCH_2CH_3$

C. $CH_3CH_2CH_2OCH_2\overset{\displaystyle O}{\overset{\|}{C}}OH$

D. $CH_3\overset{\displaystyle O}{\overset{\|}{C}}OCH_2CH_2CH_3$

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Explanation:

$CH_3-CO-OH + HO-CH_2-CH_2-CH_3 \longrightarrow CH_3-CO-O-CH_2-CH_2-CH_3 + H_2O$

Acetic acid **1 –propanol** **propyl ethanoate**

∴ ***Answer D***

77. A lab instructor is preparing 5.0 liters of a 0.10M $Pb(NO_3)_2$ (molecular mass 331)solution .What is the mass required?

A .165.5 g of $Pb(NO_3)_2$ and add 5.0 kg of H_2O.

B .165.5 g of $Pb(NO_3)_2$ and add H_2O until the solution has a volume of 5.0 liters.

C .33.1 g of $Pb(NO_3)_2$ and add H_2O until the solution has a volume of 5.0 liters.

D .33.1 g of $Pb(NO_3)_2$ and add 5.0 liters of H_2O.

Explanation: Given : Concentration (C) = 0.1M or 0.1 mol/L

Volume of solution = 5.0 L

From the relation: $C = \frac{n}{V}$

$\Rightarrow n = C \times V$ where n is the number of moles of solute $Pb(NO_3)_2$

i) $n = C \times V$

$n = 0.1\ mol/L \times 5.0L = 0.5\ mole\ of\ Pb(NO_3)_2$

ii) From the relation $n = \frac{m}{M}$

$\Rightarrow m = n \times M$

$m = 0.5\ mole \times 331\ g/mol = 165.5\ g$

Dissolve 165.5 g of $Pb(NO_3)_2$ in sufficient water by stirring the mixture constantly. Add more water until the volume of the solution becomes 5.0L

∴ ***Answer B***

78. Which of the following statements is true about the percent ionization of a weak acid?

A .It decreases with increasing concentration.

B .It increases with decreasing concentration .

C .It increases with increasing concentration

D .It decreases with decreasing concentration.

Explanation: percent ionization of a weak acid $= \frac{\text{Amount dissociated in } \frac{mol}{L}}{\text{the initial conc } mol/L} \times 100$

The percent ionization increases as more molecules of the acid ionize and the amount in the numerator increases. As the solution is diluted more molecules of the acid ionize. The percentage ionization increases with decreasing concentration as the solution is diluted.

∴ ***Answer B***

79. Which of the following is true about an open system ?

A .A system that exchanges both energy and matter with its surroundings.

B .A system that cannot exchange both matter and energy with its surroundings.

C .A system that only exchange matter with its surroundings.

D .A system that only exchange energy with its surroundings.

Explanation:

- A system that exchanges both energy and matter with its surroundings is an **open system.**
- A system that cannot exchange both energy and matter with its surroundings is an **isolated system.**
- A system that only exchanges matter with its surroundings. There is no such system.
- A system that only exchanges energy with its surroundings is a **closed system.**

∴ ***Answer A***

80. What is the value of ΔH , for the reaction S (s) ⟶ S(g)?

$S(s) + O_2(g) \longrightarrow SO_2(g)$ $\Delta H = -395\ KJ$

$S(g) + O_2(g) \longrightarrow SO_2(g)$ $\Delta H = -618\ KJ$

A .-1013 KJ B .-223KJ C .+223KJ D. +1013KJ

Explanation: The second equation must be reversed to get the target equation:

$S(s) + O_2(g) \longrightarrow SO_2(g)$ ..... $\Delta H = -395\ KJ$

$SO_2(g) \longrightarrow S(g) + O_2(g)$ ..... $\Delta H = +618\ KJ$

$S(s) \longrightarrow S(g)$ ..... $\Delta H = +223\ KJ$

Note: ΔH takes the opposite sign when the equation is reversed for the second equation.

∴ ***Answer C***

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1. Which of the following elements is found in proteins and nucleic acids but does **NOT** occur in carbohydrates?
   A. Carbon
   B .Nitrogen
   C. Oxygen
   D. Hydrogen.

***Explanation*: Nitrogen** is found in many biological compounds. It is present in proteins ,amino acids ,DNA ,RNA (all kinds)and adenosine triphosphate (ATP) as well as ADP.

∴ ***Answer B***

2. Which of the following is important to regulate the entry and exit of materials into and out of plant cells?
   A. Cell wall
   B .Cell membrane
   C. Nucleus
   D. Central vacuole.

***Explanation*:**

❖ **A cell wall** forms a semi-rigid covering around plant cells. The cell wall is composed of the complex **polysaccharide(cellulose)** and other similar compounds. The cell wall which is located outside the cell membrane is not part of the cytoplasm. Hence it is simply a non-living covering of the cell.

✓ **Cell wall** provides protection and support to plant cells.

❖ **Cell membrane** : It is also called plasma membrane. The cell membrane , a living part of a cell is a thin covering found in all plant and animals cells. The cell membrane is semi-permeable or selectively permeable .Hence it is very active in selecting substances that enter or leave the cell.

❖ **Nucleus:** The nucleus in eukaryotes is bound by a nuclear membrane. It has a dense and round body called nucleolus and thread like structure known as **chromatin (chromosomes)**.The chromatin material contain the DNA molecule responsible for the transmission of hereditary information's. The nucleus is a vital component of a cell as it controls or directs the overall activities of the cell.

❖ **Central vacuole:** This is a portion of the protoplast made distinct from the protoplast by accumulation of watery solution of various substances enclosed with in a vacuolar membrane. The vacuoles are larger in plant cells than in animals cells. They contain solution of salt and water with other materials.

∴ ***Answer B***

3. Which of the following terms is synonymous with a producer?
   A. Omnivore
   B .Herbivore
   C. Carnivore
   D. Autotroph.

***Explanation*:**

❖ **Omnivores** feed on both plant matter and animal flesh. Human beings are omnivores.

❖ **Herbivore** are obtain their food exclusively from plant tissue.
***Example* :Grazers** (which mostly feed on the grass and other ground level herbs or browers (which mostly feed on the leaves of small trees and shrubs).Cattle and sheep are grazers while camels and goats are **browers.**

❖ **Carnivores** are flesh eaters. The lion ,which is a flesh –eater is a carnivore.

❖ **Autotroph (producer)** are self feeders. They are organisms obtain energy either from the sun (photosynthetic)or from some chemicals (chemosynthetic).

✓ **Autotrophs** synthesize their own food that could also be used by hetrtrophs

∴ ***Answer D***

4. Which of the following steps in cellular respiration can take place in the absence of oxygen?
   A. Electron transport.
   B. Glycolysis
   C. Krebs cycle
   D. Acetyl-CoA formation.

***Explanation*:**

❖ **Electron transport** is the last stage of aerobic respiration which requires oxygen. The oxygen acts as the final acceptor of electrons at the end of the electro transport chain.

❖ **Glycolysis** is the first stage of aerobic respiration which can take place in the absence of oxygen. This takes place outside the mitochondrion and it is also known as the an aerobic stage of aerobic respiration. It produces 2 net ATP molecules.

∴ ***Answer B***

5. Which of the following molecules is known as the energy currency of the cell?
   A .Glucose B .Starch C. Fat D.ATP

***Explanation*:** The molecules of **glucose, starch and fat** release energy after passing through a continuous long metabolic path ways. The energy generated from these organic molecules is stored in ATP molecules.

✓ ATP is known as the immediate source of energy for all activities in living cells. ATP molecules is the energy currency of the cell.

∴ ***Answer D***

6. Which of the following processes releases oxygen to the atmosphere?
   A .Respiration.
   B .Photosynthesis
   C. Transpiration
   D. Burning of fossil fuels.

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Explanation:

- **Respiration** is an energy releasing process where glucose is broken down to 6 molecules of carbon dioxide and 6 molecules of water.
- **Photosynthesis** is the process by which green plants and cynobacteria manufacture food from carbon dioxide and water using light energy and releasing oxygen as by product.
- **Transportation** is a process by which plants lose water through their micro-scoping opening known as **stomata.**
- **Burning of fossil fuels** is a process of using fossil fuels as a source of energy in the modern world.

∴ ***Answer B***

7. The concentration of which of the following gases is on the increase in the atmosphere in recent years than before?
   A .Carbon dioxide C. Nitrogen
   B .Oxygen D. Ozone.

   Explanation: The **concentration of carbon dioxide** is an increase in the atmosphere in recent years. Because in today's world humans are continuously burning lots of fossil fuels which release a huge amount of carbon dioxide in to the atmosphere. This results global warming because of its green-house effect.

∴ ***Answer A***

8 .What is the source of the oxygen that green plants release during photosynthesis?
   A .Sugar C. Water
   B .Carbon dioxide D. Chlorophyll

   Explanation: In the light dependent stage of photosynthesis ,part the energy obtained from the sun is used to split the water molecule in to hydrogen and oxygen. This process is known as **photolysis**. The hydrogen is used to manufacture carbohydrate in the light independent stage where as the oxygen is given off as by product.

∴ ***Answer C***

9. All the living components of the cell are collectively known as what?
   A .Cytoplasm C. Cell membrane
   B . Nucleus D. Protoplasm.

   Explanation:

- **Cytoplasm :** all the living part of a cell inside the cell membrane and excluding the nucleus.
- **Nucleus :** a large dense organelle bounded by a double membrane (nuclear envelope), present in eukaryotic but not prokaryotic cells and which contain the chromatin and in which DNA replication and transcription takes place.

- **Cell membrane** : a membrane bounding the surface of all living cells, formed of a fluid lipid bilayer in which proteins carrying out the functions of enzymes ,ion pumps, transport proteins ,receptor for hormones etc. are embedded .It regulates the entry and exit of most solutes and ions.
- ***Protoplasm:*** the living matter ,the total substance of a living cell, cytoplasm and nucleoplasm in the case of eukaryotic cell.

∴ ***Answer D***

10. With which type of agricultural crops are nitrogen fixing bacteria more associated?
    A .Legumes. B .Cereals C. Vegetables. D. Fruits

    Explanation: **Legumes** are plants that contain swellings known as root nodules on their root that harbour nitrogen fixing bacteria.

✓ **Nitrogen fixing bacteria** converts the atmospheric nitrogen (N_2) in to ammonium ions that later will be converted to nitrate.

∴ ***Answer A***

11 .Which of the following is found in both plant and animal cells?
    A .Cell wall C. Chromosomes
    B .Chromoplasts D. Leucoplasts

    Explanation: **Cell wall** is found only in plant cells. A structure that is found on the outside of the cell membrane.

- **Chromoplasts** is a coloured plastid found only in plant cells which contain some red or yellow pigments.
- **Chromosomes** one of the small ,rod shaped , deeply staining bodies that become visible under the light microscope in the eukaryotic plant and animal cell nucleus at mitosis and meiosis which carry the genes in linear order.
- **Leucoplast** colourless plastid found only in plant cells that develops in to chloroplast etc.

∴ ***Answer C***

12. What is the stored form of sugar that is found in human tissues?
    A .Cellulose C. Nucleic acid
    B .Coenzyme D. Glycogen

    Explanation: **Glycogen** :a branched chain polysaccharide made up of glucose units and which acts as a storage substance invertebrate (including humans) liver and muscle.

∴ ***Answer D***

13. What happens to a human red blood cell when it is placed in a hypertonic solution?
    A .It becomes turgid
    B .It might swell and burst

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C .It might lose water and shrink.
D .It will remain unchanged.

***Explanation*: When human red blood cells are placed in hypertonic solution** (a solution which has more solute and less water than the red blood cells) water moves out of the cell by **osmosis**. This movement of water from these cells cause shrink the cells.

∴ ***Answer C***

14.Choose the final electron acceptor in the electron transport chain during aerobic respiration of eukaryotic cells.

A . H_2O
B .O_2
C. CO_2
D. NADP

***Explanation*:**

- **H_2O** is a byproduct of aerobic respiration produced at the end of electron transport chain which is the lasts stage of this path way.
- **O_2 (Oxygen)** always acts as a final electron acceptor of the electron transport chain. As oxygen accepts the electron and protons, water molecules is formed.
- CO_2 is another byproduct of aerobic respiration .Carbon dioxide is released in the link –reaction and krebs cycle of the path way.
- **NADP (Nicotine Amide Dinucleotide Phosphate)** is a molecules that is reduced to NADP H_2 in the light dependent reaction of photosynthesis which then becomes the chief source of energy for the reduction of carbon in the light independent reaction.

∴ ***Answer B***

15. Choose the cells of the human body that are results of meiosis.

A .Bone cells
B .Brain cells
C. Egg cells
D. Muscle cells.

***Explanation*: Meiosis** is a reduction division that reduces the chromosome number of the daughter cells by half. It is a type of cell division that forms gametes (sex cells) such as **egg and sperm cells** in humans and other organisms.

∴ ***Answer C***

16. Where in the plant cell does the krebs cycle (citric acid cycle) take place?

A .Nucleus
B .Cytoplasm
C. Mitochondrion
D. Chloroplast.

***Explanation*: Krebs cycle** is a cyclic series of chemical reaction that occurs in the mitochondria of plant and animals cells. It is the second stage of cellular respiration (aerobic respiration) where carbon dioxide ,reduced NAD and FAD are produced.

∴ ***Answer C***

17. Which organelle of the cell has a function of modifying proteins for secretion?

A .Golgi body
B .Ribosome
C. Food vacuole
D. Lysosome.

***Explanation*: Golgi body** consists of stacks of plate like membrane filled with cellular products. Its function is to serve in collecting ,concentrating ,modifying and packing cell products (proteins)in the form ready for transport.

- **Ribosome** are tiny spherical structures which contain RNA and protein and their function is the synthesis of all sorts of proteins.
- **Food vacuole** are small vacuole enclosing fluid and ingested food particles ,in many unicellular protists (amoeba and paramecia).
- **Lysosome** are cytoplasmic organelles filled with digestive enzymes and their function is to break down worn out organelles , cells or compounds.

∴ ***Answer A***

18.In which aspect of biodiversity are such measures like richness, abundance and taxonomic diversity in populations commonly applied?

A .Ecosystem diversity.
B .Cultural diversity
C. Species diversity
D. Genetic diversity

***Explanation*: Ecosystem diversity** is refers to a number of different types of ecosystem with a number of species richness or abundance of different species of plants and animals (taxonomic diversity).

- **Cultural diversity** is a number of different culture practiced in different part of the word in the human society.
- **Species diversity** is a number of different species of an ecosystem.
- **Genetic diversity** is the genetic variability of a single species.

∴ ***Answer A***

19. Among the following, identify the molecule that is an enzyme.

A . Cellulase
B .Starch
C. Cellulose
D. Glycogen

***Explanation*: Cellulase** is an enzyme found in herbivores which breaks down cellulose in to glucose like molecules.

- **Starch** is a polysaccharide carbohydrate found in plants and used as a store of energy.
- **Cellulose** is a structural carbohydrates used to build the cell wall of plant cells.
- **Glycogen** is a polysaccharide carbohydrate found in animals and used as a store of energy.

∴ ***Answer A***

20. What is the process by which cells like amoeba and white blood cells engulf and internalize particles such as bacteria?

A .Osmosis
B . Pinocytosis
C. Active transport
D. Phagocytosis.

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***Explanation*: Osmosis** is the passage of water molecules from a region of higher concentration (of water) to a region of lower concentration through a partially permeable membrane.

- **Pinocytosis** is a form of cell transport where the material taken up is in liquid form.
- **Active transport** is an energy consuming transport of molecules or ions across a membrane against a concentration gradient. Energy is required because the substance must be moved against its natural tendency to diffuse in the opposite direction.
- **Phagocytosis**(cell eating) material taken up is in solid form. It involves the creation of pseudopodia(extension of the plasma membrane) to enclose (engulf) large particles or even whole organisms outside the cell (as white blood cells engulfing bacteria).

∴ ***Answer D***

21. An enzyme known as catalase converts hydrogen peroxide into water and oxygen .In this reaction, which one is the substrate?
A .Catalase C. Water and oxygen
B .Hydrogen peroxide D. Water only.

***Explanation*: Catalase** is an enzyme present in both plant and animal tissues that catalyses the breakdown of hydrogen peroxide , a toxic compound produced during metabolism, in to oxygen and water.

$\Rightarrow 2H_2O_2 \xrightarrow{\text{Catalase (enzyme)}} 2H_2O + O_2$

Hydrogen peroxide substrate — Productse

∴ ***Answer B***

22.In a honey bee colony, Which one of the following is the duty of the queen?
A .Foraging C. Guarding the hive.
B .Rearing the young D. Reproduction.

***Explanation*: Foraging ,rearing the young and guarding the hive is** the duty of the worker bees.

- **Reproduction** is laying eggs is the duty of the queen ,the only fertile female.

∴ ***Answer D***

23.To which of the following molecules is most of the energy released during the Krebs cycle transferred?
A .ATP B .$FADH_2$ C. NADH D. ADP

***Explanation*:** In the **Krebs cycle** of aerobic respiration only 2ATP molecules are produced from a molecules of glucose. But 6 NADH and $2FADH_2$ are produced from the same glucose molecule. Each NADH generates 3 ATP molecules (i.e a total of 18 ATP) and

$FADH_2$ generates only 2ATP (i.e a total of only 4 ATP) .Therefore, most of the energy released in the Krebs cycle is transferred to NADH.

∴ ***Answer C***

24. Of the following which cell type is attacked by HIV ?
A .Red blood cell. C. Egg cell
B .Sperm cell. D.T- helper cell.

***Explanation*: HIV** targets cells that form part of the immune system. Its main target is a type of cell called a $CD_4 - T -$ lymphocytes .These cells are also called **T – helper cells**, because they help other cells in the immune system to mount an immune response to pathogens in the body.

∴ ***Answer D***

25. Among the following traits of the garden pea studied by GregorMendel, identify the one that did **NOT** express itself in the F1 generation?
A .Yellow pod colour. C. Round seed shape.
B .Violet flower colour D. Axial flower position.

***Explanation*: Violet flower colour, round seed shape and axial flower position** are dominant traits of garden pea that can be expressed in the hybrid F1 generation. A hybrid is produced from pure dominant and pure recessive trait. A yellow pod colour is a recessive trait that cannot be expressed in a hybrid F1 generation.

∴ ***Answer A***

26. In the Miller and Urey's simulation experiment to study organic molecules formation in primeval atmosphere of the earth, all the following gases were used except which?
A .Ammonia B .Hydrogen C. Oxygen D. Methane.

***Explanation*:** In 1953,Stanley Miller and his graduate adviser Harold Urey conducted a classic experiment, the result of which basically supported oparin's hypothesis .They assembled an atmosphere similar to the one thought exist on the primitive earth. It is widely believed that **oxygen** was totally absent in the primitive atmosphere, but it was composed primarily of nitrogen gas. It also contain significant amount of **carbon dioxide** and **water vapour**. Since the elements **sulphur,** nitrogen, hydrogen, and carbon were present , the compounds ammonia **(NH_3),methane (CH_4)and hydrogen sulphide (H_2S)** are assumed to have constituted the atmosphere.

∴ ***Answer C***

27. The bones in human hands, flippers of the whale and wings of the bird are known as what?
A .Linker fossils C. Analogous structures
B .Homologous structures. D. Vestigial structures.

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***Explanation*: Linker fossils** are fossils that serve as a linkage between some extinct organisms and present day ones.

- **Homologous structures** are structures having the same phylogenetic origin but not necessarily the same final structure of function.
  ***Example*:** Humans arm, cats forearm, whale's flipper, the wings of bats are homologous structure indication of a common ancestral origin of these organisms.
- **Analogous structures** are structures similar in function and appearance , but not necessarily in origin, show superficial similarity but this does not indicate descent from the same ancestor.
  Example: Birds wings and flies wings.
- **Vestigial structures** are rudimentary (reduced in size) structures in many organisms which have no known function. However, the same structures could be well developed and functional in other organisms. A good example is a human appendix

∴ ***Answer B***

28. Which of the following is a polysaccharide?
    A .Sucrose B .Glucose C. Cellulose D. Fructose

***Explanation*:**

- **Sucrose** is a disaccharide known as **a cane sugar**.
- **Glucose** is a monosaccharide which is the primarily source of energy for most organisms.
- **Cellulose** a structural polysaccharide carbohydrate used to build cell walls of plants.
- **Fructose** is a monosaccharide

∴ ***Answer*** C

29. What makes phospholipid molecules arrange themselves into a bilayer in a water medium?
    A .The double bonds of the fatty acids.
    B .The length of the fatty acid chains.
    C .The presence of hydrophilic and hydrophobic groups.
    D .Their high solubility in water.

***Explanation*:** The **phospholipids molecules** is unusual because the heads is **hydrophilic (water loving)** and the **tails are hydrophobic (water hating)**.The reasons why phospholipids molecules arrange themselves in to a bilayer in a water medium are:

- ✓ The fatty acid chains forming non-polar tails are water repellant (hydrophobic) and therefore project in wards away from the water to avoid the aqueous environment.
- ✓ Polar heads are water loving (hydrophilic) project out ward facing the aqueous environment.

∴ ***Answer C***

30. Which of the following plant groups is a good pioneer colonizer in primary successions that start with bare rocks?
    A .Mosses B .Lichens C. Liverworts D. Weeds.

***Explanation*: Primary successions** are named according to the conditions in which they start. A succession beginning on a very dry substartum, such as bare rock, sand dune or cooled volcanic lava is a xerosere. This succession is associated with the buildup of soil and soil moisture. In this case the pioneer organisms that start such succession are **lichens.**

∴ ***Answer B***

31. Where in the mitochondria does the krebs cycle take place?
    A .On the cristae
    B .In the matrix
    C .Between the outer and inner membranes.
    D .On the inner surface of the outer membrane.

***Explanation*:** The reaction of the **link reaction and krebs cycle** take place in the fluid matrix of the mitochondrion.
The reaction of the **electron transport chain and chemiosmosis** takes place on the inner mitochondria membrane – cristae.

- ✓ The protons (hydrogen ions) are pumped by hydrogen pumps from the matrix of the mitochondria to the inter membrane space (space between the inner and outer membrane).
- ✓ None of the stages of aerobic respiration takes place on the inner surface of the outer membrane

∴ ***Answer B***

32. Regarding seed characteristics ,suppose a heterozygous round yellow pea plant of Rr Yy genotype is selfed and produced 128 seeds, how many of the seeds are expected to be round yellow?
    A .72 B.64 C. 32 D .16

***Explanation*:** When a heterozygous round yellow pea plant is selfed.
Parental phenotype : Round yellow × Round yellow
Parental genotype : Rr Yy × Rr Yy
Gametes produced : RY ,Ry, rY , ry ×, RY,Ry,rY,ry
Fertilization :

	RY	Ry	rY	ry
RY	RRYY	RRYy	RrYY	RrYy
Ry	RRYy	RRyy	RrYy	Rryy
rY	RrYY	RrYy	rrYY	rrYy
ry	RrYy	Rryy	rrYy	rryy

✓ $\frac{9}{16}$ of the plants have round yellow seeds.

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Parental genotype : Rr Yy × Rr Yy
Gametes produced : RY ,Ry, rY , ry ×, RY,Ry,rY,ry
Fertilization :

	RY	Ry	rY	ry
RY	RRYY	RRYy	RrYY	RrYy
Ry	RRYy	RRyy	RrYy	Rryy
rY	RrYY	RrYy	rrYY	rrYy
ry	RrYy	Rryy	rrYy	rryy

✓ $\frac{9}{16}$ of the plants have round yellow seeds.

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✓ $\frac{3}{16}$ of the plants have round green seeds.

✓ $\frac{3}{16}$ of the plants have wrinkled yellow seeds

✓ $\frac{1}{16}$ of the plants have wrinkled green seeds

The off spring appeared in the ratio of **9:3:3:1.**

✓ Out of 128 seeds, the number of expected round yellow seeds is = $128 \times \frac{9}{16} = 72$

∴ ***Answer A***

33.If one regularly includes an orange in his daily diet, which deficiency disease would be avoided?

A .Night blindness
B .Pellagra
C. Beriberi
D. Scurvy.

Explanation:

- **Night blindness** is a deficiency diseases caused by the lack of vitamin A in the diet.
- **Pellagra** is a deficiency diseases caused due to the lack of niacin or **nicotinic acid** in the diet
- **Beriberi** is a deficiency diseases caused due to the lack of vitamin B **(thiamin)** in the diet.
- **Scurvy** is a deficiency diseases caused due to the lack of vitamin C **(Ascorbic acid)** in the diet. **Orange** is known for its high content of vitamin C.

∴ ***Answer D***

34. Which of the following methods of avoiding predation by animals is mimicry?

A .Resembling another organism.
B .Producing warning colouration.
C .Fighting away the enemy.
D .Running away from the enemy

Explanation: **Mimicking** is an imitation of the action or voice of a person or animal in a way that is meant to be amusing or entertaining.

∴ ***Answer A***

35.In pavlov's classical conditioning experiments with dogs, which one of the following is the conditioned stimulus?

A .The sight of food
B .The smell of food
C. The sound of the bell
D. The salivation of the dog.

Explanation:

- **The sight of food** is unconditional stimulus that naturally and automatically triggers a response.
- **The smell of food** is also unconditioned stimulus similar to the sight of food.

- **The sound of the bell** is conditioned stimulus. It is a neutral stimulus that does not initially produce the unconditioned response. But after association with unconditioned stimulus , it triggers the same response.
- **The salivation of the dog** is the unconditioned response that is unlearned response that occurs naturally to the unconditioned stimulus.

∴ ***Answer C***

36.Identify the enzyme that is used to cut DNA molecules into small pieces.

A .Urease.
B .Reverse transcriptase.
C.DNA polymerase.
D. Restriction endonuclease.

Explanation:

- **Urease** is an enzyme which catalyses hydrolyses of urea in to ammonia and carbon dioxide.
- **Reverse transcriptase** is an enzyme that converse the HIV RNA in to DNA after it infected the host cell.
- **DNA polymerase** is an enzyme that join nucleotides to form DNA.
- **Restriction endonuclease** is an enzyme that use to cleave or cut DNA molecule in to fragments.

∴ ***Answer D***

37.Suppose a lady has given birth to four boys in a row, what is the chance that her next child will be a girl?

A .100% B. 75% C.50% D. 25%

Explanation: The chance of giving birth to a boy or girl is always 50%.The female always produces the egg cell carrying X – chromosomes.Millions of sperm cells are discharged in to the female sex organ during mating.50% of these sperm cells carry Y-chromosome and the remaining 50% carry X- chromosome.The chance of the egg cell carrying X- chromosome to be fertilized by sperm cell carrying Y or X - chromosome is always the same 50%.XX results female and XY results male.

∴ ***Answer C***

38.Which of the following mechanisms can separate two genes located on the chromosome?

A .Back crossing
B .Crossing over
C. Segregation
D. Independent assortment.

Explanation:

- **Back crossing** is a technique used to identify the individual organism that show the dominant trait is whether or not homozygous or heterozygous for the trait.
- In **the prophase I of meiosis I** whilist chromosomes are associated in bivalent, chromatids from different chromosomes undergo **crossing over**. These chromatids are called **non-sister chromatids.** In this

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process the chromatids exchange equivalent sections of DNA .This separates two genes located on the same chromosome.

- **Segregation** is the separation of heritable factors (alleles) when the gametes (sex cells) are formed.
- **Independent assortment** is the inheritance of one trait independent of the inheritance of another one.

∴ ***Answer B***

39. According to Lamarck's theory of evolution , what is the origin of new structures that are necessary for evolution?
A .Mutation.
B .Gene recombination.
C .Natural selection.
D .Use – and – disuse of a body parts.
***Explanation*: Mutation** is any spontaneous change in the genetic material of an organism.

- **Gene recombination** is the regrouping of genes that regularly occurs during meiosis as a result of independent assortment of chromosomes in to new sets and crossing over. It is a source of genetic variation.
- **Natural selection** is the non-random survival of randomly varying hereditary units resulting in the evolution and maintenance of adaptive improvements.
- **Use – and disuse theory** of Lamark suggest that by continually using a structure or process , the structure or process will become enlarged or more developed.

∴ ***Answer D***

40. Suppose in a certain population, natural selection removes individuals with average height, leaving tall and short individuals behind ,which of the following modes of selection is in operation?
A .Stabilizing selection. C. Directional selection.
B .Normalizing selection. D. Disruptive selection.
***Explanation*: Disruptive selection** in which individuals at both extremes(tall and short) of a range have some advantage over those displaying the mean value (average height).

∴ ***Answer D***

41. If the DNA of the following animals is hybridized to human DNA, with which one of them would human DNA hybridized more?
A .Horse B .Chimpanzee. C .Fish D. Mice
***Explanation*:** To measure the similarity of one species DNA with another species DNA ,we use a technique called **DNA hybridization**. The technique measures the extent to which a strand of DNA from one species can bind with (or hybridise with) a strand of DNA from another species.

✓ A comparison of DNA sequences (using DNA hybridization technique)shows that is 99.9% certain that chimpanzees are humans closest relatives (98% of our DNA is the same as that of chimpanzees).

∴ ***Answer B***

42. In an ecosystem , the presence of which of the following groups is essential in order for the others to be present?
A . Autotrophs C. Decomposers
B .Herbivores. D. Omnivores.
***Explanation*: Autotrophs** are the primary producers of any ecosystem that make their own organic food (organic matter) that passes from them to **herbivores, omnivores and decomposers** through feeding and decomposition.All forms of organisms are dependent on autotrophs for their survival because they are the organism that only make the organic matter of the ecosystem.

∴ ***Answer A***

43. What are the three components of a DNA molecule?
A .Glucose ,Nitrogen and phosphorus.
B .Sugar ,phosphate group, nitrogenous base.
C .Protein , carbohydrate and fat with base.
D .Monosaccharide, disaccharide, polysaccharide.
***Explanation* :DNA** molecules is composed of nucleotides.
✓ **A nucleotides** in turn is made up of a pentose sugar (five carbon sugar),a Phosphate group and a nitrogenous base.

∴ ***Answer B***

44. What is a biome?
A .A group of interacting individuals of the same species.
B .The total life zone of the earth's surface.
C .A group of populations that interact with the physical environment.
D .A region of the earth characterized by distinctive life forms
***Explanation* :A biome** is a climatically controlled group of plants and animals of a characteristic composition and distribution over a wide area such as **tropical rain forest , tundra ,temperate grass land ,desert, savanna** etc.

∴ ***Answer D***

45. Which of the following defines ATP ?
A .Adenine + Adenosine + a phosphate.
B .Adenosine + ribose sugar+ double phosphate.
C .Adenine + ribose sugar + triple phosphate.
D .Nitrogen +sugar + mono phosphate.
***Explanation* :ATP(Adenosine tri-phosphate)** is adenine joining a five carbon sugar- ribose and three phosphate group.

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∴ *Answer C*

46.Among the following, which one can be an example of prokaryotic organisms?
A .Paramecium B .Amoeba C. Liverwort D. Bacteria.
Explanation :**Paramecium** is an eukaryotic protozoa that belongs to the kingdom protista.
- **Amoeba** are eukaryotic protozoans that belongs to the same kingdom protista.
- **Liverworts** are eukaryote plants that belongs to the phylum brayophyta of the plant kingdom.
- **Bacteria** are prokaryotes the belong to the kingdom monera.

∴ *Answer D*

47.Which alternative best explains speciation?
A .Members of a species diverge genetically and produce new species.
B .It is characterized by minor extinction events.
C .A process that only results from interspecific hybridization.
D .Two different species are able to interbreed in captivity.
Explanation :**Speciation** is a process by which one or more new species are formed from pre-existing ones.

∴ *Answer A*

48.Choose the alternative that shows the monomer and polymer of carbohydrates.
A . Nucleotide, amino acid.
B . Polypeptide, polynucleotide
C . Monosaccharide, polysaccharide.
D .Nucleic acid ,amino acid.
Explanation :**Monosaccharide** are the monomers of carbohydrate and polysaccharide are the polymers of carbohydrate.

∴ *Answer C*

49.Which of the following is expected to produce new combinations of genes?
A .Hybridization of a tall variety of "teff" with a short variety.
B ,Keeping a male goat and a female sheep in the same room.
C .Vegetative propagation of fruit trees.
D .Asexual reproduction of unicellular animals.
Explanation : **Hybridization of a tall variety of "Teff" with a short variety** produces new combination of genes. Because the alleles for tallness and the allele for shortness are brought together in the same individual that is the offspring produced by sexual reproduction.
- **Male goat and female sheep** cannot inter breed or reproduce because they belong to different species. Even if they inter breed they cannot produce fertile hybrid.

- **In vegetative propagation fruit** trees and asexual reproduction of unicellular organisms, the off spring produced are genetically identical to the parents. In asexual reproduction there is no new combination of genes, because there is no fusion of gametes of different parents. **Asexual reproduction** does not produce genetic variation.

∴ *Answer A*

50.What is implied by biological magnification?
A .Increase in the sizes of organisms with increasing trophic levels.
B .Increase in the number of organisms at higher trophic levels.
C .More concentration of chemicals in organisms of higher trophic levels.
D .Capture of small organisms by larger organisms.
Explanation : **Biological accumulation or biological magnification** is the increase in the concentration of a compound usually fat soluble pesticides such as DDT ,in the bodies of living organisms at successively higher levels in the food chain.

∴ *Answer C*

51.What is the cause of infectious diseases?
A .The parental genes. C. Carcinogenic substances.
B .Micro organisms D. Bad lifestyles.
Explanation :
- **Parental genes** may cause genetic diseases.
- **Microorganisms** the invasion of the body by micro –organisms can cause infectious diseases.
- **Carcinogenic substances** are substance that cause cancer.
- **Bad life style** such as working conditions and the content of the diet can cause the life style disease known as **human induced diseases.**

∴ *Answer B*

52.Which of the following compounds is quickly absorbed through the walls of the stomach?
A .Fat B .Protein C. Sugar D. Alcohol
Explanation :
- **Fat** is a complex large molecule. It cannot be absorbed through the wall of the body without being digested in to fatty acids and glycerol.
- **Proteins** are also complex large molecules. They have to be hydrolysed in to amino acids to be absorbed.
- **Sugars** are monosaccharide sugars can be easily absorbed since they are end products of carbohydratesdigestion .Double sugars or disaccharides cannot be absorbed.
- **Alcohol** is a simple organic molecule composed of only 2 carbon atoms,6 hydrogen and one oxygen atoms. It is absorb rapidly as it is taken in to the body producing its effect immediately.

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∴ *Answer D*

53.Coffee and tea are normally grouped into which type of substances?
A .Pulses. B .Perfumes C. Stimulants D. Grains
Explanation :**Coffee and tea** are normally grouped in to a type of drug known **as stimulants** because they contain a substance known as **caffeine** that stimulates the brain.

∴ *Answer C*

54. What does the feeding relationship in an ecosystem consists of
A .Many stages of community succession.
B . Always a network of interconnected food chains.
C . One organism feeds on another but itself eaten by another one.
D . Recycling of solar energy between the source and the trophic levels.
Explanation :The transfer of nutrients in the energy containing organic molecules produced by autotrophic organisms are the source of food (materials and energy) for hetrotrophic organisms. These organisms may in turn be eaten by other animals and in this way energy is transferred through a series of organisms, each feeding on the proceeding organism and providing row materials and energy for the next organism. These sequence of eating and being eaten is what is described in the widely known ecological concepts food chain and food web.

∴ *Answer B*

55. Which process of respiration helps to release most of the energy stored in glucose?
A .Oxidatiove phosphorylation. C. Fermentation reaction.
B . Glycolysis D. Anaerobic reaction.
Explanation :**Oxidative phosphorylation** is the production of ATP from phosphate and ADP in aerobic respiration. **Oxidative phosphorylation** occurs in mitochondria the energy being provided by steps in the electron transport chain. Most of the ATP molecules are generated from a glucose molecule in this stage of aerobic respiration.

∴ *Answer A*

56.To which biological species do Neanderthal humans belong?
A .Homo erectus C. Homo sapiens
B .Homo habilis. D. Ardi ramidus.
Explanation: **Homosapiens neanderthalensis** was present until the last ice age. It practiced ritualized burial of dead and hunter-gatherers. **Homosapiens** has dominated the neanderthalensis since the last ice age.

∴ *Answer C*

57.When trying to view an object under the microscope and after switching to the high power objective, which adjustment knob do you use in order to see a clear picture?
A .The coarse adjustment.
B .The middle adjustment
C .The fine adjustment
D .The arm of the microscope
Explanation :One of the important points of caution in focusing is don't try to focus an object directly with high power or the oil immersion objectives. First locate or focus the object with the low power , then shift to the middle or high power and use a fine adjustment knob (not a coarse adjustment knob) in order to see clear image.

∴ *Answer C*

58. Which one of the following belongs to the category of heterotrophic organisms?
A .Algae B .Fungi C. Higher plants. D. Mosses.
Explanation :
- **Algae** are photosynthetic autotrophs that can make their own food.
- **Fungi** are heterotrophic organisms that depends on autotrophs.
- **Higher plants** are photosynthetic autotrophs.
- **Mosses** are photosynthetic autotrophs.

∴ *Answer B*

59. Which of the following is the most abundant polysaccharide found in plants?
A .Glycogen B .Sucrose C. Cellulose D. Starch
Explanation :**Cellulose** is a polymer of β- glucose. Unlike starch and glycogen it has structural role. About 50% of the carbon found in plants is cellulose and it is the most abundant organic molecule on earth. It is virtually confined to plants ,although it is found in some non vertebrate animals and ancestral fungi. Its abundance is due to its being a structural component of plant cell walls, making up about 20-40% of the wall on average

∴ *Answer C*

60. Choose the alternatives that illustrates the medical application of biotechnology.
A . Bio- fuel production.
B . Biosensors for testing blood glucose level.
C . DNA finger printing technology.
D . Protein production from single cells.
Explanation: **Bio- fuel production** is the application of biotechnology in the production of bio energy.

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- **Biosensors** for testing blood glucose level is the application of biotechnology in the field of medicine.
- **DNA finger printing technology** is the application of biotechnology in the field of crime investigation.
- **Protein production from single cells** is another application of biotechnology in the nutrition science.

∴ ***Answer B***

61. Which of the following molecules is capable of mutation?
A .Nucleic acids C. Carbohydrates
B . Proteins. D. Lipids.

Explanation: **Mutation** is genetic material can be altered by natural or artificial means, and thereby can cause sudden and permanent changes. Such a change in the quantity ,structure or arrangement of the DNA on the chromosomes of organisms is known **as mutation.**

✓ It is only **nucleic acids (DNA)** that can mutate, not any other organic molecule.

∴ ***Answer A***

62. If you observe a piece of paper on which several small squares are drawn, under which of the following magnifications of the microscope do you count more number of the squares?
A .40 X B.100X C.400X D.1000X.

Explanation: More number of squares can be counted under 40 X magnification. Because this is the lowest magnification from the other given magnifications. As magnification increases the size of each square enlarges which results a decrease in the number of squares to be viewed.

∴ ***Answer A***

63. A number of viruses ,including HIV , have more than one strain. This shows that viruses are:
A .Capable of living outside their hosts.
B .Capable of mutation.
C .Capable of reproduction.
D .Obligate parasites.

Explanation: **Mutation** brings about new genetic makeup in an organism which results the existance of different strains of the species. Similarly the different strains of HIV particles is due to their capability of mutation

∴ ***Answer B***

64. Identify the statement that more correctly explains why each cell contains so many different types of enzymes? Enzymes are:
A .consumed in the process of catalysis.
B .used over and over again.

C .specific to their substrates.
D .easily synthesized in the cell.

Explanation: Every living cell contains so many different types of enzymes. Because the cell carries so many different kinds of chemical reactions. Each type of chemical reaction is catalyzed by its own specific enzyme. That is enzyme are specific, meaning that one enzyme catalyses only one type of chemical reaction. That is why every cell contains so many different types of enzymes.

∴ ***Answer C***

65. When is a behavior referred to as stereotyped? when:
A .an animal responds similarly to all stimuli.
B .all members of a species respond in the same way to a stimulus.
C .a behavior is easily earned.
D .an animal ignores a stimulus because it is familiar with it.

Explanation: An **innate behavior came** to be regarded as a behavior that appears as the animal develops apparently without having been learned. It is often stereotyped (always performed in the same way by members of a particular species) and complete (performed in its entirety)

∴ ***Answer B***

66. Which of the following is a mismatch between a biome and what it contains?
A .Savanna – large herbivores.
B .Desert – cacti.
C .Deciduous forest –evergreen trees.
D .Tropical rainforest – high biodiversity.

Explanation: **Temperate deciduous** forests grow in continental climates with summer rain fall and server winter. They are dominated by broad-leaved deciduous trees. The trees are characterized by shedding their leaves periodically. The leaves often change in colour before they fall. Deciduous forests are not evergreen.

∴ ***Answer C***

67. In an ecosystem that consists of grasses, birds ,grasshoppers and frogs, which of the following food chain is possible?
A .Grass → Frog → Grasshopper → Bird
B. Grass → Grasshopper → Frog → Birds.
C .Grass → Bird → Grasshopper → Frog.
D .Bird → Frog → Grasshopper → Grass.

Explanation: In an **ecosystem** that consists of grasses ,birds, grass hopper, the possible food chain is
Grass → Grasshopper → Frog → Bird

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Grasses are the primary **producers** of the ecosystem which makes the bases of the food chain. **Grass hoppers** feed on the grass as the primarily consumers, the frog feeds on the grass hoppers, secondary consumers (carnivore) and the bird feeds on the frog as tertiary consumer (second carnivore).

∴ ***Answer B***

68. If an animal inhales a radioactive form of oxygen, in which of the following products of the cellular respiration would the radioactivity be detected?

A .Water B .Carbon dioxide C.ATP D. NADH

Explanation: In **aerobic respiration** oxygen serves as a final electron and proton acceptor and combines with the protons (hydrogen ions) to form water. Therefore, an animal that inhaled radioactive form of oxygen, the radioactivity is detected in the water molecules which are the products of cellular respiration (aerobic respiration)

∴ ***Answer A***

69. If pieces of fresh potato are kept for some time in sugar solutions of 20%, 10%, 5% and distilled water, which piece will gain the highest percentage of weight?,

A .The one in 20% solution. C. The one in 5% solution.

B .The one in 10% solution. D. The one in distilled water.

Explanation: As the water potential of solution outside the cell increases (more dilute), the rate and amount of diffusion of water increases from the solution to the cell (in this case fresh potato piece).20% sucrose solution has the lowest water potential there for least amount of water enters the potato pieces from the solution.

✓ Distilled water has the highest water potential as compared to the other solutions. Therefore, water enters most from the distilled water in to the fresh potato pieces and this results the highest percentage of weight gain in pieces kept in distilled water.

∴ ***Answer D***

70. Suppose a lost and found baby is claimed by the four families whose blood genotypes are shown below. If the baby has blood type O, which of the families is the possible parent?

A .AO and BB B .AA and OO C.AO and BO D.AB and OO

Explanation:

❖ AO and BB

- ✓ Parental blood type A and B.
- ✓ Parental genotype $I^AI^O \times I^BI^B$.
- ✓ Gametes I^A, I^O and I^B, I^B.

✓ Fertilization

	I^A	I^O
I^B	I^AI^B	I^BI^O
I^B	I^AI^B	I^BI^O

✓ This family cannot be the parents of a child with O – blood type.

❖ AA and OO

- ✓ Parental blood type A and O
- ✓ Parental genotype I^AI^A and I^OI^O
- ✓ Gametes I^A and I^O
- ✓ Fertilization → I^AI^O
- All the children born will have A blood type.
- This family cannot be the parent of this child .

❖ AO and BO

- ✓ Parental blood types A and B
- ✓ Parental genotypes I^AI^O and I^BI^O
- ✓ Gametes - I^A, I^O and I^B, I^O
- ✓ Fertilization

	I^A	I^O
I^B	I^AI^B	I^BI^O
I^O	I^AI^O	I^OI^O

✓ One of the child from these parents could have O- blood type. Therefore, those individuals with blood type A(AO) and B(BO) could be the parents of the child with blood type O.

✓ Parents with blood type AB and O (O O) produce only children with blood type A or blood type B

∴ ***Answer C***

71. For what objective do geneticists engineer the Bt gene of Bacillus thuringiensis into crop plants? To make the plant:

A .resistant to insect pests.

B .resistant to pesticides.

C. resistant to herbicides.

D. produce food of improved quality

Explanation: **Organic farmers** use the naturally occurring soil bacterium Bacillus thuringiensis as a non-chemical insecticide. The genetic engineer have extracted the Bt gene from this bacterium and transferred it to cotton plants to make them resistant to attack by insects.

∴ ***Answer A***

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72. Among the following fossil forms of the human evolutionary ancestors, which one was the first to develop the ability to walk upright?
A .Homo erectus.
B .Homo habilis
C. Ardipithecus ramidus.
D. Australopithecus afarencis.

Explanation:

- **Homo erectus** found between 1 million and 0.5 million years ago and walked up right.
- **Homo habilis** is found between 2.5 million and 1.5 million years ago and walked up right.
- **Ardipithecus ramidus** Was bipedal walked up right and 4.4 million years old.
- **Australopithecus afarancis** is lived 3.2 million years ago and walked up right.

∴ ***Answer C***

73. Suppose a girl, after watching her mother making "enjera" on several occasions successfully did it herself, what type of learning is this?
A .Sensitization.
B .Imprinting
C. Habituation
D. Latent learning

Explanation:

- **Sensitization** is an increase in the response to a harmless stimulus when that stimulus occurs after a harmful stimulus.
- **Imprinting** is a kind of instinctive behavior in which the fixed action pattern is for newly born /hatched organisms to imprint on (or become attached to) the first thing they see that has certain general features (those of an adult of its species).
- **Habituation** is a process which results in a decreased response to a stimulus after repeated exposure to that stimulus over a period of time.
- **Latent learning** the word "latent" means "hidden" so we are taking about hidden learning or learning that is not apparent as it takes place. **Latent learning** happens when the brain aquires knowledge at a certain time ,without reinforcement , but does not use it until later, at the time when the knowledge is needed

∴ ***Answer D***

74. Assuming that each of the following is separately analyzed, which one is expected to give more variety of organic molecules?
A .A virus
B .A molecule of DNA
C.A mitochondrion
D.A polysaccharide.

Explanation:

- **A virus** composed of nucleic acid (RNA or DNA) enclosed by a protein coat.
- **A molecule of DNA** is composed of nucleotides.

- **A mitochondrion** is composed of double membrane which are made up of phospho lipids and various types of proteins.

✓ The fluid matrix of a mitochondrion consists of:

- DNA and ribosomes.
- Salts and other kind of solutes.
- Proteins
- Intermediate products of cellular respiration

∴ ***Answer C***

75. Which evolutionary evidence was very unlikely to have been used by Darwin in the formulation of the theory of evolution?
A .Comparative anatomy.
B .Comparative biochemistry.
C. Fossils.
D. Geographical distribution.

Explanation: The evolutionary evidence that was very unlikely to have been used by **Darwin** in the formulation of the theory of evolution is **comparative biochemistry**.

∴ ***Answer B***

76. Which of the following alternatives best defines gene flow?
A .The movement of genetic material from the nucleus to the cytoplasm.
B .The transmission of genetic information from the cytoplasm to the nucleus.
C . The transmission of genetic information from one population to another.
D .The movement of genes from one chromosomes to another.

Explanation: **Gene flow** is the movement of alleles between population through interbreeding.

∴ ***Answer C***

77. Which one of the following corresponds to the first step of the scientific method?
A .Analyze the data collected
B .Identify a topic to be studied.
C .Undertake the research.
D .State the results.

Explanation: In **a scientific method** what comes first is identification of the topic to be studied. This is followed by gather information.

✓ The information gathered (collected data) is then analyzed etc.

∴ ***Answer B***

78. Which of the following makes a food web of an ecosystem different from a food chain?
A . Presence of many tropic levels.
B . Clear sign of the cycling of materials.
C .Presence of many interconnected food chains.

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D .Presence of many organisms through which energy passes.

Explanation:

✓ **A food chain** is a simple and hypothetical type of feeding relationship in which an organism feeds on only one type of organisms.

✓ **Food chains** are simply attempts to demonstrate the fate of individual organisms in a particular habitat.

✓ **Food web** is most food chains connect with other chains, since most organisms are the prey of more than one predator. Food chains are linked together to form food web.

∴ ***Answer C***

79. What term is used to describe the chemical process of living organisms that involves both the joining up of monomers to produce polymers and the splitting of polymers to monomers?

A .Anabolism
C. Metabolism
B .Catabolism
D. Polymorphism.

Explanation:

❖ **Anabolism** is the constructive biochemical process in living organisms involving the formation of complex molecules from simpler ones and the storage of energy.

❖ **Catabolism** is breaking down of complex molecules by living organisms with release of energy.

❖ **Metabolism** is integrated network of biochemical reactions in living organisms, often referring to the biochemical changes occurring in the living organism or cell as a whole.

❖ **Polymorphism** is the existence within a species or a population of different forms of individuals , occurrence of different forms of, or different forms of organs in the same individuals at different periods of life.

∴ ***Answer C***

80.choose the environmental factor that decreases both with increasing altitude and latitude?

A .Temperature
C. Moisture
B .Sunlight
D. Fertility of soil

Explanation: **Temperature** is the most important factor that influences almost all the activities of the organism. In nature, there are minimum and maximum effective temperature ranges within which an organism carry on its active life and beyond which death evolutually occurs. Generally higher temperatures prevail in the equatorial areas and decreases as we go north or south of the equator towards the polar area. There is a global horizontal and vertical relationship in the

density and type of vegetation. This is due to a decrease in temperature as altitude and latitude increases.

∴ ***Answer A***

81.The carbon dioxide produced during the respiratory cycle and other processes is largely absorbed by which of the following?

A .The ozone layer
C. Plants
B .Animals
D. The sky

Explanation: **Plants or forests** are said to be natural sink of carbon dioxide because they absorb the carbon dioxide produced by respiration and other processes. That is why **deforestation** is a threat to increase in the concentration of carbon dioxide in the planet.

∴ ***Answer C***

82. Which process of life is grouped into autotrophic in some organisms and heterotrophic in others?

A .The transport system
B .The aspect of Nutrition
C. Anaerobic respiration.
D. The coordination system.

Explanation: Modes of nutrition based on the method of obtaining food , the living organisms are broadly classified as in to two major groups. These are **autotrophs and hetrotrophs**.

∴ ***Answer B***

83. How do the two strands of DNA combine to form a molecule?

A .A base of one strand pairs with a complementary base of the other strand.
B .The phosphate of one strand combines with the phosphate of the other strand.
C .The sugar of one strand combines with the complementary sugar of the other strand.
D .A ribose sugar combines with a deoxyribose sugar on the same DNA strand.

Explanation: **DNA** is a huge molecule made from two anti-parallel polynucleotide strands (chains).The nucleotides are held together by bonds between the sugar in one nucleotide and phosphate group in the next .The bases do not take part in this linkage of the nucleotide strands. The nucleotides in one strand are paired with nucleotides of the complementary strand according to the base pairing rule.

✓ **Adenine** containing nucleotide will always be opposite **thymine-**containing nucleotide

✓ **Cytosine :** containing nucleotide will always be opposite **guanine-**containing nucleotide

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∴ *Answer A*

84. Which of the following is true about the first stage of photosynthesis?
A .Light – dependent. C.ATP - driven
B .Temperature –dependent. D. Glucose – driven.

Explanation: The first phase of photosynthesis is known as light dependent. It is a process by which light energy is absorbed by special photosensitive pigments such as chlorophyll in the chloroplast. The light dependent reactions take place in the membrane of the thylakoids in the chloroplast

∴ *Answer A*

85. To which category of enzymes do the digestive enzymes that breakdown food substances in the human alimentary canal belong?
A .Intracellular enzymes. C. Extracellular enzymes.
B .Globular enzymes. D. Fibrous enzymes.

Explanation: **Intracellular enzymes** are enzymes that are produced in the cells and remain within the cell to catalyze chemical reactions there.

✓ **Globular enzymes** are enzymes that resemble a globule or ball and derived from globular proteins.

✓ **Extracellular enzymes** are enzymes that are secreted out in to cavities such as digestive tract to catalyze chemical reactions that take place there. Digestive enzymes are extracellular enzymes.

∴ *Answer C*

86. Of the following information, which one was most important to the development of the techniques of modern genetic engineering?
A .The knowledge about recessive characters.
B .The discovery of lethal genes.
C .The formation of punnett squares.
D .The knowledge about the structure of the DNA molecule.

Explanation:

✓ **The knowledge about the structure** of the DNA molecule was important to the development of the technique of modern genetic engineering.

✓ The development of three main techniques following the knowledge of the structure of DNA made genetic engineering possible.

✓ The discovery that genes can be "cut " out of a DNA molecule using enzyme called **restriction endonucleases**.

✓ The discovery that genes can be inserted ("tied") in to another DNA molecules using a ligase enzyme.

✓ Genes can be transferred in to other cells using vectors. These are usually either **plasmids or viruses**.

∴ *Answer D*

BOOKLETCODE:33

1. Which one of the following is true about an innate behavior ?
A. It is non – adaptive behavior.
B. It is learned behavior
C. Members of a species behave similarly
D. It is environmentally determined.

Explanation: The word "**innate**" literally means "inborn".

✓ **Innate behavior** is a behavior that is present (potentially) at birth or hatching.

❖ **Innate behaviors are adaptive :** they have been retained in the species by natural selection because they confer a survival advantage.

❖ They are fully functional the first time they are performed meaning that they require no learning.

❖ usually innate behaviors are fixed stereo typed movements that are the same in all individuals of the species every time they are performed. It means that all members of the species perform the behaviors in the same way.

❖ There is a key stimulus that triggers , the behavior. There is always an external environmental stimulus that ignites the innate behavior to be performed. This does not mean the stimulus changes or determines the type of the innate behavior that is going to be performed.

∴ *Answer C*

2. Which one of the following has a unidirectional flow in an ecosystem?
A .Nitrogen C . Phosphorus
B .Carbon dioxide D . Energy

Explanation: **Nitrogen, carbon dioxide** and **phosphorus** are nutrients of any ecosystem or they are the materials of ecosystem. These materials are found in limited amount in any ecosystem. So that life can continue in the ecosystem these materials have to be **re-used** , over and over again, meaning that they are **recycled**. There is no unidirectional flow of nutrients in any ecosystem.

✓ In the ecosystem, the movement of **energy** is not cyclic but unidirectional.

∴ *Answer D*

3. If , after dividing plant materials into two groups, an experimenter has applied a fertilizer to group A plants and not to group B plants ,What are group B plants called?
A .Control group C . Test group
B . Experimental group D . Variable group.

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Explanation: **A control group** acts as a **"standard"** for comparison. It is used to "isolate" the factor we are investigating and show that changes are due to this factor.

- **Experimental group** is the group in an experiment which is being experimented on in order to compare with the control group.
- **Test groups** and **experimental groups** are the same. They are different phrases used to name the same thing.
- There is no group called **variable group** in any scientific experiment.

∴ Answer A

4. The cells of which group of micro- organisms can be described as prokaryotic?
   A .Viruses C .Algae
   B .Protozoa D . Bacteria

Explanation: **Viruses** are not cellular and thus are not prokaryotic or eukaryotic.

- **Protozoa** are unicellular eukaryote organisms that lack cell wall. Most of them are motile(able to move) and include organism such as Amoeba, plasmodium (that cause malaria) and paramecium.
- **Algae** are mixed group of simple plant like eukaryote organisms, largely found in marine or fresh water habitats.
- **Bacteria** are unicellular prokaryotic organisms that lack true nucleus separated from the rest of the cell by membrane. The DNA of the bacterium forms a continuous loop that is intermingled with the cytoplasm.

∴ Answer D

5. In the scientific method, which one of the following is used to test a hypothesis?
   A .Observation C .Experiment
   B .Theory D . Prediction

Explanation: **Observation**: the foundation of all modern science is observation. **Observations** can be made directly through the sensory systems, mainly, vision, hearing, taste, smell and touch.

- **Theory** is hypothesis that has with stood many tests and has been shown to allow predictions to be made or it is a tested hypothesis.
- **Experiment** is component of a scientific method used to test any hypothesis. Experiment proves whether the hypothesis is correct or incorrect.
- **Prediction** is an educated guess as to how the biologist think his/her experiment turn out.

∴ Answer C

6. Which one of the following is largely made up of phospholipids?
   A .Cell wall C . Nucleus
   B .Cell membrane D . Chromosomes

Explanation: **Cell wall** is rigid wall or structure surrounding the cells of bacteria fungi and plants. **Plant cell walls** are made of **cellulose fibers..**

- **Cell membrane** that encloses the content of all living cells and defined the boundary between the contents of the cell and the surroundings. Cell membranes' are made almost entirely of **proteins and lipids.** The most common lipids are **phospholipids..**
- **A nucleus** is the most important organelle found in the cytoplasm of the cell. It controls the activities of the cell.
- **Chromosomes** are thin thread like structures found in the nucleus mainly composed of DNA bound to basic protein called **his tones.**

∴ Answer B

7. How many daughter cells are produced from a single mitotic division?
   A .Two B .Four C . Six D . Eight

Explanation: **Mitosis** is the process by which a cell nucleus divides to produce **two daughter** nuclei containing identical sets of chromosomes to the parent cell. Division of the whole cell to form two daughter cells usually follows the nuclear division.

∴ Answer A

8. Which of the following are rod- shaped bacteria?
   A .Cocci C . Spirilla
   B .Spirochetes D . Bacilli.

Explanation:

- **Cocci (singular coccus)** ar spherical shaped bacteria.
- **Spirochetes** is spiral or corkscrew shaped bacteria.
- **Spirilla** is another name given to spirochaes.
- **Bacilli (Singular bacillus)** is rod shaped bacteria.

∴ Answer D

9. Which one of the following refers to a change in a gene?
   A . Mutation C .Protein synthesis
   B .Replication D .Transcription.

Explanation:

- **Mutation** is any spontaneous change in the genetic material of an organism. Mutations could be **chromosomal mutation or gene mutation or point mutation.**
- **Replication** is a process by which a DNA molecule (the genetic material) makes an exact copy of itself. This is the basis of all methods of reproduction and the basis of passing on genetic information from one generation to the next.

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❖ **Protein synthesis** is the process whereby proteins are synthesized (are made) on the ribosome's of cells.
❖ **Transcription** is the means by which the genetic information in DNA is converted into a single stand of messenger RNA (m RNA) that carries the information to the ribosome's where protein is synthesized.

∴ ***Answer A***

10. Which one of the following disciplines is studied in ethology ?
A .Heredity
B .Physiology
C . Reproduction
D . Behavior

Explanation:
❖ **Heredity** is the study of all those characteristics of organisms that are governed by the genetic materials (DNA) derived from the parents.
❖ **Physiology** is the way in which organisms or parts of organisms function.
❖ **Reproduction** is the process by which plant and animal cells or organisms produce off spring .
❖ **Behavior** is a co-ordinated response of an organism to an internal or external stimulus. The science of behavior is also known as the science of ethology.

∴ ***Answer D***

11. What are bacteriophages?
A .Bacteria
B .Viruses
C . Protozoa
D .Fungi

Explanation: **Bacterio phages** are viruses , that use bacteria to replicate its genetic material (the virus particles).

∴ ***Answer B***

12. What is the plant growth substance which causes them to bend towards the light?
A .Starch
B .photosynthesis
C . Auxin
D . Chlorophyll.

Explanation:
❖ **Starch** is a polysaccharide carbohydrate which is a polymer of ∝-glucose. It is a major fules store in plants, but is absent from animals where the equivalent is **glycogen**.
❖ **Photosynthesis** is the process by which green plants, algae and some bacteria manufacture organic compounds using chlorophyll (a green pigment),sun light ,carbon dioxide and water.
❖ **Auxin** is one of a class of plant growth substances . **Auxin** is made continually in growing shoot and root tips. It regulates the rate of elongation of cells in the growing region behind the shoot tip and is involved in phototropic curvature (bending of the shoot)of the shoot towards, the direction of light in response to one directional light.

❖ **Chlorophyll** is a photosynthetic green pigment found in the cells of plants and algae that traps sun light energy in the process of photosynthesis.

∴ ***Answer C***

13. The unicellular organism, Euglena , swims using its flagellum towards areas of increased light intensity. What is this kind of response to stimulus called?
A .Negative kinesis
B .Positive kinesis
C . Negative phototaxis
D. Positive phototaxis.

Explanation:
❖ **Negative kinesis** is a response in which the organism changes its rate of movement in response to changes in the intensity of the stimulus and slows down its speed of movement.
❖ **Positive kinesis** is a response in which the organism changes its rate of movement in response to changes in the intensity of the stimulus and increases its speed of movement.
❖ **Negative photo taxis** is a response of an organism where the organism move away from the greatest intensity of light.
❖ **Positive photo taxis** is a response of an organism where it moves towards the direction of the greatest intensity of light.

∴ ***Answer D***

14. What does an ornithologist study?
A .The biology of mammals
B . The biology of frogs and reptiles
C . The biology of fishes
D . The biology of birds.

Explanation:
❖ The biology of mammals or a branch of biology that studies mammals is called **mammalogy**.
❖ **Ornithology** is the branch of vertebrate zoology that is concerned with the study of **birds**.
❖ **Ornithologist** is the scientist that study birds.

∴ ***Answer D***

15. Which one of the following best defines modern biology?
A .The study of animals
B .The study of plants and animals
C .The study of all living things.
D . The study of transfer of energy.

Explanation:
❖ The study of animals is called **zoology**.
❖ The study of plants is called **Botany**.
❖ The study of all living things is **biology**.
❖ Energy transfer in living thing is studied in **Ecology**.

∴ ***Answer C***

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16.In aerobic respiration of cells, in which cellular part does the Krebs cycle (citric acid cycle) take place?

A .Chloroplasts
B .Mitochondria
C . Nuclei
D . Lysosomes.

Explanation:

- **Chloroplast** is a photo synthetic plasmid containing chlorophyll and other photosynthetic pigments. It is found in all photosynthetic plant cells. It is a site of photosynthesis.
- **Mitochondria** are the sites of the reactions of aerobic respiration. They are surrounded by two membranes. The inner membrane that is folded in to cristae to increase the available surface area and the outer membrane.
- ✓ The different stages of aerobic respiration including **Krebs cycle** and **electron transport** take place in mitochondria.
- **Nuclei** are found in all eukaryotic cells containing the genetic information (DNA) and hence controlling the cell activities.
- **Lysosomes** are found in most eukaryotic cells. They are surrounded by a single membrane and are simple sacs that contain **digestive enzymes** that breaks down cellular waste and debris.

∴ ***Answer B***

17. Which alternative correctly matches the polymer with its monomer?

A .Carbohydrate ⟶ amino acid.
B .Protein ⟶ monosaccharide
C . Nucleic acid ⟶ nucleotide.
D . Lipid ⟶ fat

Explanation:

- The polymer of **carbohydrate** is made from glucose monomers (monosaccharide).
- The polymer **protein** is made from amino acid monomers.
- The polymer **nucleic acid** (DNA and RNA) is made from nucleotide monomers.
- **Lipids** are made from fatty acids and glycerol molecules.

∴ ***Answer C***

18. Which term better captures the concept of variety of organisms present in a whole continent as well as all the genes found in them?

A .Niche
B .Biodiversity
C . Ecosystem
D . Biome

Explanation:

- **Niche** is the functional role of an organism in a community in terms of the habitat it occupies, its interactions with other organisms and its effect on the environment.

- **Biodiversity** is the number of different species present in a given environment(species diversity), the genetic diversity within a species and the number of different ecosystems present in a given environment (ecological diversity).
- **Ecosystem** is a unit made up of all the living and non-living components of a particular area that interact and exchange materials with each other.
- **A biome** is a geographical area or regional area with a specific climate, specific soil type and animals and plants that are adapted in similar way to abiotic conditions within the area.

∴ ***Answer B***

19.Choose the name of the researcher/ scientist who introduced the term cell for the first time?

A .Aristotle.
B .Anton Van Leeuwenhoek
C . Robert Hooke
D . Robert Brown.

Explanation:

- **Aristotle, 384-287 B.C** Initiated the study of morphology and classification of animals and plants.
- **Anton Van LeeuWenholk,1632-1723**: produced a simple microscope from a hand lens, and discovered bacteria growing in fusions of plant materials.
- **Robert Hooke, 1635 – 1703** :Introduced the term "Cell " describing the structure of cork.
- **Robert Brown , 1773-1858** :Asserted the presence of nucleus in cell and discovered the Brownian movement in protoplasm.

∴ ***Answer C***

20.What is the total magnification power of a microscope with a 10 X ocular lens and a 25 X objective lens?

A .2.5 X
B .35 X
C .250 X
D .2,500 X

Explanation: The magnification power of a microscope is equal to the magnification of the ocular lens (eye piece lens) times the magnification of the objective lens.
The magnification power of this particular case
= 10 × Ocular lens and 25 X Objective lens.
= 250X

∴ ***Answer C***

21.Which of the following molecules requires a template molecule for its synthesis?

A .Cellulose
B .Starch
C . Protein
D . Glycogen

Explanation:

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- **Cellulose** is a polymer of β– glucose joined by β– 1 -4 glycosidic bonds formed by condensation reaction and a major component of plant cell walls.
- **Starch** is a polymer of α - glucose joined by α - 1,4 glycosidic bonds and it is a mixture of amylose and amylopectin.
- **proteins** are polymers of amino acid and macro molecules. Protein molecules are synthesized by a process of protein synthesis on the mRNA (messenger RNA)that serves as **a template.**
- **Glycogen** is a storage carbohydrate in animal cells. It has similar structure to that of amylopectin of starch, but there are more α –1, 6 links, making it much more highly branched.

∴ ***Answer C***

22.Which of the following is a process by which cells take in fluid by means of vesicles?

A .Pinocytosis
B .Endocytosis
C . Osmosis
D . Phagocytosis.

Explanation:

- **Pinocytosis ("cell drinking")** – material taken up is in liquid form. Vesicles formed are small. It involves the ingestion of smaller particles(but particles too large to cross the membrane by other methods).
- **Endocytosis** is a process in which large particles are engulted by a cell.
- **Osmosis** is the process by which water moves across a partially permeable membrane from a region of its highest concentration to a region of lowest concentration (from a weak solution to a strong solution).
- **Phagocytosis** is a process in which large materials (solid particles)are taken up by a cell involving the creation of pseudopodia (extension of the plasma membrane) to enclose large particles or even whole organisms outside the cell.

∴ ***Answer A***

23. In the life cycle of sexually reproducing organisms, which of the following processes restores the chromosome number from haploid to diploid state?

A .Fertilization
B .Meiosis
C . Chromosome replication
D . Mitosis.

Explanation:

- **Fertilization** is a process by which usually two haploid cells(haploid egg and haploid sperm nucleus) fuse together and produces a diploid zygote, hence it restores the diploidity of chromosome numbers in sexual reproduction.

- **Meiosis** is a reduction division which reduces the chromosomes number of the daughter cells by half or reducing cells from diploid to haploid in sexual reproduction. It result four daughters that are different from one another and the mother cell or original cell.
- **Chromosome replication** is a process in which every chromosomes makes its own exact copy during cell division (meiosis and mitosis).
- **Mitosis** is a kind of cell division also known as **equation division** where the daughter cells formed receive the same number of chromosomes as that of the original cell. The two daughter cells are identical to one another and to the original cell.

∴ ***Answer A***

24.The hormone insulin balances the level of blood glucose in the body. This is an example of which of the following properties of life?

A .Metabolism
B .Respiration
C . Homeostasis
D . Response.

Explanation:

- **Metabolism** is an integrated network of biochemical reactions in living organisms often referring to the biochemical changes occurring in the living organism or cell as a whole.
- **Respiration** is any or all of the process used by organisms to generate metabolically usable energy, chiefly in the form of ATP from the oxidative break down of food stuffs.
- **Homeostasis** is the maintenance of constant internal environment by an organism.
- **Response** is a charge in an organism or part of an organism that is produced as a reaction to a stimulus.

∴ ***Answer C***

25. All the following carbohydrates are polysaccharides **EXCEPT** which?

A Glucose B .Cellulose C . Starch D . Glycogen

Explanation: **Starch , cellulose** and **glycogen** are polysaccharides made from a number of monosaccharide's.

- **Glucose** is a simple sugar or monosaccharide that composes all of the above three carbohydrates.

∴ ***Answer A***

26. If an enzyme is provided with its normal substrate plus another molecule having the same shape and size as the substrate, which one of the following would happen?

A . Allosteric inhibition.
B .Competitive inhibition
C . Irreversible inhibition
D . End product inhibition.

Explanation: **Allosteric inhibition** is a kind of inhibition of enzymes activity where some enzymes known as allosteric enzymes that possess a receptor site called **allosteric site** other than the active site and the

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substance present in the cell (allosteric inhibitor) binds to this allosteric site causing a change in the structure of the active site resulting slowing down of the rate of the reaction.

- **Competitive inhibition** is a type of inhibition where the competitive inhibitor molecules are often similar in shape to the substrate molecules.
- **Irreversible inhibition** is an inhibition where irreversible inhibitor substances bind strongly to enzymes, usually by a covalent bond, permanently altering the structure of the enzyme molecule and in activating it.
- **End product inhibition** is an inhibition where the end product inhibits the enzyme controlling the first stage of the reaction in series of enzyme controlled reactions.

∴ ***Answer B***

27. Which one of the following sugars do babies get from the milk of their mothers?

A .Maltose B .Sucrose C . Lactose D . Fructose

Explanation: **Maltose** is a disaccharide made from two molecules of glucose that mainly occurs as a breakdown product during digestion of starch by an enzyme called **amylase**.

- **Sucrose or cane sugar** is the most abundant disaccharide in nature. It is made from a glucose and a fructose monosaccharide. **Sucrose** is mainly common in plants.
- **Lactose** is a disaccharide made from a glucose and a Galactose monosaccharide's found exclusively in milk and is important energy source for young mammals. It is also known as **milk sugar**.
- **Fructose** is a simple sugar or a monosaccharide found in sweet fruits and honey.

∴ ***Answer C.***

28. Which one of the following is the major killer of AIDS patients?

A . Pneumonia C . Anaemia

B .Opportunistic infections. D . Malnutrition.

Explanation:

- **Pneumonia** is a **"Respiratory disease"** affecting the air ways of lungs. The organisms are carried in tiny drop lets through the air .It is caused by bacteria known as **pneumococci.**
- **Opportunistic infections** is an infection that result when the immune system is weakened or destroyed following the development of HIV infection to AIDS.
- **Anaemia** is a deficiency disease caused by the lack of iron which is the important component of hemoglobin which carries oxygen in red blood cells.

- **Malnutrition** is a condition where the body suffers or its health is affected, if too little food is eaten (under nutrition) or too much food is taken in (over nutrition) , or any one element of the diet is lacking.

∴ ***Answer B***

29. Among the following discoveries in biology which one is the latest of all?

A .The law of heredity.
B .The cell theory
C . The double helix nature of the DNA
D. The binomial system of classification

Explanation:

- Gregor Mendel, 1822-1884 put forward the law of **inheritance or heridity.**
- In 1839 Matthias Schleiden and Theodor Schwann introduced an idea known as **the cell theory.**
- Watson and crick, 1962 explain the **double helix structure of DNA.**
- Carolus Linnaeus, 1707-1778 expounded the **binomial system of classification.**

∴ ***Answer C***

30. In Pavlov's classical conditioning experiment on dogs, which one of the following is the unconditioned response?

A .The sound of the bell before presenting food.
B .The slight of the food from far away.
C .The salivation of the dog at the sight of food.
D .The salivation of the dog at the sound of the bell.

Explanation:

- The **condition stimulus** or the **neutral stimulus** is the stimulus that does not initially produce the unconditioned response. In Pavlov's classical conditioning experiment on dogs, the sound of the bell is the conditioned (neutral) stimulus.
- The **unconditioned stimulus** is the stimulus that unconditionally, naturally and automatically triggers a response. For example, the sight of food from far away results unconditionally secretion of saliva or feeling hunger, a response.
- The **unconditioned response** is the unlearned response that occurs naturally to the unconditioned stimulus. For example the dogs salivates or secrete saliva at the sight of food.
- The **conditioned response** is the response to previously neutral stimulus after the association of unconditional stimulus with different stimulus. For example, the salivation of the dog at the sound of the bell that previously does not produce the response.

∴ ***Answer C***

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31. It is said that a goat mother accepts and nurses as her own kid any young that she smells at a certain critical period, and rejects all others. This is an example of:

A . habituation
B .imprinting
C . insight learning
D . sensitization.

Explanation:

- **Habituation** is a process which results in a decreased response to a stimulus after repeated exposure to the stimulus over a period of time.
- **Imprinting** is a type of instinctive behavior in which the fixed action pattern is for newly born (hatched organism to imprint on (or become attached to) the first thing they see that has certain general feature .
- **Insight learning** involves finding solutions to problems that are not based on actual experience (as with trial and error) button trials occurring mentally .Often the solution is learned suddenly, such as when a person have been trying to solve a problem for a period of time and suddenly the solution appears almost "out of nowhere".
- **Sensitization** is an increase in response to a harmless stimulus when the stimulus occurs after a harmful stimulus.

∴ ***Answer B***

32. What percentage of the amount of carbon 14 originally present in a fossil would be left after its second half life?

A .75% B .50% C .25% D. 12.5%

- ***Explanation:*** Only 25% of the original carbon 14 was decayed.This is less than 50%.There fore the decaying process of carbon 14 in the fossil wood is within the first- half life.
- If 50% of the original carbon 14 was remained in the fossil wood, it means that only 1 half- life was passed.
- If 25% the original carbon 14 was remained it indicates that only 2 half- life was passed.
- If 12.5% the original carbon 14 was remained it shows that only 3 half-life was passed.

∴ ***Answer C***

33. Which one of the following modes of natural selection was responsible for the fast increase in the number of the black form of moths in Europe during industrial Revolution?

A .Stabilizing selection
B .Disruptive selection
C . Normalizing selection
D . Directional selection

Explanation:

- **Stabilizing selection** is a type of natural selection where in the stable environment, individuals at both ends of the range of values for a feature are the least well adapted.
- **Disruptive selection** is, in effect , the converse of stabilizing selection.

- **Normalizing selection and stabilizing selection** are two different biological phrases explaining the same phenomenon.
- **Directional selection** is a type of natural selection where a feature may show a range of values. Individuals at one extreme could have a disadvantage where as those at the other extreme have an advantage. Individuals with advantageous features are selected against those with disadvantageous feature and the frequency of these individuals will increase over time.

✓ The black form of moths were advantageous and those with pale form were disadvantageous. Therefore, during industrial revolution where the environment appeared black, the black form moths were selected against the pale form moths.

∴ ***Answer D***

34.Among the following infectious human diseases, identify the one that is caused by a virus.

A .Yellow fever B .Pneumonia C . Cholera D . Typhoid fever.

Explanation: **Yellow fever** is known as arbovirus . It is caused by arthropod- born viruses, that is viruses carried by arthropod- vectors such as thicks, mosquitoes etc.

- **Pneumonia** is a **"respiratory disease"** affecting the air ways of the lungs. The organisms are carried in tiny droplets through the air. It is caused by bacteria known as **pneumococci.**
- **Cholera** is a bacterial infection that affects the intestinal tract. It is caused by bacteria called **vibro cholera.**
- **Typhoid fever** is a bacterial infection of the intestinal tract that occasionally also affects blood stream.

∴ ***Answer A***

35. In plant hybridization experiment, which one of the following is the best way to prevent self- pollination?

A .To remove the stamens.
B .To cover flowers with bags.
C. To cross- pollinate flowers.
D . To make reciprocal crosses.

Explanation: In **plant hybridization experiment**, to hybridize two different varieties of a species, pollen grains must be transferred from the stamen of one plant to the stigma of the other plant. The stamens of the plant that is going to be pollinated must be removed before they mature in order to prevent self pollination and making the hybridization successful.

∴ ***Answer A***

36.Among the following ,which one is a learned behavior?

A .Reflex action
B .Habituation
C. Imprinting
D . Fixed action pattern.

Explanation:

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- **Reflex action** is an innate behavior (in born behavior) which is there at birth of the organism.
- **Habituation** is a learned behavior which develops as an organism is exposed to the same stimulus repeatedly that is harmless.
- **Imprinting** is an innate behavior that was explained in question 32.
- **Fixed action pattern** is a series of reactions originally triggered by a key stimulus in the response of the organism. It is a characteristics of instinctive behavior.

∴ ***Answer B***

37. Among the following choices, identify the ones that have identical DNA fingerprints?
A .Parents and children
B .Non-twin sibs
C . Fraternal twins
D . Monozygotic twins.

Explanation: **Genetic finger printing (DNA finger printing)** is a technique for comparing the DNA of different people.

- The DNA finger prints of parents and children are different because children receive two different sets of DNA from two different parents.
- **Non- twin sibling** have slightly different DNA finger prints as they are produced from two different egg cells and sperm cells carrying slightly different sets of DNA.
- **Fraternal twins** have slightly different DNA finger prints as they are produced from two different egg cells and two different sperm cells similar to non-twin siblings.
- **Monozygotic twins or identical twins** have the same or identical DNA finger prints because they are always produced from just one sperm cell and egg cell containing identical sets of DNA that formed a zygote that cleaved into two identical embryos.

∴ ***Answer D***

38. All other factors remaining the same ,which of the following decreases the size of a population?
A .Birth rate that is greater than death rate.
B .Emigration rate that is less than immigration rate.
C .Immigration rate that is greater than death rate.
D .Emigration rate that is greater than death rate.

Explanation: If all factors remained the same:

- and birth rate is greater than the death rate, that is more are born than die, the population size will increases.
- and emigration rate is less than immigration rate, that is more enter than leave, the population size will increase.
- and immigration rate is greater than death rate that is more enter than die, the population size will increase.

- and the emigration rate is greater than death rate , that is more leave than die the population size will decrease.

∴ ***Answer D***

39. According to the theory of evolution as proposed by Charles Darwin, which one of the following is the most important driving force of evolution?
A .Mutation
B .Over- reproduction
C . Genetic recombination
D . Natural selection.

Explanation:

- **Mutation** is any spontaneous change in the genetic material of an organism. It is the ultimate source of variation in all living organisms.
- **Over- reproduction** is all species tend to reproduce more off spring that can possibly survive.
- **Genetic recombination** is the process in sexually reproducing organism by which DNA is exchanged between homologous chromosomes by chromosomes pairing and crossing over at meiosis during gamete formation.
- **Natural selection** is the driving force "behind evolution. It is the process that brings about changes (overtime) in populations that can eventually lead to speciation (the formation of new species).

∴ ***Answer D***

40. Which of the following is expected to have a positive effect on the biodiversity of Ethiopia?
A Diverse ecology
B .Extensive deforestation
C. Overgrazing by livestock
D. Logging and fuel wood removal

Explanation:

- **Diverse ecology** has a positive effect on biodiversity, because diverse ecology means an environment with different ecosystem having a number of species diversity, many ecological niches occupied by variety of organisms and leading to genetic variability of the species.
- **Extensive deforestation** is a continuous removal of the vegetation which results a reduction in species diversity.
- **Over grazing by live stock** reduces the availability of forage and Woody plants species for other animals resulting a reduction in species varieties of plants and animals. Therefore, reduces biodiversity.
- **Logging and fuel wood removes trees** of variety of species reducing species diversity and there for biodiversity of an area.

∴ ***Answer A***

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41. Which one of the following terms is most appropriate to collectively refer to all the plants, animals and micro- organisms found in Munesa forest in Ethiopia?

A .Ecology
B .Ecosystem
C .Population
D . Community.

Explanation: **Ecology** is the study of the relationships of organisms to one another and to their surroundings.

- **Ecosystem** : A unit made up of all the living and non-living components of a particular area that interact and exchange materials with each other.
- **Population** is a group of organisms of the same species occupying in a particular space.
- **Community** is a general term covering any naturally occurring group of different organisms living together in a certain environment and interacting with each other.

∴ ***Answer B***

42. What is the final electron acceptor in the electron transport chain of cellular respiration?

A .O_2 B. H_2 C. $NADP^+$ D. NAD^+

Explanation: At the end of electron transport chain of aerobic respiration, the electrons combine with protons (hydrogen ions) and with oxygen to form molecules of water. Because of this , oxygen is known as the terminal electron acceptor.

- H_2 :as the protons accumulated in the inter membrane spaces pass through ATP – syntheses in the process of production of ATP , they (hydrogen ions or protons) combines with electrons to form hydrogen atoms that produce water molecules as by products.
- $NADP^+$ (**Nicotine amide dinucleotide phosphate**) is an oxidized form Of NADP that is used to transport hydrogen ions in the electron transport chain in the light phase (light reaction) of photosynthesis.
- NAD^+ is an oxidized form of NAD in electron transport chain in aerobic respiration used to transport hydrogen ions.

∴ ***Answer A***

43. What is biomass?

A .The number of species in a given biome.
B .The total weight of living matter in a given area.
C .The density of a species in a given area.
D .The total energy found in plants and animals in an ecosystem.

Explanation:

- The number of species in a given biome is a **biodiversity.**
- The total weight of living matter in a given area is the **biomass.**

- The density of a species in a given area is a biotype mass.
- The total energy found in plants and animals in an ecosystem in the biomass energy.

∴ ***Answer B***

44. Which of the following is the right vector to transfer alien genes to bacteria through genetic engineering?

A .Plasmids B .Snails C . Mosquitoes D . Bacteria.

Explanation: **Plasmids** are extra chromosomal circular DNA found within bacteria cells that replicates independently of chromosomal DNA. Plasmids are widely used as vectors to transfer genes to bacteria cells in the genetic engineering.

- **Snails** are animals that belong to phylum mollusca in the animal kingdom. Snails are involved as hosts in the life cycle of the worm that causes Bilharia a water born disease.
- **Mosquitoes** are insects that transmit plasmodium parasite (protozoa) from an infected person to a healthy one. They are animal vectors.
- **Bacteria**, large group of mostly microscopic, unicellular organisms that lack distinct nucleus

∴ ***Answer A***

45. Why are T – lymphocytes more vulnerable to HIV infection? Because they posses:

A .thin cell membrane.
B .gp 120 on their surface.
C . HIV receptor protein
D . large pores in their cell membrane.

Explanation: **T – lymphocytes** or also known as **T – helper** cells are more vulnerable to HIV infection because they have HIV receptor protein known as CD_4 on their surface membrane that can easily bind to the gp 120 (glyco protein 120) on the lipid membrane of the HIV - particle which allows the virus to enter the cell.

∴ ***Answer C***

46. In a given ecosystem , members of which one of the following feeding groups release nutrients that are locked up in organic molecules?

A .Herbivores C . Carnivores.
B .Producers. D . Decomposers.

Explanation:

- **Herbivores** are animals which feed on seeds, fruits, leaves and other parts of plants. They are all primary consumers in a food chain.
- **Producers** are organisms that can manufacture their own food from simple inorganic substance.
- **Carnivores** are animals that feed on other animals.

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❖ **Decomposers** are organisms (bacteria and fungi) that feeds up on dead organisms breaking them down in to simpler substance. Decomposer release nutrients in to the soil that will be available for living producers to make organic matter.

∴ ***Answer D***

47. Suppose in a monohybrid cross 80 F2 plants were produced, what is the number of the plants that are expected to have the dominant and the recessive phenotypes?
A .70 dominant + 10 recessive
B .60 dominant + 20 recessive
C .50 dominant + 30 recessive
D . 40 dominant + 40 recessive.

Explanation: In monohybrid cross where two F_1 plants are crossed the F_2 plants always exist in the ratio of 3:1 .3 plants with dominant trait to 1 plant with a recessive trait. That is ¾ of the 80 plants (60) will show the dominant feature and ¼ of the 80 plants(20 plants) will show the recessive feature.

∴ ***Answer B***

48. Which of the following contains phosphate group in its molecule?
A .Nucleotide
B .Protein
C . Cellulose
D . Simple sugar.

Explanation:

❖ **Nucleotides** are the building blocks of Nucleic acids(DNA and RNA) .Each nucleotide is composed of a five- carbon sugar (ribose or deoxyribose) , a nitrogenous base and a phosphate group.
❖ **Proteins** are made from their building blocks amino acids which contain an amino group and the carboxylic group.
❖ **Cellulose** a structural carbohydrate made from simple sugars.
❖ **Simple sugar** are monosaccharide's which consist 3 to 6 carbon atoms.

∴ ***Answer A***

49. What is the collective name for plants that are well adapted to live in very dry environments?
A .Epiphytes.
B .Xerophytes
C . Mesophytes.
D . Hydrophytes.

Explanation:

❖ **Epiphytes:** any plant growing up on or attached to another plant merely for physical support.
❖ **Xerophytes :** are plants adapted to grow in dry condition or in physiologically dry habitat.
❖ **Mesophytes:** are plants that are adapted to grow under adequate conditions of water supply.

❖ **Hydrophytes:** Plants found growing in water or in extremely wet areas.

∴ ***Answer C***

50. The mature fruits of a flowering plant burst open when touched, forcefully ejecting the seeds contained in the pods. Why is this advantageous to the plant?
A .It attracts insects that aid in pollination.
B .It aids in the dispersal of the species.
C .It prevents seed germination.
D .It can cause genetic changes to occur.

Explanation: In some matured fruits the tension in the two halves of the pod results a forceful or violent movement that ejects or thrown the seeds to the land at some distance from the parent plant. This is known as **self dispersal mechanism.**

✓Dispersal avoids overcrowding and competition for food and light. It also allows colonization (invasion) of new habitats.

∴ ***Answer B***

51. In which of the following biomes are epiphytes typically present as characteristic elements?
A .Tundra
B .Boreal forest
C . Cold desert
D . Tropical rainforest.

Explanation:

❖ **Tundra** is a biome that covers the far northern part of north America ,Europe and Asia. It is a biome where water is not available for most of the year because of its frozen soil. Tundra is covered with lichens and mosses and some migrating animals.
❖ **Boreal forest or taiga** is located south of the tundra and occupies large portion of Canada, Alaska and Russia. The biome is dominated by coniferous evergreen trees. It is also inhabited by many mammals ,birds, insects and arachnids.
❖ **Cold desert** is characterized by low rain fall (or no rain fall) , Cold temperature and poor soil.
❖ **Tropical rainforest** is a biome found near the equator. It is wet, warm and a biome with poor and thin soil. There are many varieties of plants and animals and the **epiphytes** are common. It is the richest biome in terms of biodiversity.

∴ ***Answer D***

52. Which of the following is more likely to ensure the conservation of all the rest?
A . Soil conservation
B .Plant conservation
C .Wildlife conservation
D . Watershed conservation.

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Explanation: **Plant conservation** is more likely to ensure the conservation all of the other natural resources. Plants or vegetations are the home of wild life. If we conserve vegetations we conserve the wild life.

∴ ***Answer B***

53. Which of the following statements is true about bacteria?
A .Bacteria found in the human body are all parasitic.
B .All bacteria are harmful because they spoil foods.
C .All bacteria cause diseases to plants, animals and humans in some ways.
D .Some bacteria living in the human body make vitamins needed by the body.

Explanation: **Bacteria** are ubiquitous, meaning found ever where. Some of these are harmful in that they cause diseases to human (including animals)and plants and spoil food. Many of them are useful, in recycling nutrients in the ecosystem, living as symbiotic in the human body providing some useful organic substance (example Vitamins') and treatment of sewage etc.

∴ ***Answer D***

54. Which group of organisms is found at a lower level of evolution according to the modern principles of biology?
A .protests. B .Fungi C . Animals D . Plants

Explanation: **Protests** are eukaryotic single celled protozoa and algae.

- **Fungi** with the exception of yeast most of the fungi are multicellular.
- **Animals:** The body of the simplest animals is organized from various tissues. More complex animals have more complex body organization that leads to organ – system level.
- **Plants:** Similar to animals the simplest plant body is composed of various tissues. In higher plants the body organization proceeds up to organ level. There is not organ system organization in plants.

∴ ***Answer A***

55. How do animals obtain nitrogen?
A .By absorbing nitrates and ammonia from the soil.
B .From the protein in the organisms they consume.
C .By absorbing nitrogen gas from the atmosphere.
D .Through a mutuality relationship with nitrogen- fixing bacteria.

Explanation: Plants make proteins that contain nitrogen by absorbing substances that contain nitrogen from the soil such as **nitrates** and ammonium ions through their root. Animals obtain nitrogen as they feed on plant materials or plant products that contain nitrogen(proteins).

∴ ***Answer B***

56. Which of the following organelles are likely to be more abundant in active cells such us the muscle cells of human heart?
A .Lysosomes. C . Mitochondria
B .Chromosomes D . Golgi bodies.

Explanation:

- **Lysosomes** are found in most eukaryotic cells. They are surrounded by a single membrane and a simple sacs that contain digestive enzymes that break down cellular waste and debris.
- **Chromosomes** are thin thread like structures compose of DNA (the genetic materials) and protein his tone. They are found in the nucleus of eukaryotic cells.
- **Mitochondria** often thought, as the power house of the cell and the sites where energy releasing process known as **respiration** take place. They are found more abundantly in active cells like muscle cells.
- **Golgi bodies** consists of a number of flattened membrane bound sacs in which proteins are modified.

∴ ***Answer C***

57. Which of the following pairs are both organelles concerned with energy transformation?
A .Nucleus and nucleolus.
B .Mitochondria and nucleus.
C . Chloroplast and vacuole
D .Chloroplasts and mitochondria.

Explanation:

- **The nucleus** is the largest organelle found in all eukaryotic cells. It contains the genetic information of the cell (DNA).
- **Mitochondria** are the sites of most of the reaction of aerobic respiration (energy releasing process).Most of the energy locked in organic food required for the activities of the organism are generated here in the mitochondria. That is why it is called the **power house of the cell.**
- **Chloroplasts** are photosynthetic plastids containing chlorophyll and other photosynthetic pigment. It is found in all plant cells and algae. Chloroplasts are the sites of photosynthesis where the radiant energy (energy) from the sun is converted(transformed) in to chemical energy locked in organic compounds.
- **Vacuole** is a spherical fluid- filled organelle of variable size found in plant and animal cells bound by a single membrane and functioning as a compartment to separate variety of materials from the cytoplasm.

∴ ***Answer D***

58. How is it possible for different species to coexist in an ecosystem?
A .Each species has a specific biome of its own.

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B .Each species occupies a different niche within the ecosystem.
C .There is no competition for resources among species in an ecosystem.
D. All resources are equally shared among species in an ecosystem.

Explanation: **Population** of many different organisms are able to live in the same area (coexist) because each exploits a different habitat in that area or ecosystem. Sometimes, different organisms can even share the same habitat because they make different demands on that habitat. This means that they occupy different niches.

∴ ***Answer B***

59.If it is known that the total amount of DNA in a cell contains 300 nucleotides and adenine contributes to 80 of these, how many nucleotides go to guanine?

A .35 B .70 C .80 D .140

Explanation: In DNA molecules **adenine** always pairs with **thymine** and **guanine** with **cytosine**. The number of Adenine is always equal to the number of thymine and similarly the number of guanine is equal to that of cytosine.

✓ If a DNA molecule contains 300 nucleotides and 80 of the nucleotides contain adenine, the number of thymine that bind to the adenine is also 80.

= 80 adenine + 80 thymine = 160 nucleotide bases.
=300 nucleotides base – 160 nucleo tide bases
= 140 nucleotide base (guanine and cytosine).
= the number of guanine is equal to the number of cytosine.
= $\frac{140}{2}$ = 70 guanine and 70 cytosine.

∴ ***Answer B***

60. Which one is the correct direction of transfer of genetic information in most living things?

A .Protein ⟹ DNA ⟹ mRNA.
B . DNA ⟹ Protein ⟹tRNA
C. Protein ⟹ RNA ⟹ DNA
D .DNA ⟹ RNA ⟹ Protein

Explanation: The DNA inherited by an organism controls the activities of each cell by specifying the synthesis of enzymes and other proteins.

✓ **Gene (DNA)** does not build protein directly but instead dispatches instruction in the form of RNA , which intern programs protein synthesis. Cells are thus governed by a chain of command:

DNA ⟹ RNA ⟹ Protein

∴ ***Answer D***

61. Which of the following are **NOT** included in lipids?

A .Nitrogenous bases C .Steroids
B .Waxes. D . Phospholipids.

Explanation:

- **Nitrogenous bases** are nitrogen containing organic bases and are the components of nucleotides that make nucleic acids (DNA and RNA).
- **Waxes** are one of a group of lipids with a very high molecular weight; they are Esters of long chain alcohol with fatty acids.
- **Steroids** are commonly classified as lipids because their solubility characteristics are similar to those of fats ,oils ,waxes and phospholipids.
- **Phospholipids** are lipids containing phosphate group. The commonest type is formed when one of the three- OH groups of glycerol combines with phosphoric acid instead of a fatty acid.

∴ ***Answer A***

62. Sodium ions pass from a region where they are at lower concentrations to a region where they are found at higher concentration in human cells. This is an example of which of the following processes?

A .Osmosis C . Passive transport
B .Simple diffusion D .Active transport.

Explanation:

- **Osmosis** is the passage of water molecules from a region of its higher concentration to a region of its lower concentration across partially permeable membrane.
- **Simple diffusion** is the movement of molecules or ions from a region of higher concentration to a region of lower concentration down a diffusion gradient.
- **Passive transport** is a type of movement of particles from a region of higher concentration to a region of lower concentration without expenditure of energy in such as simple diffusion, osmosis and facilitated diffusion.
- **Active transport** is an energy consuming transport of molecules or ions across a membrane against a concentration gradient, that is from a region of lower concentration to a region of higher concentration.

∴ ***Answer D***

63.Choose the structure that is usually present only in the cells of animals.

A .Vacuole B .Cell wall C . Nucleus D . Centriole .

Explanation: **Vacuole** is a spherical fluid filled organelle of variable size found in plant and animals cells, bound by a single membrane.

- **Cell wall** is a rigid wall surrounding the cells of bacteria, fungi, algae and plants.

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- **A nucleus** is the largest organelle found in all eukaryotic cells of plants and animals. It contains the genetic information of the cell(DNA).
- **Centriole:** are two small cylindrical bodies , which are oriented at right angles to each other, in animal cells at the intrphase stage of cell division in a special region of the cytoplasm just outside the nucleus.

∴ ***Answer D***

64. What is the best term that refers to the maximum number of individuals that a given environment can support over a long period of time without degradation?

A .Density
B .Climax community
C . Carrying capacity
D . Limiting factor.

Explanation: **Density** refers to the number of living organisms occupying a given area at a given period.

- **Climax community** is the final or stable community in a succession of natural plant communities in one area under a particular set of conditions.
- **Carrying capacity** is the maximum number of individual organisms a given area of land is capable of supporting (carrying) at a particular period.
- **Limiting factor** is any factor in the environment that alone governs the behavior of an organism or system by being above or below a certain level.

∴ ***Answer C***

65. Which of the following is true about both DNA and RNA?

A .Both are single stranded.
B .Both have a five carbon sugar.
C .Both are polymers of amino acids.
D .Both contain the same four nitrogenous bases.

Explanation: DNA and RNA are nucleic acids made from monomer of nucleotides.

DNA	RNA
Double stranded	Single stranded.
Contain five carbon sugar.	Contain five- carbon sugar.
Polymers of nucleotides.	Polymer of nucleotide.
Contains base Adenine, thymine ,guanine and cytosine.	Contain base adenine, Guanine ,cytosine no thymine but uracil.

∴ ***Answer B***

66. A field biologist who studies the behavior of primates in the forest is most likely expected to frequently use which method of data collection?

A .Experiment.
B .Animal dissection
C . Observation
D. Comparative biochemistry.

Explanation: **Experiment** is an important component of a scientific method used to test a hypothesis.

- **Animal dissection** is cutting in to different parts of the animals to study its various structures.
- **Observation** is the foundation of all modern science is observation. Our entire view of the natural worded depends on the accurate recording of sensory data (data obtained by seeing , tasting , touching ,smelling and hearing)and the organization of these data in to general concepts.
- **Comparative biochemistry** is one of the evidence used by evolutionists to suggest that organisms that possess similar biochemical are closely related ancestrally.

∴ ***Answer C***

67. Which one of the following types of mutations is responsible for sickle cell anaemia?

A . Addition of a base pair.
B .Substitution of a base pair .
C . Deletion of a base pair
D . A shift in the reading frame of the genetic code.

Explanation: **Addition of base pair or addition mutation** is a kind of point mutation (gene mutation) where an extra base is added in to a gene. It is more significant mutation than substitution mutation. The reason for this is , it doesn't just alter the triplet in which mutation occur. Because there is one extra base, the whole sequence after the point of the mutation is altered.

- **In substitution of a base pair or substitution mutation :** one base is replaced by a different base. A substitution of just one base in the sixth triplet of the gene coding for one of the four polypeptide chins in the hemoglobin molecule alters the triple from **GAG to GTG** .This results in the amino acid valine replacing glutamine in the polypetide chain. The different hemoglobin molecule formed results in the condition known as **sickle cell anaemia.**
- **Deletion of base pair or deletion mutation** is a point mutation where a base is **"missed out"** during replication of DNA .It is more significant than substitution mutation in that it results a frame shift mutation.

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❖ **A shift in the reading frame of the genetic code** is the result of addition or deletion mutation

∴ ***Answer B***

68. If one keeps on adding a substrate to a fixed amount of an enzyme, when does the reaction rate reach a plateau?
A .When all the substrate is converted to products.
B .When all the active sites of the enzyme are occupied.
C .When most of the enzyme molecules are consumed.
D .When much of the products are accumulated.
Explanation: A small number of substrate molecule means few collisions between the substrate and the enzyme molecule to form enzyme substrate complex. Increasing the concentration of the substrate means more collisions and more enzyme- substrate complexes. So, the overall rate of the reaction is increased. Increasing the substrate concentration beyond a certain point will have no effect on the activity of the enzyme because all the active sites are occupied all the time.

∴ ***Answer B***

69. In Ethiopia, which of the following groups of the vertebrates is the most diverse in terms of species number?
A .Mammals B .Birds C . Reptiles D . Amphibians.
Explanation: In Ethiopia there are 277 species of **mammals** out of which 22 are endemic.
❖ In Ethiopia there are 86 species of **birds** out of which 27 are endemic (found only in Ethiopia).
❖ **Reptile** : In Ethiopia there are 78 species out of which 3 are endemic.
❖ **Amphibians** there are 63 species of amphibians out of which 17 are endemic.

∴ ***Answer B***

70. Which one of the following objective and ocular combinations of the light microscope, respectively, gives the largest field of vision?
A . 10 X and 10 X C . 40 X and 12 X
B .4X and 10 X D. 100 X and 10 X
Explanation: **Magnification of a microscope** is the magnification power of the objective lens times the magnification power of the ocular lens.
❖ $10\,X \times 10X = 100X$
❖ $4X \times 10X = 40\,X$
❖ $40\,X \times 12X = 480X$
❖ $100X \times 10\,X = 1000X$

∴ ***Answer D***

71. Which of the following land biomes occupies a large part of *Brazil*, parts of equatorial west Africa and Southeast Asia?
A .Acacia forest.
B .Deciduous temperature forest
C .Coniferous forest.
D .Tropical rainforest.
Explanation: **Tropical forest** is one of the most complex and species – rich ecosystem in the world. It is found in south and central America (particularly around Amazon basin, in Brazil)west and equatorial Africa and south east Asia.

∴ ***Answer D***

72. Which property of water keeps the bottoms of lakes and the life there in from being frozen during cold winters in the temperate regions?
A .Its highest density a 4°C.
B .Its high heat of vaporization.
C .Its high boiling point.
D .Its high latent heat of fusion.
Explanation:
❖ **The density of water** decreases below 4 °C and ice there fore tends to float.It is the only substance whose solid form is less dens than its liquid form. Since ice floats , it forms at the surface first and the bottom last. If ponds froze from the bottom upwards, fresh water life could not exist in temperate or arctic climates.
❖ **Latent heat of vaporization or high heat of vaporization** is a measure of the heat energy required to vaporize liquid, which is used to overcome the attractive forces between its molecules so that they can escape as gas. A relatively large amount of energy is needed to vaporize water. This is due to the hydrogen bonding.
❖ **Latent heat of fusion** is a measure of the heat energy required to melt a solid, in this case ice with its high heat capacity water in the form of ice requires relatively large amounts of heat energy to melt it. Conversely, liquid water must lose a relatively large amount of heat energy to freeze.

∴ ***Answer A***

73. Which of the following phases of mitosis if blocked would produce a cell with twice as many chromosomes as the mother cell?
A . Interphase. C . Anaphase.
B .Prophase D . Telophase.
Explanation:
❖ **Interphase :** the non-dividing cell is said to be in the interphase stag. At this stage the nucleus is clearly visible as a distinct membrane bounded organelle and the cell is carrying out all the activities .

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- **Prophase :** at this stage of mitosis the nuclear membrane dis-integrates, centroils are formed , spindles are extended , chromomosomes become short, thick and double threaded (sister chromatids) and form bivalents.
- **Anaphase:** the chromosomes that formed the bivalents (the homologous pairs) separate (segregate)each moving to the opposite pole of the cell .
- **Telophase:** The two sets of chromosomes, having reached their respective poles, become enclosed in new nuclear membranes as spindle fibres , disappear and this is followed by the division of the cytoplasm in to two equally halves (cytokinosis) forming two identical daughter cells. If cytokinesis fails a single cell with double chromosome number as the mother cell is produced.

∴ ***Answer D***

74. Which one of the following is the mode of reproduction in bacteria?
A .Mitosis. C . Conjugation
B .Binary fission D . Lysogenization.

Explanation:

- **Mitosis** is the process by which a cell nucleus divides to produce two daughter nuclei containing identical sets of chromosomes to the parent cell. This followed by the division of the whole cell to form two daughter cells.
- **Binary fission** is a type of sexual reproduction which occurs in single celled organisms such as **bacteria, amaeba, paramecium** etc. It is the result of mitotic cell division.
- **Conjugation** is a pristine type of sexual reproduction where to conjugates(two different organisms) unite temporally and exchange their genetic material,
  Examples : paramecias, fungi and some bacteria.
- **Lysogenization** is a type of viral reproduction where the virus injects its genetic materials(DNA) in to the host cell and the DNA remains incorporated(integrated) in the DNA of the host cell.

∴ ***Answer B***

75. If AIDS patients are given antiretroviral drugs that are targeted against the protease enzyme, which one of the following stages of the HIV life cycle would be inhibited?
A .Entry into the host cell.
B .Reverse transcription.
C .Integration of viral DNA into host DNA.
D .Assembly of viral parts into a whole virus.
Explanation: **Protease** is an enzyme that cuts protein chain in to parts for a new HIV particle, which can bud from the host cell and infect

other cells. Therefore , anti retroviral drugs that inhibits the action of protease enzyme known as protease inhibitors prevent the assembly of the new virus particles.

∴ ***Answer D***

76. Which of the following is true about those bacteria which dwell in hot springs?
A .They are not metabolically active.
B .They use non- protein enzymes.
C .They get energy of activation from the spring heat.
D .Their enzymes do not denature at that temperature.
Explanation: **Bacteria** that require high temperature (around 60°C) for growth are known as **thermophilic bacteria.** These bacteria survive such conditions because their enzymes are not denatured at this temperature.

∴ ***Answer D***

77. Choose the disease that is caused by what is known as the "droplet infection".
A .Flu B.AIDS C .Cholera D .Malaria.

Explanation:

- **Flu or Influenza** is respiratory disease caused by viral infection and transmitted by droplet infection.
- **AIDS** is caused by the infection of HIV particles. It is transmitted by sexual intercourse and blood to blood contact.
- **Cholera** is caused by drinking contaminated water and it is the result of bacterial infection.
- **Malaria** is caused by protozoan's called **plasmodium** and transmitted by **female anopheles mosquito.**

∴ ***Answer A***

78. Why is it that the typical diploid chromosomes number of many organisms including human beings is an even number?
A .It is only a coincidence.
B .Chromosomes duplicate before cell division.
C .Both parents contribute equal number of chromosomes.
D .Meiosis reduces chromosomes number.
Explanation: The **diploid number of chromosomes** of many organisms arise from the fusion of gametes (sex cells) from different parents having the same number of chromosome(haploid) that form the zygote with diploid number of chromosomes. The zygote continue to divide to form an embryo which develops in to a full adult organism with diploid number of cells. The diploid number of chromosomes in an organism is the result of the sum of two haploid cells(gametes).

∴ ***Answer C***

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79. Which of the following helps to keep the biological membranes in a fluid state?
A .Waxes B .Phospholipids C . Cholesterol D . Water.
Explanation: **Phospholipids** form the bulk of the cell membrane (biological membrane). Membranes also contain **cholesterol**. Like unsaturated fatty acids , cholesterol disturbs the close packing of phospholipids, and keep them more fluid. This can be important for organism, living at low temperature where membranes can solidify.
∴ ***Answer C***

80. With which of the following hibernation is most associated in animals?
A .Migration
B .Daily biological cycle
C . Annual biological cycle.
D .Availability of food in the store.
Explanation: The term "biological clock" is used to describe some internal regulatory mechanism that controls various cyclical responses in living things. Both plants and animals show yearly, monthly, daily and other cyclical changes that are genetically programmed.
The daily rythms are called **circadian rythms**. The annual biological clocks or the cycles of response they occur annually is called **circannual(yearly)rhythm**. Many animals show circannual rhythms in behaviors such as migration from one region to another and hibernation remaining inactive or seasonal sleep.
∴ ***Answer C***

81. If, due to incomplete dominance, the F1plants from a cross of red flowered x white flowered parents are pink, which of the following ratios are expected in the F2 generation?
A .1 red: 2 pink : 1 white C . 1red: 1 pink : 1 white
B .2 red: 1 pink : 1 white D . 3 red : 1 pink : 1 white.
Explanation: In complete dominance or co-dominance is a condition where neither of the alleles are nor dominant or recessive. Both alleles expressed equally producing an intermediate phenotype different from that of the parents.
If a plant producing red flower and another plant with white flower are cross- bred all of the F1 appear pink
✓ R –represent an allele for red colour and r –for white.
✓ parents – Red flowered plant × white flowered plant
✓ parental geno types RR × rr
✓ meiosis (gamete formation) R R r r
✓ Gametes R ,R × r, r

✓ Fertilization – F_1

	r	r
R	Rr	Rr
R	Rr	Rr

All of the plants produced are pink flowered plants.
✓ If the F_1 are selfed.
Pink flowered plant × pink flowered plant.
✓ Genotype Rr × Rr
✓ Meiosis R R r r
✓ Gametes R, r R, r
✓ Fertilization – F_2

	R	r
R	RR	Rr
r	Rr	rr

✓ ¼ of the plants have red flowers (RR).
✓ 2/4 of the plants have pink flower (Rr).
✓ ¼ of the plants have white flower(rr).
✓ Therefore, the plants appeared in the ratio of 1 Red:2pink :1 white
∴ ***Answer A***

82. From which of the following sources did Charles Darwin get the idea that selection can change living organisms? From:
A .the work of plant and animal breeders.
B .his observation of the Galapagos birds.
C .the evolution theory of Lamarck.
D .the book by Malthus.
Explanation: The 14 species that **Darwin** encountered in the Galapagos Islands differ greatly in size and other features such as break morphology. The variations in break morphology reflect differences in feeding habits. Darwin and other scientists suggested that soon after the islands were formed they were invaded by a flock of finches from the main land, which occupied different ecological niches and eventually evolved in to different species as a result of different selection pressure in the different islands.
∴ ***Answer B***

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83. Wood lice increase their movement in different directions in response to increased light intensity. This is an example of which of the following?
A .Positive taxis
B .Negative taxis
C. Kinesis
D .Phototropism.

Explanation: **Taxis** is a response in simple animals in which the animal moves along the gradient of intensity of the stimulus towards the greatest intensity of the stimulus(**a positive taxis**) and sometimes away from the greatest intensity (**negative taxis**).

- **Kinesis** again a response in simple animals in which a change in the intensity of the stimulus brings about a change in the rate of movement not a change in the direction of movement. For example, wood lice increase their rate of movement in bright light. This increases the probability that they will move into darkness, where it is usually more humid and they will lose less water.
- **Phototropism** is the growth response of plants where shoots are growing toward the direction of the light.

∴ ***Answer C***

84. Whic one of the following is most important for speciation to occur?
A .Hybridization among populations.
B .Isolation of the gene pool.
C .Migration between populations.
D .Free flow of genes between populations.

Explanation: As long as two population that are in the process of evolution or process of formation of distinct **gene pool** are able to inter breed, they are unlikely to evolves into distinct species. They must somehow go through a period when they are prevented from inter breeding. That is prevention of gene flow between the two gene pools that are becoming distinct. All forms of speciation involve isolating mechanisms that prevent different population from interbreeding (gene flow)for a period of time.

∴ ***Answer B***

85. Which of the following is a threat to the survival of small populations?
A .Habitat destruction.
B . Breeding in captivity.
C . Disease resistance.
D .Absence of competition.

Explanation: If **a habitat** that provides shelter, food and a place to breed to any population of organisms is threatened or destroyed the populations of that particular habitat .

∴ ***Answer A***

86. Which of the following descriptions is true about the action of selective pressures in populations? Selective pressure that acts:
A .Around the mean value is stabilizing.
B .Around the mean value is disruptive.

C .At both ends of the distribution is directional.
D .At one ends of the distribution is stabilizing.

Explanation: Refer question 33 in 2003 year.

∴ ***Answer A***

87. Which one is a biotic factor that can influence a plant?
A .The PH of the soil.
B . A pollinating animal.
C .CO_2 concentration.
D .The amount of radiant energy

Explanation: **Biotic factors** are the effects of other organisms of the same species or different species.

✓ **A biotic factors** are the effects of factors in the physical environment (light, temperature, carbon dioxide, concentration, oxygen concentration, PH of the soil, physical space etc).

- **The PH of the soil** is a biotic factor in the physical environment.
- **A pollinating animal** is biotic factor where one organism affects another organism.
- **CO_2 concentration** is abiotic factor.
- **The amount of radiant energy** is abiotic factor

∴ ***Answer B***

88. Why is it that the leaves and the soft young stems of plants that have started wilting become stiff again when they are provided with water ?This is because of:
A .fast intake of minerals.
B .the cooling effect of water.
C . increased turgor pressure
D . increased rate of photosynthesis.

Explanation: The leaves and soft stems of plants wilt due to loss of turgidity in their cells that is caused by inadequate water absorption from the soil.

✓ When there is enough amount of water in the soil, the plant absorbs this water through its root and transport it up through the stem to the leaves. As more water enters ,the cells of the leaves and soft young stems. by osmosis the cells again become turgid and stift resulting a total siftness in the plant which is the result of increase in turgor pressure of every cell.

∴ ***Answer C***

89. Animals without close evolutionary relations are sometimes seen to have similar structures adapted for the same function. Which evolutionary principle is illustrated by this observation?
A .Convergent evolution
B .Divergent evolution

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C . Sympatric evolution
D .Allopathic evolution.

Explanation:

- **Convergent evolution** occurs when different organism that are not related evolutionary occupy similar niches. The selection pressure on the population are the same and so similar structural adaptations evolve over time.
- **Divergent evolution** occurs when a basic type " **diverges**" along different lines because of different selection pressure in different environment.
- **Sympatric speciation (evolution)** is a speciation that need not involve physical separation. The two diverging populations may inhabit the same area , but be prevented from breeding in a number of ways.
- **Allopathic evolution** , the species become isolated by same physical feature which we call it geographical isolation.

∴ ***Answer A***

90. If it is known that the total number of the purine bases account for 50% of a DNA molecule and if each of the remaining bases are known to have the same proportion, what proportion is accounted for by Thymine alone in the same molecule?

A .25% B .50% C. 75% D. 100%

Explanation: There are two families of nitrogenous bases :**pyrimidines** and **purines** .A pyrimidine is characterized by a six membered ring made of carbon and nitrogen atoms. The members of the family are **cytosine (C) ,thymine (T) and Uracil (U)** .In the second family the purines, a five membered ring is fused to the pyrimidine type of ring. The purines are **adenine (A) and guanine (G).**

If the total number of purine bases (**adenine and guanine**) account for 50% of DNA molecule.The remaning base are known to have the same proportion. Therefor ,the pyrimidine bases (thymine, cytosine) composes 50% of the total bases in the DNA molecule.

- ✓ **Adenine** accounts 25% of the purine bases in this particular DNA.
- ✓ The number of adenine is always equal to the number of thymine.

Therefore, thymine accounts 25% bases in this DNA molecule

∴ ***Answer A***

91.Which of the following is based on sufficient supporting data?

A .Experimentation C . Theory
B .Prediction D .Hypothesis.

Explanation:

- **Experimentation :**hypothesis are often tested by experimentation.
- **Prediction:** an educated guess as to how the biologist thinks his/her experiment will turn out. **Prediction** is made based on the hypothesis.

- **Theory :** A hypothesis that has with stood many tests and has been shown to allow predictions to made is known as a **theory**. A theory may generate such confidence through its predictive ability to be known as a law. Some special hypothesis have been transformed to theories as a result of being tested by repeated observation and accumulation of supporting evidence (data).
- **Hypothesis** could be defined as a suggested explanation of certain observed phenomena. Scientific hypothesis need to be tested since they are assumptions of tentative explanations.

∴ ***Answer C***

92. What is the term used to express the ability of a microscope to show details clearly?

A .Enlargement C .Magnification
B .Reduction. D .Resolution.

Explanation:

- **Magnification** refers to multiplication of the size of the object so that it looks several times bigger than its actual size.
- **Resolution** refers to the scattering of parts of the object so that the details are seen.

∴ ***Answer D***

93.In what ways are the modes of nutrition of animals and saprophytes similar?

A .No enzymes are produced.
B .Digestion is external to their cells.
C .They build up proteins from light.
D .They are autotrophic organisms.

Explanation: **Saprophytes** feed by a method known as saprobitic **nutrition**. They feed on dead matter and digest the parts of the dead animals and plants by secreting enzymes on to the dead matter. The enzyme digest the organic molecules in to simpler ones and the micro-organism absorb these products of digestion. This is known as **extracellular digestion**.

- ✓ In **animals digestion** occurs outside their cells, in the gut, that is extracellular digestion similar to saprophytes.

∴ ***Answer B***

94. Which of the following is **NOT** an important unifying theme in biology?

A .The evolution of life.
B .The diversity and unity of life
C .The interdependence of living organisms.
D .The variation in climatic factors.

∴ ***Answer D***

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1.How many different kinds of amino acids are there for protein synthesis?

A .Twenty
B .Twenty – four
C . Twenty -six
D .Thirty- two

Explanation: There are 20 amino acids are used to make all the different proteins.

∴ ***Answer A***

2.Which of the following compounds is an important component of the bacteria cell wall?

A .Chitin
B .Peptidaglycan
C .Cellulose
D. Pectin

Explanation:

- **Chitin** is the major component of the cell wall of fungi.
- **Peptidaglycan** is an important component of the bacterial cell wall.
- **Cellulose** make the bulk of the cell walls of plants and algae.
- **Pectin** is a group of highly variable complex carbohydrates found in middle- lamella of plant primarily cell wall.

∴ ***Answer B***

3.Which one of the following is **NOT** recycled between organisms and the environment in an ecosystem?

A .Energy
B .Carbon
C .Nitrogen
D. Phosphorus.

Explanation: **Energy** is not recycled in any ecosystem. Energy flows through the ecosystems and is continually lost as heat from the ecosystem and has to be replaced as light. Nutrients, however, are recycled. There is a finite amount of each nutrient, such as **carbon, nitrogen and phosphorus** in the ecosystem and so the same atoms must constantly be re- used , over and over .

∴ ***Answer A***

4. As was shown by Gregore Mendel in garden pea, what percentage of the F_2 generation of a monohybrid cross has the recessive phenotype?

A .75%
B. 50%
C.25%
D.12.5%

Explanation: In a monohybrid cross breeding of garden pea conducted by Gregore Mendel; when a pure plant showing the dominant trait was cross- bred with another pure plant showing the recessive trait, all of the off spring (F_1) showed the dominant trait. When again these offspring were selfed, their off spring (F_2) appeared in the ratio of 3:1 .3 plants having the dominant trait to one plant having the recessive trait. It means that ;3/4 of the plants have the dominant trait(75%).

¼ of the plants have the recessive trait(25%).

∴ ***Answer C***

5.During seasons of reproduction, the males of some species of birds produce colourful feathers to attract females. What do ethologists call this method of communication in animals?

A .Visual
B .Chemical
C .Auditory
D .Touch

Explanation:

- Communication by **visual display** is common in many animals. These animals have evolved a variety complex actions that function as signals when seen by the other individual. These actions are displayed by the male or female animal in courtship behavior to attract one of the sex as motivational stimulus during seasons of reproduction. The motivation signals (stimulus) could be coloration changes increase the size of parts of the body. Plumage of birds (changing the colour of feathers), flufting of feather etc.
- **Chemical:** Many insects use chemical sex attractants instead of mating call. Female silk worm moth, for example, released a sex attractant so powerful that males are attracted to a single female from distance of two kilometers or more.

These chemical sex attractants are also known as **pheromones.**

- **Auditory:** This is communication via sound. The croaking sounds of frogs are species specific and are used for calling mates. Bird songs function primarily as a species recognition signals or are used for calling mates (male or female bird)or as a display that attracts females to the male and contributes to the synchronization of their reproductive drives.
- **Touch:** This is a mechanical means of attracting the opposite sex. For example, male spiders approach the web of a female sitting in the center of the web and pluck a thread of the web at a species- specific frequency.

∴ ***Answer A***

6.What do we call the substance upon which an enzyme acts?

A .Product
B .Substrate
C .Activation energy
D. Enzyme- substrate complex.

Explanation:

- **Product** is a substance that is produced at the end of enzyme catalyzing biochemical reaction in living organisms.
- **Substrate** is a molecular or part of it on which the enzyme acts in chemical reaction catalyzed by enzymes.

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❖ **Activation energy** is the extra energy required to start a reaction between substance in enzyme catalyzing reaction. The speed of a reaction depends on the activation energy necessary to initiate it.
Enzyme substance complex is a complex formed as the enzyme binds to a substrate molecule temporarily in enzyme catalyzing reactions to produce the product.

∴ ***Answer B***

7. Which one of the following is **NOT** a mutation?
A .DNA replication to form two daughter DNAs.
B .Gain of an extra chromosome by a cell.
C .Deletion of a base pair from DNA.
D .Loss of chromosomes by a cell.

Explanation:

❖ **DNA replication to form two daughter DNA** is a process that occurs in every dividing cells in both mitosis and meiosis so that the genetic materials of the parent cell (original cell) passes to the daughter cells. It is because of this DNA replication that characteristics pass from parents to off spring or from one generation to the next. It is not a mutation.
Mutations are rare phenomena.
❖ Cells can gain extra chromosomes due to non-disjunction of chromosomes at meiosis .This is known as **chromosomal mutation.**
❖ **Errors can occur during replication of DNA** which might result deletion or missing out of a single base in a sequence of bases in a molecule of DNA .This is known as a **deletion mutation** which is a type of **point mutation.**
❖ **Loss of chromosome by a cell** reduces the chromosome of a cell and it is a type of chromosomal mutation.

∴ ***Answer A***

8. What do geneticists call the genotype in which the two alleles of a pair are identical?
A .Dominant
B .Recessive
C .Homozygous
D. Heterozygous

Explanation:

❖ **Dominant:** When two alleles of a pair are different only one of the allele expressed itself. The allele that expressed itself is called the **dominant allele.**
❖ **Recessive:** When two alleles of a pair are no identical, only one of the allele is expressing itself. The other remain hidden or masked by the dominant allele. The allele the remains hidden or masked is called the **recessive allele.**
❖ **Homozygous:** is a trait containing two alleles that are identical like RR –homozygous round or rr- homozygous wrinkled.

❖ **Heterozygous** is a trait containing two alleles that are different for a particular characteristic . For example :R r – for round seed shape and Tt for plant height.

∴ ***Answer C***

9. Which of the following corresponds to the beginning step of a scientific work?
A .Testing hypotheses
B .Making observation
C .Conducting experiments
D .Drawing conclusions.

Explanation: Any **scientific investigation** is started by making observation which is the foundation of all modern science.
✓ **Observations** provide the raw material, which leads to the formulation of a **hypothesis** which could be defined as a suggested explanation of certain observed phenomena. This is followed by testing the formulated hypothesis. **Hypothesis** are often tested by experimentation (conducting experiment).
✓ The results obtained after experimentation are analyzed and conclusions are drawn based on the experimental result analysis or
Observation → Hypothesis → Experimentation → Conclusion

∴ ***Answer B***

10. Which tool of the biologist is more suitable for culturing bacteria in the laboratory?
A .Test tubes
B .Microscopes
C. Petri dishes
D .Beakers.

Explanation:

❖ **Test tubes** are glass ware used in chemistry and biology laboratory to carry out some chemical reactions or test chemical reaction.
❖ **Microscopes** are scientific instruments which makes very small objects look bigger so that more details can be studied.
❖ **Petri dish** are round dishes made from glass or from plastic. They are used in many different ways, but usually to culture some organisms such as bacteria.
❖ **Beakers** are glass ware jars which are used in chemistry and biology laboratories.

∴ ***Answer C***

11. Which of the following is the best collective name for all bacteria with spherical shapes?
A .Cocci
B .Bacilli
C . Spirochaetes
D .Streptococci.

Explanation:

❖ **Cocci** are spherical shaped bacteria.
❖ **Bacilli** are (singular bacillus)are rod shaped bacteria.
❖ **Spirochaetes** (Singular – Spirillum) are spiral shaped bacteria.

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❖ **Streptococci** are cocci in chain.

∴ ***Answer A***

12. How many carbon atoms are contained in a single molecule of sucrose?
A .6 B . 12 C.24 D.48

Explanation: **Sucrose** is a disaccharide made from two monosaccharide's, each contain 6 carbon atoms. Therefore , sucrose contain 12- Carbon atoms.

∴ ***Answer B***

13. Which of the following statements is in agreement with the modern cell theory?
A .Cells come from nothing
B .Cells come from existing cells.
C .Cells come from non-living material.
D. Cells arises by means of spontaneous generation.

Explanation: The modern cell theory states that all cells come from pre-existing cells by cell division. There is no spontaneous generation of cells. Cells do not arise spontaneously from non-living material.

∴ ***Answer B***

14. What is the term that refers to all parts of the earth where living things are found?
A .Population C. Biosphere
B .Ecosystem D. Environment

Explanation:

❖ **Population** is a group of closely related organisms that normally interbreed and live together.
❖ **Ecosystem** is a natural unit in which energy flows from the source to producers, consumers and decomposers, and nutrients circulate from the soil through plants animals and back to the soil through the action of micro-organisms.
❖ **Biosphere** :This is the part of the earth that contains life.
❖ **Environment** is the surrounding that consists of all things external to the organism that in any way act up on it.

∴ ***Answer C***

15. In which kingdom of life are the unicellular eukaryotes grouped?
A . Monera C. Plantae
B . Protista D. Animalia

Explanation:

❖ **Monera:** This kingdom include organisms the cells of which are prokaryotic.
❖ **Protista:** According to some biologist kingdom protista contains only unicellular eukaryotes. of the kingdom protest are sub kingdom protozoa and sub kingdom algae.

❖ **Plantae:** Kingdom plantae includes organisms that are eukaryotic and multicellular.
❖ **Animalia:** They are multicellular eukaryotes.

∴ ***Answer B***

16. Select the function that living things are **NOT** capable of performing.
A .Maintain their internal body environment.
B .Pass genetic information to their offspring.
C .Respond to other organisms found in their surroundings.
D .Determine the amount of radiation reaching the environment.

Explanation: Some of the characteristic common to all living things are:

✓ **Homeostasis :** Is maintenance of a constant internal environment by an organism. It enables the cells to function more efficiently.
✓ **Reproduction :** The life span of organisms is limited, but they all have the ability to perpetuate life through reproduction then by ensuring the survival of the species. The resulting offspring have the same general characteristics' as the parents. This is the resulting of passing genetic information (DNA) which contain in the coded information to the off spring.
✓ **Response to stimuli:** Living things have the ability to respond to changes in both the internal and external environments and ensure that they maximize their chance of survival.

∴ ***Answer D***

17. Which factor has a more negative effect on the functions of enzymes than the others?
A .Neutral PH.
B. Very high temperature.
C . Optimal amount of salt concentration.
D .Optimal amount of substrate concentration.

Explanation:

❖ The PH scale is a measure of the hydrogen ion concentration of a solution or other liquid system. The PH scale ranges from 0 to 14. The majority of enzymes in most mammals function most efficiently within the PH range of 6.0 - 8.0 ,that is around the neutral PH.
❖ **The effect of temperature :** since heating increases the motion of molecules, the molecules of both the substrate and enzyme move faster increasing the chance of their collision.
✓ Consequently, the probability of a reaction to occur increases. For most reactions, the rate roughly doubles for every 10°C rise.However,if the temperature is increased above the optimum level, then a decrease in the rate of the reaction occurs despite the increasing frequency of collisions.

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This is because the secondary and tertiary structures of the enzyme get disrupted, and the enzyme is said to be **denatured.**

- There is no factor known as optimal salt concentration affecting the enzyme activity.
- **The effect of substrate concentration**: For a given concentration, the rate of the enzyme increases with increasing substrate concentration.

∴ ***Answer B***

18. Which group of organisms has a system of protein synthesis in which transcription and translation take place at separate times?
A .In all eukaryotic organisms.
B .In multicellular animals only.
C .Only in prokaryotic organisms.
D .In both prokaryotic and eukaryotic organisms.
Explanation: In prokaryotic transcription and translation are coupled; mRNA can be translated by ribosomes at one end of its molecule while it is still being transcribed from the DNA at the other end. Prokarytic mRNA does not need post- transcriptional processing.
In all eukaryotic transcription and translation are separated. Transcription occurs in the nucleus where as translation takes place in the cytoplasm. Eukaryotic mRNA are modified before leaving the nucleus

∴ ***Answer A***

19. To which genus of human- like organisms does Lucy belong?
A .The genus **Homo.**
B .The genus **Ardipithecus.**
C .The genus **Sahelanthropus.**
D .The genus **Australopithecus.**
Explanation :

- **The genus Homo** existed some 2 million years ago including a number of extinct species such as **Homoerectus, Homohabillis** and **Homo neander thalensis.** The only existing species of the genus is Homo sapiens – the modern man.
- **The genus Ardipithecus** existed between 4- 6 million years ago. One of the species of this genus is another famous fossil of Ardipithecus ramidius also known as Ardi found in Ethiopia and Kenya.
- **The genus Sahelan thropus** existed between 6 -7 million years ago.
- **The genus Australopithecus** existed some 3 -4 million years ago consisting the famous species Australopithecus afaransis also known as **Lucy.**

∴ ***Answer D***

20. What is the role of the worker honey bee just after it emerges?
A .Forage for nectar, pollen and water.

B .Clean out dirty honey comb.
C .Guard the hive.
D .Build honey comb.
Explanation : The roles of a worker bee at different stage of development are as follows:

- ✓ a newly emerged worker bee cleans out dirty honey comb and feed larvae.
- ✓ between 1 week and 2 weeks- **build** honey comb and receive food from incoming foragers.
- ✓ between 2 weeks and 3 weeks- **guard** the hive and remove corpses.
- ✓ between 3 week and death – **forage** for nectar, pollen and water.

∴ ***Answer B***

21. Which of the following pairs of elements are found in all carbohydrates in addition to the element carbon?
A .Nitrogen and oxygen
B .Hydrogen and oxygen.
C. Hydrogen and nitrogen.
D . Phosphorus and nitrogen.
Explanation : All **carbohydrates** contain the elements carbon, oxygen and hydrogen.

∴ ***Answer B.***

22. When an enzyme is denatured by heat or extreme PH, which one of the following does it lose?
A .The peptide bond.
B .Primary structure
C. Secondary structure.
D. Tertiary structure.
Explanation : When the enzyme is **denatured** by heat of higher temperature, it loses its **secondary structure and tertiary structure** which are important in maintaining the enzyme shape and active site that provides the enzyme its biological characteristic.
✓Significant change in PH or extreme PH can affect or denature an enzyme molecule by:

- breaking ionic bonds that hold the tertiary structure in place.
- altering the charge on some of the amino acids that form the active site.

∴ ***Answer D***

23. Which of the following ideas in the cell theory was contributed by Rudolf Virchow?
A .All plants are made up of cells.
B .All animals are made up of cells
C. Cells are the structural unit of life
D. Cells come from pre-existing cells
Explanation : An essential part of the cell theory is the idea, first proposed in 1855 by **Robert Virchow** that new cells come from pre-existing cells.

∴ ***Answer D***

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24. When the muscle cells are in short supply of oxygen, which of the following compounds would be accumulated in them?
A .Ethanol
B .Acetic acid
C. Lactic acid
D. Carbon dioxide

Explanation : The process in which a cell releases energy in the glucose in the presence of oxygen is called **aerobic respiration**. The process of aerobic respiration involves many separate chemical reactions that occur in a definite sequence.
The following equation summarizes the many reactions of aerobic respiration of glucose:

$$C_6H_{12}O_6 + 6O_2 \xrightarrow{\text{Enzyme}} 6CO_2 + 6H_2O + 36/38\ ATP$$

✓ Many animals carry out **anaerobic respiration** (respiration without the use of oxygen)at certain times. For example, athletes us an enormous amount of energy during a race. They cannot obtain enough oxygen to respire by normal aerobic respiration, and they do not have any special stores of energy. In such occasion the muscle cells" **switch**" to anaerobic respiration. In this case glucose is broken down without oxygen to produce large quantities of **lactic acid** and very little energy. It is sometimes called **lactic acid fermentation** and summarized by the following chemical equation:

Glucose $\xrightarrow{\text{Enzymes}}$ 2 lactic acid + 2 ATP

✓ There is another type of anaerobic respiration also known as **alcoholic fermentation** which occurs in micro- organisms like yeast and is summarized as follows:

Glucose $\xrightarrow{\text{Enzymes}}$ 2 molecules of Ethanol+2molecules of carbon dioxide +2 ATP

∴ ***Answer C***

25. Which one of the following is the main source of the greenhouse gases that are concentrating in the atmosphere of the earth?
A .Burning of fossil fuels.
B .Plants growing in green houses.
C .Photosynthesis by aquatic plants.
D .Respiration by animals and plants.

Explanation : The term **green house** first used in the 1930s, is meant to compare the heat- blocking action of atmospheric gases to the glass that covers a green house.
Some of the green houses are:
✓ **Carbon dioxide** produced from burning fuel.
✓ **Methane** :Produced from bacteria decay activity and natural feed lot.

∴ ***Answer A***

26. Which of the following fossils is the nearest to the common ancestor of the hominids and the apes?
A .Homo habilis
B .Homo erectus.
C. Ardipithecus ramidus.
D. Australopithecus afarensis.

Explanation :According to currently available fossil evidence and the evolutionary tree of humans and the great apes, the nearest common ancestor of apes and humans existed some 6 million years ago.
✓The extinct species of genus home, **Homo habilis and Homo erectus** existed between 1 million and 2 million years ago.
✓**Ardipethicus** existed between 4 and 5 million years ago.
✓**Australopithecus** afarans is existed between 3 and 4 million years ago.

∴ ***Answer C***

27. Which one of the following is referred to as the First law of Mendel?
A .The occurrence of alleles in pairs.
B .The dominance of one allele over the other.
C .The equal contribution of alleles by both parents.
D .The separation of alleles during gamete formation.

Explanation : These are **Mendel's explanations** of his experimental results sometimes known as the laws of Mendel in old biology text books.
✓Every trait is controlled by two **heritable factors** which we now call them genes or alleles.
✓If the two alleles in an individual are different one is dominant + (will be expressed in the organisms appearance or physiological and the other is recessive (cannot be expressed unless the individual has two copies of the recessive allele. Dominant alleles masks the expression of the recessive allele.
✓ Organisms get the alleles from their parents, one from each parent.
✓ **The alleles** or the heritable factors separate when the gametes (sex cells) are formed ,each gamete therefore contains only one allele controlling the trait. This is **Mendel's law of segregation.**

∴ ***Answer D***

28. Which one of the following types of microscope is the best to show the details of the surface of an object?
A .Optical microscope.
B .Compound microscope.
C. Scanning electron microscope
D. Transmission electron microscope.

Explanation :
✓ **Compound microscope** is a microscope in which the magnifying power of two convex lenses, the eye piece (ocular lens) and the objective lens are used to produce a magnified image of small objects.

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